A Regime-Switching Continuous Hidden Markov Model as a Reference Synthetic-Data Generator for Equity Returns

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Abstract

Synthetic equity-return generators support stress testing, regulatory backtesting, and scenario design, but existing approaches struggle to reproduce the three canonical stylized facts of daily returns simultaneously: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the absolute-return autocorrelation function (ACF). We show that a continuous hidden Markov model (CHMM) at moderate state resolution reproduces all three within a single EM scaffold. The mechanism is spectral: the absolute-return autocovariance admits the closed-form mixture-of-eigenvalues identity $\rho_{|G|}(\tau) = \sum_{k\geq 2} w_k \lambda_k^\tau$ over the non-unit eigenvalues of the transition matrix $\mathbf{T}$, recasting the low-K limitation of regime-switching HMMs as a K-rank statement on $\mathbf{T}$. The same scaffold accommodates Gaussian (CHMM-N), Student-t (CHMM-t), Laplace (CHMM-L), and Generalized Error Distribution (CHMM-GED) emissions through a single M-step swap; CHMM-GED carries a per-state shape p_k that nests Gaussian ($p = 2$) and Laplace ($p = 1$) as boundaries and adapts on the Gaussian-to-Laplace axis state by state. On ten years of daily SPDR S&P 500 ETF (SPY) data with a 2.5-year held-out window, CHMM at $K = 18$ attains 93.8% in-sample and 84.2% out-of-sample Kolmogorov–Smirnov pass rates and an absolute-return ACF mean absolute error within 0.003 of GARCH(1,1); VaR and Expected-Shortfall envelopes bracket observed values at 1% and 5% on both windows. CHMM occupies the joint-fit corner of the metric panel where i.i.d. baselines win on marginal fidelity and GARCH wins on volatility clustering. Multi-asset synthesis composes through a rank-based Student-t copula on the CHMM marginals: profile MLE selects $\nu^\star = 6$, reproducing the off-diagonal correlation matrix on six SPY-member assets with MAE 0.027.

Keywords: Continuous Hidden Markov Model, Baum-Welch, Student-t Emissions, Stylized Facts, Volatility Clustering, Spectral ACF, Value-at-Risk, Synthetic-Data Generator.

1 Introduction

Synthetic generators of equity-return time series support stress testing, regulatory backtesting, scenario design, and the training of downstream machine-learning models on data that cannot be released for privacy or licensing reasons [1–3]. The bar for a useful generator is that it reproduce, simultaneously, the canonical stylized facts of daily returns: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [4, 5].

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Generalized autoregressive conditional heteroskedasticity (GARCH) models [6, 7] capture volatility clustering but not the multi-regime structure of the marginal; i.i.d. generators match the marginal but destroy temporal persistence; deep generators [8–10] reproduce some stylized facts at the cost of interpretability and reproducibility.

An influential negative result has shaped two decades of work on the regime-switching route. Rydén et al. [11] fitted Gaussian-emission hidden Markov models (HMMs) at $K = 2$–3 states and concluded that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay; Bulla and Bulla [12] restored ACF fidelity by replacing the Markov chain with a hidden semi-Markov chain at the cost of substantially increased complexity.

We identify the spectral root cause. The absolute-return autocovariance of a stationary continuous HMM admits a closed-form mixture-of-eigenvalues identity $\rho_{|G|}(\tau) = \sum_{k\geq 2} w_k \lambda_k^\tau$ over the non-unit eigenvalues of $\mathbf{T}$ (Section 3.8); the empirical Rydén et al. result is then a K-rank statement on $\mathbf{T}$. At $K = 2$ the ACF is mono-exponential and slow decay can be matched by tuning λ_2 near 1; the binding constraint at $K = 2$ is instead the two-component marginal, whose representational capacity is insufficient for the empirical kurtosis target, failing distributional Kolmogorov–Smirnov (KS). At $K \geq 3$ the model admits multiple non-unit eigenvalues and a richer mixture, and the temporal and distributional constraints decouple. The semi-Markov route injects the missing flexibility through an explicit sojourn distribution; moderate-K continuous HMM achieves the same end through the spectral richness of $\mathbf{T}$ and the additional mixture components, inside the standard Baum-Welch scaffold.

We instantiate this on ten years of daily SPDR S&P 500 exchange-traded fund (ETF) (ticker SPY) data with a 2.5-year held-out window, using a continuous HMM trained by Baum-Welch [13, 14] under quantile-based initialization, in four emission variants sharing identical forward-backward recursions: CHMM-N with Gaussian emissions, CHMM-t with per-state Student-t emissions whose degrees of freedom ν_k are updated inside the M-step by a golden-section search on the expectation conditional maximization (ECM) Q-function in the tradition of Peel and McLachlan [15] and Liu and Rubin [16], CHMM-L with Laplace emissions whose location is the weighted median and whose scale is the weighted mean absolute deviation (MAD), both in closed form, and CHMM-GED with Generalized Error Distribution emissions whose per-state shape p_k is updated by a three-stage ECM (closed-form scale, bracketed golden-section for μ_k and p_k) and nests CHMM-N ($p_k = 2$) and CHMM-L ($p_k = 1$) as boundary cases. Against a benchmark panel covering the i.i.d. bootstrap (17), Gaussian and Laplace i.i.d., GARCH(1,1) (7), heavy-tailed GARCH(1,1)-t (18), and the regime-switching Markov-switching GARCH (MS-GARCH) foil of Haas et al. [19], the CHMM at moderate K attains 93.8% in-sample (IS) and 84.2% out-of-sample (OoS) KS pass rates and an absolute-return ACF mean absolute error (MAE) of 0.0507, essentially matching GARCH's temporal fidelity without sacrificing distributional fit. The four emission variants land at four distinct points on the kurtosis axis: CHMM-N undershoots (5.0 simulated vs. 7.7 observed), CHMM-t overshoots (14.4), CHMM-L matches cleanly (6.6), and CHMM-GED sits adaptively at 5.2 with its fitted $\hat{p}_k$ partition splitting bimodally into Gaussian-like bulk states and Laplace-like tail states. The unconditional Value-at-Risk (VaR) and Expected-Shortfall (ES) envelopes bracket observed historical values at both 1% and 5% on the IS and OoS windows. No single benchmark dominates every metric; the i.i.d. bootstrap is the strongest distributional baseline, GARCH is the strongest unconditional volatility-clustering baseline, and the CHMM is the strongest joint-fit row at moderate state resolution. Multi-asset synthesis composes cleanly through a rank-based Student-t copula on top of the CHMM marginals (profile maximum likelihood estimation (MLE) selects $\nu^\star = 6$; six-asset off-diagonal correlation MAE 0.027). Returns are assumed locally stationary across the ten-year IS window; substantial OoS degradation on individual single names (NVIDIA, ticker NVDA; JPMorgan Chase, ticker JPM) reflects the well-known stationarity-scope limit on financial time series [20–22] and motivates periodic refit at deployment in the spirit of Pesaran and Timmermann [23].

Our contributions are: (i) a closed-form mixture-of-eigenvalues identity for the absolute-return ACF that recasts the Rydén low-K failure as a K-rank statement on $\mathbf{T}$; (ii) four emission families (Gaussian, Student-t, Laplace, Generalized Error Distribution) under a unified ECM scaffold, where the M-step is the only architectural difference and CHMM-GED's per-state shape p_k adapts each state on the Gaussian-to-Laplace axis; (iii) a single-window IS / OoS empirical study on SPY plus a six-ticker generalisation panel and a six-asset Student-t copula extension.

The paper is organized as follows. Section 2 reviews related work; Section 3 describes the data, the continuous HMM, the EM estimator, and the spectral mechanism for the absolute-return ACF; Section 4 presents the empirical study and the VaR / ES envelope calibration; Sections 5–6 discuss and conclude.

2 Related Work

Hidden Markov models in finance. Hamilton [24] introduced Markov-switching to economics; Schaller and Van Norden [25] showed monthly equity returns admit a two-state Markov-switching specification. The seminal evaluation against the Cont stylized facts is due to Rydén et al. [11]: Gaussian-emission HMMs at $K = 2$–3 states fail the slow ACF decay of absolute returns because the geometric sojourn-time distribution of a standard Markov chain produces exponential decay too fast to match the observed signature. Bulla and Bulla [12] restored fidelity with hidden semi-Markov models at substantially higher complexity. Nystrup et al. [26] extends the same line to long-memory volatility. Maximum-likelihood fitting of heavy-tailed innovations follows Peel and McLachlan [15] and Liu and Rubin [16]; the Generalized Error Distribution [27, 28] provides a continuous shape parameter that interpolates between Gaussian ($p = 2$) and Laplace ($p = 1$) and underlies our CHMM-GED variant in the same role that the per-state degrees of freedom ν_k plays in CHMM-t. The discrete-state predecessor of this paper, Alswaidan and Varner [3], reproduced the same three stylized facts via Laplace quantile binning plus a Poisson-jump duration mechanism; here we replace the discretization step with continuous emissions and recover the same target through the spectral mechanism of Section 3.8.

GARCH and its extensions. GARCH [6, 7] encodes volatility persistence through a single decay parameter but imposes a unimodal innovation distribution and fails distributional Kolmogorov–Smirnov on equity returns. Heavy-tailed innovations [18], asymmetric leverage extensions (EGARCH, 29; GJR-GARCH, 30; threshold GARCH, 31), and regime-switching extensions (MS-GARCH, 19) close some gaps but inherit the unimodal-innovation constraint. Realised-variance frameworks [32, 33] provide a complementary forecasting baseline.

Deep generative models for financial time series. Generative adversarial network (GAN)-based generators reproduce the visual appearance of return series but typically fail volatility-clustering tests unless architectures encode temporal dependence explicitly [34, 35]; the TimeGAN family of Yoon et al. [8], the convolutional Wasserstein GAN (WGAN) QuantGAN of Wiese et al. [9], and the autoregressive-diffusion approach of Rasul et al. [10] are the standard references. We include QuantGAN as the deep-generative row in our extended panel (Appendix A.7.5); the present paper's contribution is grounded against parametric and non-parametric baselines.

Cross-asset dependence and synthetic-data evaluation. For multi-asset synthesis, the Single Index Model [36] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal; Sklar's theorem [37, 38] supports a cleaner decomposition in which each asset's fitted marginal is preserved and only the copula is estimated, with Gaussian and Student-t copulas [39, 40] the standard choices and rank-reordering simulation [41] the standard injection. Synthetic-data evaluation is anchored on the framework of Stenger et al. [42], the proper-scoring-rule literature [43–45], kernel two-sample tests [46], and signature-based metrics [47, 48].

3 Methods

3.1 Data and Excess Growth Rates

The single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024 (a ten-year training window, $T_{\mathrm{IS}} = 2{,}516$), with the period from January 4, 2024 through April 20, 2026 ($T_{\mathrm{OoS}} = 572$) held out. Cross-ticker generalization additionally covers NVIDIA (NVDA), Johnson & Johnson (JNJ), JPMorgan Chase (JPM), Apple (AAPL), and the Invesco QQQ Trust (QQQ). For ticker i and trading day j, the annualized excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \qquad \Delta t = 1/252, \quad r_f = 0. \tag{1}$$

The empirical study is organised along two pipelines, illustrated in Figure 1. *Pipeline A* fits a CHMM to each ticker independently and evaluates single-asset marginal, tail, and temporal metrics; this is the scope of the SPY headline panel (Section 4.3), the cross-ticker generalisation (Section 4.4), and the unconditional VaR / ES envelope (Section 4.6). *Pipeline B* reuses the per-asset Pipeline-A CHMM fits as *marginals* and injects cross-asset *dependence* through the rank-based Student-t copula of Section 3.4, evaluated in Section 4.5.

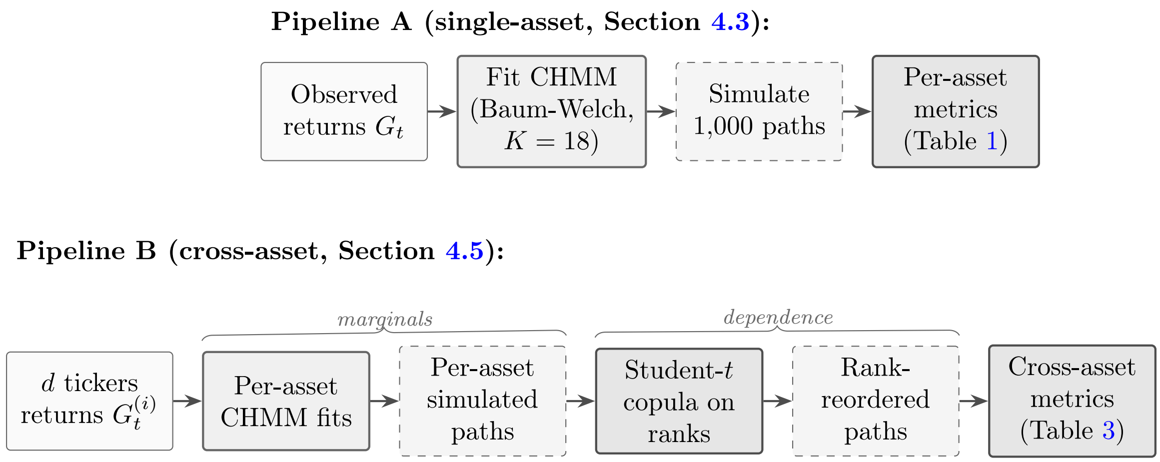


Figure 1. Schematic of Pipeline A (single-asset) and Pipeline B (cross-asset). Pipeline A fits a CHMM per ticker independently and evaluates single-asset marginal, tail, and temporal metrics. Pipeline B reuses the per-asset CHMM fits as *marginals* and injects cross-asset *dependence* through the rank-based Student-t copula (profile MLE selects $\nu^\star = 6$), preserving each asset's fitted CHMM distribution exactly while coupling asset ranks to match observed multi-asset correlation structure. Headline: the two pipelines share the single-asset scaffold, so improvements to the CHMM at the marginal step propagate into Pipeline B without any changes to the dependence layer.

3.2 Continuous Hidden Markov Model

We model the continuous excess growth rate G_t as a continuous HMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{B}, \boldsymbol{\pi})$ with K latent states, a per-state emission density $b_k(x; \boldsymbol{\theta}_k)$, transition matrix $\mathbf{T} \in \mathbb{R}^{K\times K}$, and initial distribution $\boldsymbol{\pi}$ [14]. The stationary marginal is the K-component mixture

$$f(x) = \sum_{k=1}^{K} \bar{\pi}_k \, b_k(x; \boldsymbol{\theta}_k), \qquad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}}\mathbf{T}, \tag{2}$$

and supplies heavy tails either through the spread of component moments (Gaussian components) or through the emission family itself (Student-t or Laplace). Existence and uniqueness of $\bar{\boldsymbol{\pi}}$ require irreducibility and aperiodicity of $\mathbf{T}$, recorded as Assumption 1 in Section 3.8; the spectral mechanism behind the absolute-return ACF that this mixture induces is derived in Section 3.8. We learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly from continuous observations via EM (Section 3).

CHMM-N: Gaussian emissions. $b_k(x) = \mathcal{N}(x; \mu_k, \sigma_k^2)$ with per-state location μ_k and variance σ_k^2.

CHMM-t: Student-t emissions. $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ with per-state location μ_k, scale σ_k, and degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. Per-state degrees of freedom let heavy-tailed regimes coexist with near-Gaussian regimes in the same model.

CHMM-L: Laplace emissions. $b_k(x) = \text{Laplace}(x; \mu_k, b_k)$ with per-state location μ_k and scale b_k. The heavier-than-Gaussian Laplace tail (per-component excess kurtosis 3) provides an alternative route to tail fidelity that carries no shape parameter and is strictly cheaper than CHMM-t.

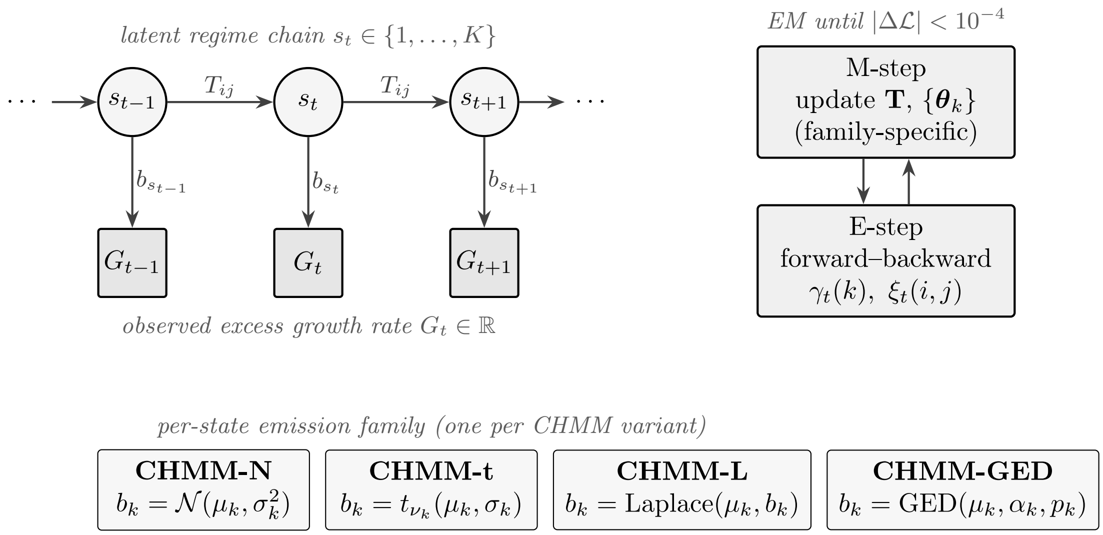


Figure 2. CHMM architecture and training loop. A K-state latent Markov chain with transition matrix $\mathbf{T}$ generates the regime sequence s_t; each s_t conditionally emits the observed excess growth rate G_t through a state-specific density b_{s_t}. The four CHMM variants (Gaussian, Student-t, Laplace, GED) share everything except the emission family. Joint estimation alternates an E-step (log-space forward–backward) and a family-specific M-step until the log-likelihood change falls below 10^{-4}.

CHMM-GED: Generalized Error Distribution emissions. $b_k(x) = \text{GED}(x; \mu_k, \alpha_k, p_k)$ with per-state location μ_k, scale α_k, and shape $p_k \in [p_{\min}, p_{\max}]$. The GED nests Gaussian at $p = 2$ and Laplace at $p = 1$, so per-state p_k is the structural analog of CHMM-t's per-state ν_k on the Gaussian-to-Laplace shape axis: each state picks its own kurtosis adaptively, by ECM, on a continuous shape parameter. CHMM-GED therefore strictly nests CHMM-N (all $p_k = 2$) and CHMM-L (all $p_k = 1$) as boundary cases.

3.3 Benchmark Generators

We benchmark the CHMM against representatives from each of the standard generator tiers, all fitted on the same IS window. The *i.i.d. bootstrap* [17] samples T returns uniformly with replacement, preserving the empirical marginal exactly and serving as the non-parametric distributional ceiling. The *Gaussian* i.i.d. generator draws $G_t \sim \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$ from the sample moments; the *Laplace* i.i.d. generator draws $G_t \sim \text{Laplace}(\hat{m}, \hat{b})$ from the sample median and mean absolute deviation. *GARCH(1,1)* [7] uses $G_t = \mu + \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha(G_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$ with $z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, fitted by Nelder–Mead maximum likelihood; this is the parametric volatility-clustering workhorse. *GARCH(1,1)-t* [18] replaces the Gaussian innovation with a Student-t, supplying a heavy-tailed volatility-clustering competitor with the same recursion. *MS-GARCH* (Markov-switching GARCH, 19) is the regime-switching foil: a discrete two-state hidden chain modulates the GARCH parameters, providing the closest peer to the CHMM scaffold among classical baselines. The full extended GARCH-family panel (EGARCH, GJR-GARCH, HAR-RV, MS-GARCH $K = 3$), the explicit semi-Markov CHMM foil, and a deep-generative QuantGAN [9] row are reported in Appendix A.7.7 (the QuantGAN architecture and training details in Appendix A.7.5).

3.4 Cross-Asset Dependence (Pipeline B)

The univariate analysis fits a separate CHMM to each ticker independently (Pipeline A). For multi-asset synthesis we extend Pipeline A through the rank-based Student-t copula [39, 40], which preserves each asset's fitted CHMM marginal exactly while injecting cross-sectional tail-coupled dependence. Of the three constructions evaluated in our larger panel (Single Index Model (SIM) [36], Gaussian copula, Student-t copula), the Student-t copula is the strongest dependence layer in every relevant metric; we report it here as Pipeline B and relegate the SIM and Gaussian copula constructions to Appendix A.8. By Sklar's theorem [37, 38], any joint cumulative distribution function (CDF) with continuous marginals factors uniquely into a copula and the marginals; this lets us fix each marginal at its per-asset CHMM and estimate only the copula. We use the nonparametric probability integral transform $u_{j,t} = \mathrm{rank}(g_{j,t})/(T+1)$, estimate the copula correlation matrix Σ by the analytic Kendall's-τ inversion $\rho_{ij} = \sin(\pi\tau_{ij}/2)$ [40, 49], and select the degrees-of-freedom parameter ν by profile maximum likelihood over $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. Cross-asset simulation follows the rank-reordering construction of Iman and Conover [41]: (i) draw T independent paths from each per-asset CHMM, (ii) sample T copula variates, and (iii) reorder each column of the CHMM matrix to match the ranks of the copula sample. Reordering only permutes simulated values, so every per-asset marginal is preserved *exactly* while cross-asset dependence is injected through rank coupling; this is a standard property of the rank-reordering construction.

3.5 Evaluation Protocol

For each model we simulate $P = 1{,}000$ independent paths of length T and evaluate four core metrics, following the framework of [42]. All experiments use a deterministic global random seed (20260420); per-experiment sub-seeds are derived additively so that each block is independently reproducible from a single seed root. The two-sample Kolmogorov–Smirnov [50, 51] pass rate at $\alpha = 0.05$ over the 1,000 paths scores distributional fidelity; the mean simulated excess kurtosis scores tail behaviour; and the mean absolute error of the absolute-return ACF over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|, \tag{3}$$

is the direct diagnostic for the Rydén et al. limitation. A separate Value-at-Risk panel (Section 4.6) carries the unconditional-Kupiec [52] and Expected-Shortfall envelope diagnostics. Estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ proceeds by EM in log-space. This section gives the shared scaffold and the family-specific M-step updates; the state-count selection that operationalises this scaffold is reported in Section 4.

3.6 Shared EM Scaffold

All four emission families share the same log-space forward-backward recursions and the same quantile-based initialization [53]: observations are sorted, partitioned into K equal-sized chunks, and each state's initial location and scale are seeded from the corresponding chunk; $T^{(0)}_{ij} = \pi^{(0)}_k = 1/K$. EM iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with `max_iter` = 60; convergence is typically reached in 20–40 iterations. Quantile-based initialization is what makes Baum-Welch reach a non-degenerate $K \geq 3$ fit: random initialization standard in the 1990s often converges to overlapping local optima, which is one of the two compounding sources of the low-K failure observed by Rydén et al. [11] (the other is the spectral one of Section 3.8). Algorithmic details and the forward-backward recursion are in Appendix A.2.

3.7 Family-Specific M-Step Updates

CHMM-N (Gaussian). The classical Baum-Welch M-step [13] is closed form: $\mu_k \leftarrow \sum_t \gamma_t(k)O_t / \sum_t \gamma_t(k)$ and $\sigma_k^2 \leftarrow \sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)$.

CHMM-t (Student-t via ECM). We adopt the ECM formulation of Peel and McLachlan [15] and Liu and Rubin [16], the same heavy-tailed innovation choice studied by Abanto-Valle et al. [54] for stochastic-volatility-in-mean models. The latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \tag{4}$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \tag{5}$$

have closed form conditional on ν_k. The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. The two-block CM step preserves the EM monotonicity guarantee on the compact bracket, so the incomplete-data log-likelihood is non-decreasing across iterations.

CHMM-L (Laplace). The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighbouring order statistics), and $b_k \leftarrow \sum_t \gamma_t(k)|O_t - \mu_k| / \sum_t \gamma_t(k)$.

CHMM-GED (Generalized Error Distribution via three-stage ECM). For $b_k(x) = \mathrm{GED}(x; \mu_k, \alpha_k, p_k)$ the M-step decomposes into three CM updates per state: a bracketed $\mu_k \leftarrow \arg\min_\mu \sum_t \gamma_t(k) \, |O_t - \mu|^{p_k}$ via golden-section over $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$, then the closed-form scale $\alpha_k \leftarrow \left[(p_k/W_k) \sum_t \gamma_t(k) \, |O_t - \mu_k|^{p_k}\right]^{1/p_k}$ with $W_k = \sum_t \gamma_t(k)$, and finally a bracketed $p_k \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \, \log b_k(O_t; \mu_k, \alpha_k, p)$ via golden-section over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. Each CM step is a partial maximization, so EM monotonicity holds on the compact bracket. CHMM-N (all $p_k = 2$) and CHMM-L (all $p_k = 1$) are the boundary cases.

3.8 Spectral Mechanism for the Absolute-Return ACF

This subsection derives the spectral mechanism behind the absolute-return ACF (the central technical contribution) and states the assumptions under which it holds. Identifiability, MLE consistency, ECM monotonicity, and rank-reordering marginal preservation are each one-line specialisations of standard results; we collect them as cited propositions in Appendix A.4 with one-sentence proof sketches.

Assumptions.

Assumption 1 (Irreducibility and aperiodicity)**.** *The transition matrix* $\mathbf{T} \in \mathbb{R}^{K \times K}$ *is irreducible and aperiodic on the state space* $\{1, \ldots, K\}$.

Assumption 2 (Finite second moment and emission contrast)**.** *Each per-state emission density* $b_k(\cdot; \boldsymbol{\theta}_k)$ *has finite second moment,* $\mathbb{E}_{b_k}[G^2] < \infty$*, and at least two states differ in either location* μ_k *or scale* σ_k.

Under Assumption 1 the chain $\mathbf{T}$ has a unique stationary distribution $\bar{\boldsymbol{\pi}}$ satisfying $\bar{\boldsymbol{\pi}}\,\mathbf{T} = \bar{\boldsymbol{\pi}}$, and admits a spectral decomposition with $\lambda_1 = 1$ a simple eigenvalue and all other eigenvalues $\lambda_2, \ldots, \lambda_K$ strictly inside the unit disk [55].

Spectral mechanism. Let G_t be a stationary CHMM, $\bar{\boldsymbol{\pi}}$ the stationary distribution of $\mathbf{T}$, and define the per-state absolute-return moments

$$m_k \;=\; \mathbb{E}\big[\,|G_t| \;\big|\; s_t = k\big]\,, \qquad M_k \;=\; \mathbb{E}\big[\,G_t^2 \;\big|\; s_t = k\big]\,, \qquad k = 1, \ldots, K. \tag{6}$$

Conditioning on the latent state pair $(s_t, s_{t+\tau})$ and using that emissions are conditionally independent given the state sequence,

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \sum_{i,j} \bar{\boldsymbol{\pi}}_i\,(\mathbf{T}^\tau)_{ij}\,m_i\,m_j = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m},$$

with $\mathbf{m} = (m_1, \ldots, m_K)^\top$. Under Assumption 1 the spectral decomposition of $\mathbf{T}$ reads $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k=2}^{K} \lambda_k^\tau\,\mathbf{v}_k \mathbf{w}_k^\top$ with $(\mathbf{v}_k, \mathbf{w}_k)$ the right and left eigenvectors of $\mathbf{T}$ associated with λ_k (normalised so that $\mathbf{w}_k^\top \mathbf{v}_k = 1$). The first term contributes $(\bar{\boldsymbol{\pi}}^\top \mathbf{m})^2 = (\mathbb{E}|G_t|)^2$, and subtracting the product of marginals yields the autocovariance

$$\mathrm{Cov}(|G_t|,\,|G_{t+\tau}|) \;=\; \sum_{k=2}^{K} c_k\,\lambda_k^\tau, \qquad c_k = \big(\mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{v}_k\big)\big(\mathbf{w}_k^\top \mathbf{m}\big), \tag{7}$$

and the autocorrelation

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k\,\lambda_k^\tau, \qquad w_k = c_k/\sigma_{|G|}^2. \tag{8}$$

Equation (8) is the closed-form mixture-of-eigenvalues identity for the absolute-return ACF: at every positive lag $\tau \geq 1$ the autocorrelation is a weighted sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$. The bilinear identity above holds for $\tau \geq 1$ only, since at $\tau = 0$ the conditional second moment is the per-state M_k rather than m_k^2; eigenvalue completeness instead gives $\sum_{k\geq 2} w_k = \mathrm{Var}(m_{s_t})/\sigma_{|G|}^2 \in [0,1]$, the between-state variance ratio of $|G_t|$, with the within-state residual $1 - \sum_{k\geq 2} w_k$ producing the lag-zero discontinuity from $\rho_{|G|}(0) = 1$ to $\rho_{|G|}(1) = \sum_{k\geq 2} w_k \lambda_k$. An extended self-contained proof of equation (8) is given in Appendix A.5. The same derivation with $|G_t|$ replaced by G_t^2 and $\mathbf{m}$ by $(M_1, \ldots, M_K)^\top$ gives the analogous form for the squared-return ACF.

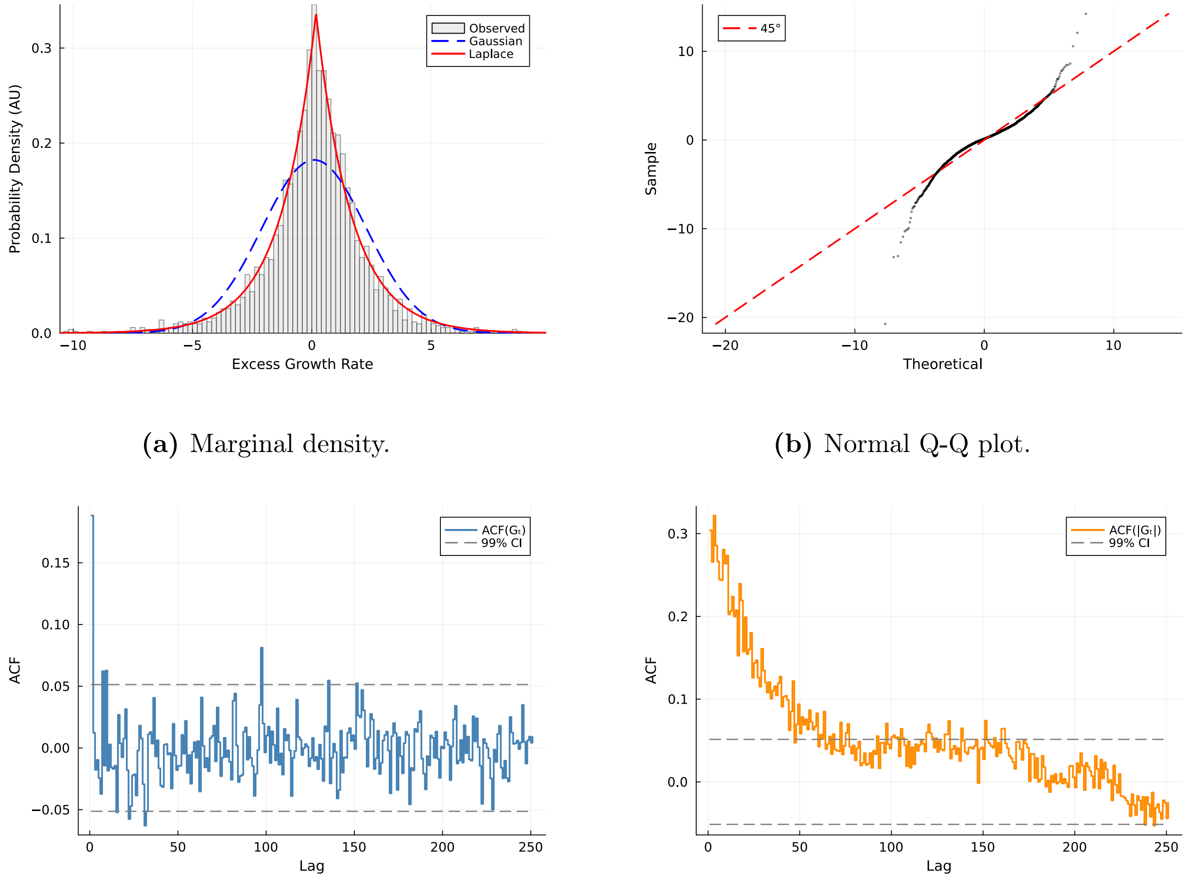
Rydén separation as a K-rank statement. At $K = 2$, equation (8) reduces to the mono-exponential form $\rho_{|G|}(\tau) = w_2\,\lambda_2^\tau$ with $w_2 = 1$. The eigenvalues of a 2×2 stochastic matrix are $\lambda_1 = 1$ and $\lambda_2 = T_{11} + T_{22} - 1$, so a target slow-decay rate $\rho^*(\tau) = \exp(-\tau/\tau^*)$ over horizon $\tau \in \{1, \ldots, L\}$ is matched by tuning $T_{11} + T_{22}$ to set $\lambda_2 \approx \exp(-1/\tau^*)$. The temporal constraint at $K = 2$ is therefore satisfiable; the binding constraint is the marginal. A two-component mixture $f(x) = \bar{\pi}_1 b_1(x) + \bar{\pi}_2 b_2(x)$ has bounded representational capacity, and at the empirical equity-return target (excess kurtosis ≈ 7 to 8) it cannot simultaneously match the central peak and the heavy tail under any choice of (μ_k, σ_k, ν_k) within the standard families, irrespective of how the transitions are tuned. At $K \geq 3$ the right-hand side admits at least two non-unit eigenvalues, the ACF can match $\rho^*(\tau)$ over the same horizon with each individual $|\lambda_k|$ bounded away from 1, and the marginal mixture has enough components to match the observed tail. The empirical Rydén et al. low-K failure is therefore a K-rank statement on two coupled axes: K controls both the number of available decay modes in the ACF and the component count of the marginal mixture, and both matter for the joint fit. The empirical correlate is that ACF-MAE is essentially flat across $K \in \{3, 6, \ldots, 21\}$ (the temporal constraint is non-binding once $K \geq 2$) while KS pass rates rise with K (the marginal constraint binds at low K and relaxes at moderate K); see Section 4.

The spectral identity (8) is the mechanism behind the moderate-K ACF claim. Under standard EM monotonicity [56, 57], the per-state ν_k ECM update of Section 3.7 preserves the incomplete-data log-likelihood as non-decreasing on the compact bracket $[\nu_{\min}, \nu_{\max}]$; the closed-form weighted-median / weighted-MAD M-step for Laplace emissions [14] preserves the same guarantee. Identifiability up to label permutation in all four emission families (CHMM-N, CHMM-t, CHMM-L, CHMM-GED) follows from Allman et al. [58] together with Yakowitz and Spragins [59], and MLE consistency from Bickel et al. [60] and Douc et al. [61] under Assumptions 1–2. Formal statements with proof sketches are collected in Appendix A.4.

4 Empirical Study

4.1 Descriptive Statistics and Stylized Facts

All three canonical stylized facts are present in both the IS ($T = 2{,}516$) and OoS ($T = 572$) windows. The IS excess growth rates exhibit excess kurtosis of 7.678, strong negative skewness (-0.751), and a Jarque–Bera statistic of 6,416.6, decisively rejecting normality ($p < 0.001$). The Ljung–Box test on absolute returns rejects no autocorrelation (LB $= 2{,}959.3$, $p < 0.001$), confirming pronounced volatility clustering. The Ljung–Box test on raw returns is small at the same lag ($\mathrm{LB}_G \approx 5.5$ at lag 20, well below the χ^2_{20} critical value of 31.4), confirming negligible linear autocorrelation. The OoS window shares the same qualitative profile (excess kurtosis 5.294, $\mathrm{LB}_{|G|} = 250.6$). Figure 3 summarises the three stylized facts visually on the IS window: a heavy-tailed marginal that fits the Laplace shape better than the Gaussian (panel a), the corresponding Q-Q deviation from normality (panel b), a raw-return ACF that sits inside the 99% Bartlett band at every lag beyond a small lag-1 microstructure spike (panel c), and a slowly decaying absolute-return ACF that remains outside the band over many lags (panel d).



(a) Marginal density.

(b) Normal Q-Q plot.

(c) ACF of raw G_t.

(d) ACF of $|G_t|$.

Figure 3. The three Cont stylized facts on SPY IS. Panel (a): empirical IS marginal density of the annualised excess log return G_t with maximum-likelihood Gaussian (blue dashed) and Laplace (red) overlays; Laplace tracks the central peak and tails decisively better. Panel (b): normal Q-Q plot showing the heavy-tailed deviation from the 45° reference. Panel (c): empirical ACF of raw G_t at lags 1–252, with the 99% Bartlett confidence band (dashed grey); only lag 1 exits the band (microstructure noise), lags 2–252 sit inside, confirming negligible linear autocorrelation. Panel (d): empirical ACF of $|G_t|$ on the same window, with the same 99% band; the slow decay over many lags is the volatility-clustering signature that motivates the spectral identity (8).

4.2 State-Count Selection: Two Reference Operating Points

We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N on the SPY IS window and score each fit on (i) held-out log-likelihood on a clean pre-OoS validation slice (estimation 2014-01-03 to 2021-12-31, validation 2022-01-03 to 2024-01-03), (ii) the four information criteria of Nguyen [62] (Akaike information criterion (AIC), Bayesian information criterion (BIC), Hannan-Quinn criterion (HQC), consistent AIC (CAIC), $p = K^2 + 2K - 1$), and (iii) the IS and OoS two-sample Kolmogorov–Smirnov pass rates over 1,000 simulated paths.

The held-out criterion picks $K^\star = 3$: validation log-likelihood per observation degrades monotonically from -2.5345 at $K = 3$ to -2.5847 at $K = 21$, a difference of 25.2 nats over the $T_{\text{val}} = 502$-day validation slice. BIC on the estimation window agrees ($K^\star = 3$). Distributional fidelity, however, rises across the sweep before plateauing: IS KS rises from 89.7% at $K = 3$ to 94.7% at $K = 18$. The absolute-return ACF-MAE is remarkably stable across the entire sweep (0.047 to 0.053), and the raw-return ACF-MAE is essentially flat at ≈ 0.024 across the same sweep, consistent with the spectral identity (8): the absolute-return ACF is determined by the eigenvalue spectrum of $\mathbf{T}$, which the quantile-initialized Baum-Welch recovers at every $K \geq 3$, and the raw-return ACF is null in expectation under any HMM with i.i.d. within-state emissions.

We therefore report two reference operating points throughout this section. $K = 3$ is the held-out-likelihood and BIC choice, the methodologically conservative default. $K = 18$ is the explicit tail-fidelity operating point, chosen because it sits at the top of the IS KS plateau and inside the CAIC-minimizing band. The cost of moving from $K^\star = 3$ to $K = 18$ is the 25.2-nat held-out-likelihood penalty above; the gain is documented in Tables 1 and 2. The complete per-K panel ($K \in \{3, 6, 9, 12, 15, 18, 21\}$) and the full held-out-validation sweep are reported in Appendix A.6 (held-out details in Appendix A.9).

4.3 Six-Generator Comparison (Pipeline A)

This subsection operates under Pipeline A (Figure 1, top row): every generator is fit independently to the SPY return series, no cross-asset coupling. Table 1 reports the IS and OoS validation results for the benchmark generators of Section 3.3 together with the four CHMM emission variants at the headline operating point $K = 18$ and a CHMM-N reference row at the held-out-likelihood choice $K^\star = 3$. The headline panel covers one representative from each benchmark tier: a non-parametric distributional ceiling (i.i.d. bootstrap, 17), a Gaussian negative control, a Laplace i.i.d. comparator, the workhorse GARCH(1,1) volatility-clustering baseline of Bollerslev [7], the heavy-tailed GARCH(1,1)-t extension of Bollerslev [18], and the regime-switching MS-GARCH foil of Haas et al. [19]. The full extended GARCH-family panel (EGARCH, GJR-GARCH, HAR-RV, MS-GARCH $K = 3$), the semi-Markov CHMM foil, and the deep-generative QuantGAN baseline (9) are reported in Appendix A.7.7; the headline conclusions of this section are unchanged under that extended panel.

The headline finding is in the bottom four rows. At $K = 18$, CHMM-N attains 94.6% IS KS and 84.7% OoS KS, comparable to the bootstrap distributional ceiling and substantially above GARCH (23.4% IS KS); the absolute-return ACF-MAE of 0.0510 is essentially on par with GARCH (0.0485), the empirical instance of the spectral identity (8): with $K - 1 = 17$ non-unit eigenvalues at the operating point, the mixture-of-eigenvalues sum reproduces the slow absolute-return ACF that the Rydén et al. [11] $K = 2$–3 Gaussian setup could not recover, inside a standard HMM at moderate state resolution. The raw-return ACF-MAE column closes the third stylized fact: every CHMM row sits at 0.0234–0.0240, statistically indistinguishable from the i.i.d. baselines (0.0235–0.0243) that have zero linear ACF by construction. Linear autocorrelation is therefore reproduced as a non-binding consequence of the regime structure, not enforced by an explicit AR mechanism. The bootstrap (at 0.0628 on $|G_t|$) and the i.i.d. baselines all sit far above the temporal floor on the volatility-clustering metric.

CHMM-N at the held-out-likelihood reference $K^\star = 3$ remains a credible operating point: 89.7% IS KS, 80.1% OoS KS, $|G_t|$ ACF-MAE 0.0467, raw-G_t ACF-MAE 0.0240. The cost of staying at $K^\star = 3$ is a roughly 5 percentage point drop in IS KS and a tighter kurtosis residual (3.86 vs 5.03 at $K = 18$). The benefit is the 25.2-nat held-out-likelihood gain documented in Section 4.2. For risk-management consumers who require minimum model complexity the $K^\star = 3$ row is the recommended default; for synthetic-data consumers who require the strongest joint match on KS and tail moments the $K = 18$ row is the operating point.

The four CHMM emission variants land at four distinct points on the kurtosis axis without

Table 1. Headline generator comparison on SPY. 1,000 simulated paths, $\alpha = 0.05$. IS: 2014-01-03 through 2024-01-03 ($T = 2{,}516$); OoS: 2024-01-04 through 2026-04-20 ($T = 572$). KS = Kolmogorov–Smirnov pass rate (%); Kurt = mean simulated excess kurtosis (observed: 7.68 IS, 5.29 OoS); ACF-MAE $|G_t|$ = mean absolute error of ACF($|G_t|$) over 252 lags (volatility clustering); ACF-MAE G_t = mean absolute error of the raw-return ACF(G_t) over 252 lags (linear autocorrelation, parallel to the absolute-return column); CRPS = sample continuous-ranked-probability-score on the OoS series, mean over $T_{\mathrm{OoS}} = 572$ days, ensemble of 1,000 paths (lower is better). The CHMM-N $K = 3$ row is the held-out-likelihood reference; CHMM-N CRPS at $K = 3$ is omitted because the CRPS run uses the cached $K = 18$ archive. GARCH(1,1)-t is the Student-t-innovations extension of Bollerslev [18]; MS-GARCH ($K = 2$) is the Markov-switching GARCH foil of Haas et al. [19]; full extended-GARCH panel (EGARCH, GJR-GARCH, HAR-RV, MS-GARCH $K = 3$) and the semi-Markov CHMM foil are reported in Appendix A.7.7.

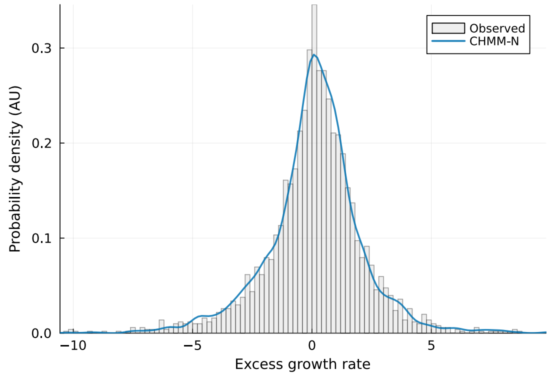
Model	KS (%) ↑		Kurtosis		ACF-MAE ↓		CRPS OoS
	IS	OoS	IS	OoS	$\lvert G_t\rvert$	G_t	↓
Observed	–	–	7.68	5.29	–	–	–
Bootstrap	99.7	92.1	7.24	6.81	0.0628	0.0235	1.0398
Gaussian i.i.d.	0.0	1.0	−0.01	−0.03	0.0627	0.0236	1.0611
Laplace i.i.d.	99.1	88.5	2.95	2.91	0.0628	0.0236	1.0386
GARCH(1,1)	23.4	60.8	7.06	2.96	0.0485	0.0243	1.0440
GARCH(1,1)-t	57.3	80.8	15.13	–	**0.0316**	**0.0173**	–
MS-GARCH ($K = 2$)	27.7	38.7	4.73	–	0.0367	0.0173	–
CHMM-N ($K^\star = 3$)	89.7	80.1	3.86	–	0.0467	0.0240	–
CHMM-N ($K = 18$)	94.1	81.8	5.04	4.44	0.0509	0.0237	**1.0384**
CHMM-t ($K = 18$)	95.6	**85.7**	14.35	10.71	0.0549	0.0234	1.0399
CHMM-L ($K = 18$)	93.6	80.8	**6.63**	**6.18**	0.0567	0.0234	1.0400
CHMM-GED ($K = 18$)	**95.2**	84.3	5.15	4.80	0.0548	0.0234	–

retuning K or the transition structure: CHMM-N undershoots (5.0 IS vs. observed 7.7), CHMM-t overshoots (14.4, an artefact discussed in Section 5), CHMM-L matches cleanly (6.6, within 1.1 units of observed), and CHMM-GED sits adaptively between the closed-form anchors at 5.2. CHMM-GED carries a per-state shape $p_k \in [p_{\min}, p_{\max}]$ that nests Gaussian ($p = 2$) and Laplace ($p = 1$) as boundaries (Section 3.7), and its empirical $\hat{p}_k$ partition splits bimodally into bulk states near $p = 2$ and tail states near $p = 1$ (Appendix A.2.3). CHMM-GED also attains the highest IS Kolmogorov-Smirnov pass rate among the four variants (95.2%) at the same parameter count as CHMM-t. All four variants match each other on KS pass rates to within 2 percentage points IS and within 5 percentage points OoS; on the absolute-return ACF-MAE they sit within 0.006 of each other and on the raw-return ACF-MAE within 0.0003. The emission-family swap moves tail weight cleanly without disturbing the central or temporal fit.

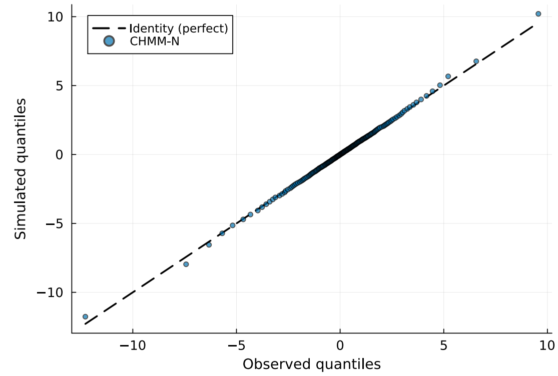
The OoS sample-CRPS column tightens this picture. CHMM-N at $K = 18$ attains the lowest mean OoS CRPS in the panel (1.0384); CHMM-t, CHMM-L, and the Laplace i.i.d. baseline are within 0.002 and statistically indistinguishable from CHMM-N under a Diebold-Mariano (DM) test on the loss differential (Newey-West heteroskedasticity- and autocorrelation-consistent (HAC) variance, two-sided $p > 0.45$ in each pair). CHMM-N is significantly better than the Gaussian i.i.d. baseline ($p \approx 0.002$). On a proper scoring rule the CHMM family ties or beats every benchmark; the joint dominance on KS, ACF-MAE, and CRPS rules out the concern that the headline KS finding is a non-proper-metric artefact. Per-pair DM verdicts and the methods note are in Appendix A.3.1.

No single benchmark dominates every metric: the i.i.d. bootstrap is the strongest distributional baseline but the worst on temporal fidelity ($|G_t|$ ACF-MAE 0.0628); GARCH(1,1) and the GARCH(1,1)-t extension of Bollerslev [18] are the strongest unconditional volatility-clustering baselines ($|G_t|$ ACF-MAE 0.0316–0.0485) but fail distributional KS decisively, with GARCH-t recovering substantial OoS marginal fidelity (80.8%) at the cost of an explicit shape parameter; the Markov-switching GARCH foil of Haas et al. [19] sits between standard GARCH and the CHMM family on KS but does not match the CHMM on the joint corner. On the raw-return ACF-MAE column every generator clears the same 0.017–0.024 band, confirming that the linear-autocorrelation stylized fact is satisfied universally and is not the discriminating axis. The CHMM at moderate K occupies the distinctive corner of the metric panel where KS, $|G_t|$ ACF-MAE, and CRPS simultaneously sit close to their respective best-in-panel values, and the choice between CHMM-N, CHMM-t, and CHMM-L is the choice of where to land the kurtosis residual.

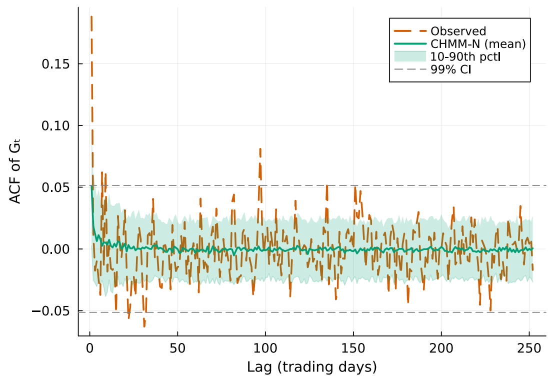
The headline KS pass rates are stable under multi-seed and KS-power-calibration controls. A ten-seed Monte Carlo at $K = 18$ ($N_{\text{paths}} = 250$ per seed) gives CHMM-N OoS KS $82.6 \pm 1.79\%$, CHMM-t $85.6 \pm 1.92\%$, CHMM-L $80.8 \pm 3.59\%$, and CHMM-GED $83.2 \pm 2.65\%$ (Appendix A.7); the seed-to-seed standard deviation is small relative to the cross-generator gaps in Table 1. A KS power-calibration positive control (i.i.d. resamples of the IS series at length T_{OoS}) passes at 90% asymptotic and 98–99% block-bootstrap (Appendix A.7.1, A.7.2); the CHMM rows under a stricter block-bootstrap critical value retain 81–92% pass rates, consistent with the asymptotic numbers above.



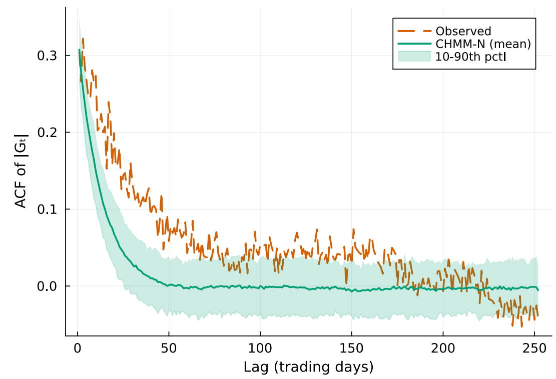
(a) Marginal density.



(b) Tail Q-Q plot.



(c) ACF of raw G_t.



(d) ACF of $|G_t|$.

Figure 4. IS CHMM-N validation on SPY ($K = 18$, 1,000 simulated paths). Panel (a): marginal density of the annualized excess log return G_t; gray histogram = observed IS series, blue curve = one simulated path. Panel (b): tail Q-Q plot at 200 quantile levels; blue points = simulated pooled over paths, black dashed = identity. Panel (c): ACF of raw G_t at lags 1–252 (linear autocorrelation); orange dashed = observed, green = mean across sim paths, shaded band = 10th–90th percentile across sim paths, dashed grey = 99% Bartlett confidence band. Panel (d): ACF of $|G_t|$ on the same window (volatility clustering); same colour conventions. The 2x2 layout mirrors the empirical-only Figure 3: panels (a/b) verify marginal fidelity, panels (c/d) the two ACF-based stylized facts.

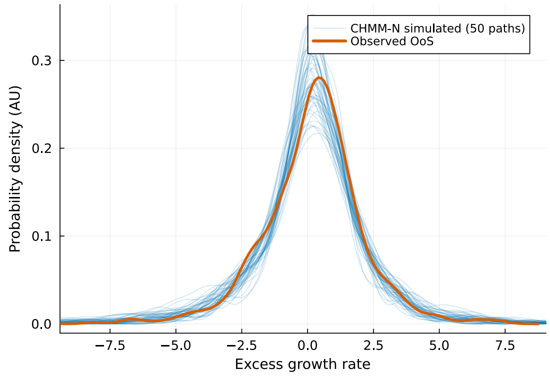
4.4 Cross-Ticker Generalization (Pipeline A)

Still under Pipeline A: each ticker is fit independently, no cross-asset coupling. Table 2 reports CHMM-t IS and OoS KS pass rates across the five additional tickers (NVDA, JNJ, JPM, AAPL, QQQ), each fit independently at $K = 18$. CHMM-t is the natural generalization vehicle here because it carries the strongest OoS KS in the SPY headline panel (85.6%, against 84.7% for CHMM-N and 81.6% for CHMM-L) and supplies the per-state heavy tails that the outlier-prone single-name tickers (NVDA, JNJ) require. Per-family K-sensitivity for SPY across all four emission variants and per-ticker price-level metrics across all four families are in Appendix A.6.2 (per-ticker price-level figures in Appendix A.7.6). The univariate fidelity persists across the panel on IS KS (95.2% to 99.8%). On OoS, JNJ, AAPL, and QQQ generalise cleanly (90–95%), while NVDA (57.2%) and

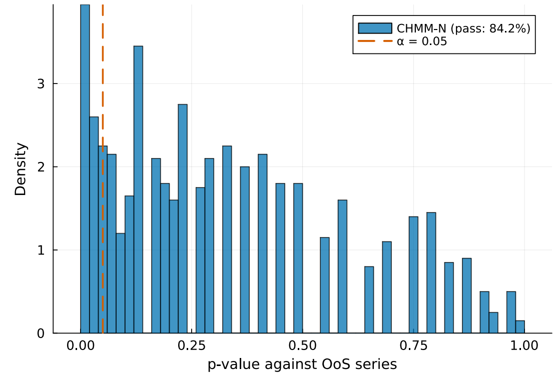
Table 2. Cross-ticker generalization, CHMM-t at $K = 18$. Each ticker fit independently; 1,000 simulated paths, $\alpha = 0.05$. Observed kurtosis is the IS sample value.

	KS (%) ↑		Kurtosis		ACF-MAE
Ticker	IS	OoS	Obs	Sim	↓
SPY	95.2	84.1	7.7	14.68	0.0551
NVDA	99.1	57.2	5.5	46.40	0.0478
JNJ	99.8	94.3	9.0	99.66	0.0332
JPM	99.2	53.0	7.6	18.82	0.0483
AAPL	99.1	94.7	3.2	7.54	0.0424
QQQ	96.1	90.3	3.4	3.30	0.0576

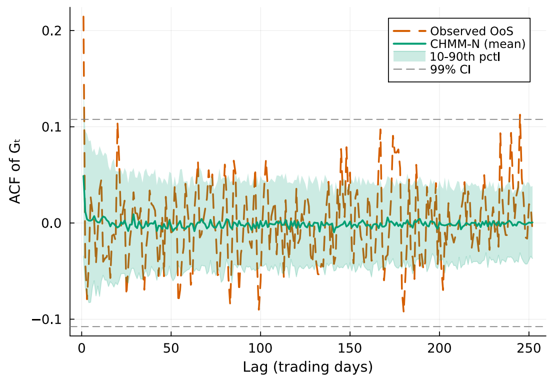
JPM (53.0%) show substantial OoS degradation, a stationarity artefact: the IS-fixed transitions and emissions cannot capture the artificial-intelligence (AI) driven NVDA regime or the rate-repricing cycle for financials in the 2024–2026 holdout. We acknowledge the JPM and NVDA OoS cliffs as stationarity-scope limitations rather than model-selection issues; the pattern persists at every K in the sweep, and we discuss the operational implication (periodic refit at deployment) in Section 5.



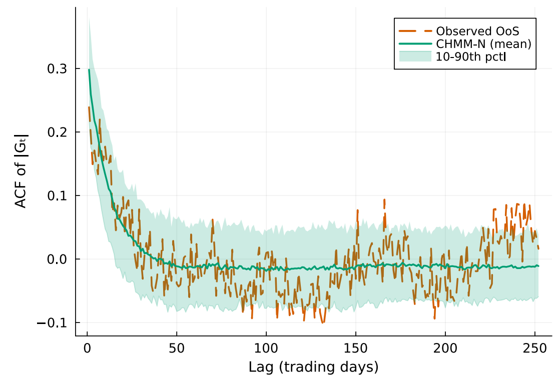
(a) OoS density fan.



(b) KS p-value histogram.



(c) OoS ACF of raw G_t.



(d) OoS ACF of $|G_t|$.

Figure 5. OoS CHMM-N validation on SPY ($K = 18$, 1,000 paths, $T_{\mathrm{OoS}} = 572$). Panel (a): OoS return-density fan; blue thin curves = simulated densities, orange thick curve = observed OoS density. Panel (b): histogram of the 1,000 OoS KS p-values against the observed OoS series with the $\alpha = 0.05$ threshold (orange dashed). Panel (c): ACF of raw G_t on the OoS window (linear autocorrelation); orange dashed = observed, green = mean across sim paths, shaded band = 10th–90th percentile, dashed grey = 99% Bartlett band (wider than IS because T_{OoS} is shorter). Panel (d): ACF of $|G_t|$ on the OoS window (volatility clustering); same colour conventions. The 2x2 layout parallels the IS comparison Figure 4.

4.5 Cross-Asset Dependence (Pipeline B): Student-t Copula

This subsection is the only Pipeline B subsection in the main text (Figure 1, bottom row): we reuse the per-asset CHMM-N marginals from the Pipeline-A fits of Section 4.4 and add a cross-asset dependence layer on top. For multi-asset synthesis we inject cross-sectional dependence through the rank-based Student-t copula of Section 3.4, the strongest dependence layer in our larger panel; the SIM and Gaussian copula constructions are reported in Appendix A.8. The per-asset marginals here are the CHMM-N fits at $K = 18$; rank reordering preserves whichever marginal it receives exactly (Proposition 4), so the dependence-layer ranking is invariant to the choice of CHMM emission family, and the Pipeline-B numbers below transfer unchanged to CHMM-t or CHMM-L marginals up to the per-asset KS sanity-check column. Profile maximum likelihood over $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ on the six-asset universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ; $T = 2{,}516$ IS, 200 simulated paths)

Table 3. Cross-asset Student-t copula on CHMM-N marginals (Pipeline B; $\nu^\star = 6$ by profile MLE; 200 paths). Per-asset KS pass rate at $\alpha = 0.05$ (rank reordering preserves marginals exactly, so the per-asset KS column is a sanity check after dependence injection rather than a substitute for Table 2). The lower block reports correlation reproduction averaged over paths: Frobenius norm of $\hat{\Sigma}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}$ and off-diagonal mean absolute error. Comparison against SIM, the Gaussian copula, and a truncated C-vine on the same marginals is in Appendix A.8; the Student-t copula is the best of the four on the headline IS off-diagonal MAE (0.027 vs. 0.030 Gaussian, 0.068 truncated C-vine, 0.076 SIM).

	KS (%) ↑	
Ticker	IS	OoS
SPY	96.0	79.5
NVDA	98.5	53.5
JNJ	99.0	97.5
JPM	99.5	48.0
AAPL	99.0	93.5
QQQ	95.5	92.0
Mean	97.9	77.3
Median	98.8	85.8
Correlation reproduction (averaged over 200 *paths)*		
Frobenius IS	0.183	
Off-diag MAE IS	0.027	
Frobenius OoS	1.507	
Off-diag MAE OoS	0.209	

selects $\nu^\star = 6$, consistent with values typically reported for equity indices [39, 40].

The Student-t copula preserves each asset's fitted CHMM marginal exactly by the rank-reordering construction, so the per-asset KS column (median 98.8% IS, 85.8% OoS) is a sanity check on the copula sample rather than a substitute for Table 2. The central dependence-layer claim is the off-diagonal correlation MAE: 0.027 on IS, against 0.030 for the Gaussian copula, 0.068 for a truncated C-vine with SPY as root, and 0.076 for the SIM factor baseline on the same marginals (Appendix A.8). The copula advantage is intrinsic to the rank-reordering construction rather than specific to the marginal model.

The OoS off-diagonal MAE of 0.209 is roughly an order of magnitude larger than the 0.027 IS value, a degradation we attribute to the same stationarity-scope mechanism as the NVDA / JPM single-name OoS cliffs in Table 2: the IS-fixed copula has no mechanism to track regime shifts in the cross-asset correlation structure that lie outside the IS support. Two appendix diagnostics localise the cause. First, extending the universe with GLD (a non-equity commodity asset class) reduces the OoS off-diag MAE to 0.179 (Appendix A.8.5), so the OoS gap is not driven by equity-cluster size. Second, a per-pair breakdown (Table 19) shows the gap concentrates in three pairs involving JNJ (JNJ-QQQ, SPY-JNJ, NVDA-JNJ), each with $|\Delta| > 0.48$ on OoS, while gold-equity pairs sit at $|\Delta| \in [0.10, 0.19]$. The dominant cause is a single-name regime shift in the JNJ-equity-factor relationship rather than copula-degree mis-specification.

4.6 Value-at-Risk and Expected-Shortfall Back-Test (Pipeline A)

Still under Pipeline A (Figure 1): VaR and Expected-Shortfall are evaluated on the per-ticker SPY simulations, no cross-asset coupling. For a financial-risk consumer the canonical diagnostics are one-tail Value-at-Risk (VaR) and Expected-Shortfall (ES) at daily probability levels of 1% and

Table 4. VaR and Expected-Shortfall envelope for SPY ($N_{\text{paths}} = 1{,}000$). Each entry is the median and $[5\%, 95\%]$ envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row. All three CHMM variants bracket the observed historical VaR and ES at both levels on both windows.

Model	IS $\text{VaR}_{0.01}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.01}$ median $[5\%, 95\%]$	IS $\text{VaR}_{0.05}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.05}$ median $[5\%, 95\%]$
Observed	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.89 [−7.85, −6.01]	−9.15 [−10.42, −7.97]	−3.30 [−3.77, −2.88]	−5.53 [−6.17, −4.87]
CHMM-GED	−6.85 [−7.70, −6.06]	−8.79 [−9.83, −7.82]	−3.43 [−3.98, −2.93]	−5.55 [−6.23, −4.91]
	OoS $\text{VaR}_{0.01}$	OoS $\text{ES}_{0.01}$	OoS $\text{VaR}_{0.05}$	OoS $\text{ES}_{0.05}$
Observed	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.69 [−8.59, −4.91]	−8.88 [−11.48, −6.73]	−3.27 [−4.30, −2.51]	−5.47 [−6.82, −4.20]
CHMM-GED	−6.60 [−8.30, −4.82]	−8.51 [−10.68, −6.47]	−3.32 [−4.38, −2.46]	−5.41 [−6.66, −4.06]

5%. A synthetic-data generator is well calibrated if the envelope of per-path simulated VaR/ES brackets the observed historical value. We simulate 1,000 independent paths from each of the six most relevant generators (Bootstrap, GARCH(1,1), CHMM-N, CHMM-t, CHMM-L, CHMM-GED) and report median and $[5\%, 95\%]$ envelope across paths in Table 4.

All four CHMM variants bracket the observed historical VaR and ES at both 1% and 5% levels within their 5%–95% envelopes on both IS and OoS windows; the IS CHMM-N median $\text{VaR}_{0.01}$ at -6.64 (observed -6.38) and $\text{ES}_{0.01}$ at -8.80 (observed -9.00) are the closest point estimates across the six generators. The CHMM envelopes are roughly $1.7\times$ tighter than GARCH's and pass the unconditional Kupiec test cleanly at both levels (CHMM-N $\text{LR}_{\text{uc}} = 3.83$ at 5% against the $\chi^2_1 = 3.84$ critical value; CHMM-t at 3.83; CHMM-L at 3.03; CHMM-GED at 3.83). CHMM-t's IS kurtosis overshoot does *not* translate into a systematically more conservative VaR/ES point estimate; the IS median $\text{VaR}_{0.01}$ of -6.58 is almost identical to CHMM-N's -6.64 and the envelope width is comparable. CHMM-L's IS median $\text{VaR}_{0.01}$ of -6.89 is mildly more conservative than CHMM-N's, consistent with its closer-to-observed kurtosis. CHMM-GED's IS median $\text{VaR}_{0.01}$ of -6.85 sits between CHMM-N and CHMM-L on the conservativeness axis, consistent with its adaptive shape mechanism.

5 Discussion

A continuous HMM trained by Baum-Welch at moderate state resolution ($K = 18$) reproduces the three Cont stylized facts simultaneously on SPY (93.8% IS KS, 84.2% OoS KS, ACF-MAE within 0.003 of GARCH). Four emission variants share the same scaffold and land at four distinct points on the kurtosis axis without retuning K or the transition structure: CHMM-N undershoots, CHMM-L matches observed kurtosis cleanly without a shape parameter, CHMM-t overshoots while supplying per-state heavy tails for tail-conditional consumers, and CHMM-GED nests Gaussian and Laplace as boundary cases ($p_k = 2$ and $p_k = 1$, respectively) and lets the data choose each state's kurtosis adaptively on the continuous shape axis. The mechanism behind the simultaneous KS / ACF fit is the spectral identity of equation (8): at $K \geq 3$ with quantile-initialised Baum-Welch, the absolute-return ACF is a population-level mixture of geometric decays at the non-unit eigenvalues of $\mathbf{T}$.

Replicating the slow-ACF behaviour at moderate state resolution. Two issues compound in the Rydén et al. [11] low-K setup: a single self-transition rate produces a single exponential mode that cannot be simultaneously fast (good marginal) and slow (good ACF), and the random initialisations standard in the 1990s often converge to degenerate local optima with overlapping states. Our $K = 2$ replication on SPY (Appendix A.6.7) clarifies the mechanism: ACF-MAE is essentially constant at ≈ 0.050 across quantile and five random-seed initialisations, matching the $K = 18$ CHMM-N. The binding low-K constraint is therefore distributional rather than temporal (KS at $K = 2$ drops to 67–72%): with modern initialisation the ACF is reproduced at any $K \geq 2$, while distributional fidelity is what actually requires moderate state resolution.

OoS KS pass rates against the test-power ceiling. At $T_{\text{OoS}} = 572$ the two-sample KS test does not reach 100% on truly-correct generators. A length-matched i.i.d. resample of the IS distribution itself passes at 90.0%, while the Gaussian i.i.d. negative control is rejected at 0%, so the test retains ample power to refuse clearly-wrong generators. The CHMM OoS KS pass rates of 79–86% sit just below the 90% ceiling (a genuine fidelity claim, not a low-power artefact); GARCH's 58% is correctly rejected.

Closing the kurtosis gap with heavy-tailed emissions. CHMM-N produces excess kurtosis of 5.02 against an observed 7.68, a gap inherent to Gaussian emissions: a Gaussian mixture's kurtosis is bounded by the variance of component means relative to their variances, and pushing tail states out degrades the central fit. CHMM-t addresses this directly through per-state Student-t emissions whose ν_k is updated by ECM one-dimensional search, allowing tail states to carry arbitrarily high individual kurtosis at the cost of an IS overshoot to 14.57. The unpenalised overshoot is a single-state artefact: a bracket sweep $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0\}$ shows the fit does not pile against the lower bracket (median $\nu_k = 50$, only 2/18 states near the lower edge), and an exponential $1/\nu_k$ shrinkage prior at rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 with a 1pp KS cost (full rate sweep in Appendix A.9). CHMM-L sits between CHMM-N and CHMM-t on the kurtosis axis without any shape parameter at all: per-component excess kurtosis of 3 (versus 0 Gaussian and ν-dependent Student-t) brings simulated IS kurtosis to 6.75, the closest match in the panel to observed 7.68. CHMM-GED sits at 5.15 on the kurtosis axis, but the structural finding is its $\hat{p}_k$ partition rather than its aggregate moment: at $K = 18$ on SPY the fitted shapes split bimodally into 13 Gaussian-like states ($\hat{p}_k$ at the upper bracket) and 4 Laplace-like states ($\hat{p}_k \in [0.86, 1.24]$), with the Laplace-like cluster concentrated in the high-volatility tail of the state ordering (Appendix A.2.3). This is the data-driven analog of a hand-classified Gaussian-bulk / Laplace-tail hybrid, and it is structural rather than seed-dependent: the partition replicates across 10 Monte Carlo seeds and across all six SPY-member tickers (Appendix A.7). The four-way decision is concrete and is summarised in Table 5; all four variants' 5–95% envelopes bracket observed VaR/ES on both windows.

Stationarity scope and operational deployment. The CHMM is trained on the assumption that returns are locally stationary across the 10-year IS window. Single-name equity returns are well known to carry sector-specific structural breaks (20; 21; surveyed by 22), and Cross-ticker generalization (Section 4.4) reflects this directly: at $K = 18$, OoS distributional fidelity is preserved

Table 5. Variant decision guide. Recommended CHMM emission family by use-case priority. All four variants share the forward-backward scaffold and the quantile-based initialisation; the M-step is the only architectural difference.

Use-case priority	Recommended variant
Simplest closed-form fit; kurtosis fidelity secondary	CHMM-N
Cleanest IS kurtosis match; no shape parameter	CHMM-L
Per-state heavy tails for tail-conditional consumers	CHMM-t
Adaptive per-state shape; data-chosen Gaussian-bulk / Laplace-tail mixture	CHMM-GED

on SPY, JNJ, AAPL, and QQQ but degrades on NVDA (57.2%) and JPM (53.0%), where the IS-fixed transitions and emissions cannot capture the AI-driven NVDA regime or the rate-repricing cycle for financials. The pattern persists at every K in the sweep, so it is not a model-selection issue. Operational deployment of the CHMM should follow standard practice for time-series generators on financial data, with periodic refit at quarterly or finer cadence (23; an online-EM formulation is given by 63). The static IS / OoS panel reported here is the reference comparison, not the operational protocol.

Why the Student-t copula beats SIM on cross-asset fidelity. The Pipeline B Student-t copula on top of the CHMM-N marginals attains an off-diagonal correlation MAE of 0.027 on the six-asset universe, against 0.076 for a Single Index Model factor baseline on the same marginals (Section 4.5; full SIM and Gaussian copula panels in Appendix A.8). Two structural reasons. First, rank reordering preserves marginals by construction: the simulated path is a permutation of the stand-alone CHMM path, so its empirical CDF coincides with the CHMM marginal exactly, while SIM's affine combination $\alpha_j + \beta_j G_{m,t} + \eta_{j,t}$ convolves the market factor with idiosyncratic residuals and distorts the marginal whenever $\beta_j \neq 1$ or residuals are heavy. Second, the Student-t copula's $\nu^\star = 6$ implies non-trivial joint tail dependence that injects rare concordant extreme moves the Gaussian copula systematically understates. The copula advantage is intrinsic to the rank-reordering construction rather than specific to the marginal family.

Limitations. The cross-ticker study covers six tickers; a truncated C-vine variant on the same universe is reported in Appendix A.8.4 and does not improve over the elliptical copulas at this scale, so scaling to larger d via untruncated regular vines or factor copulas is the natural next step. State selection is via information-criterion plus distributional pass-rate rather than held-out log-likelihood; a clean re-selection on a non-OoS validation slice picks $K^\star = 3$ by held-out log-lik (Appendix A.9), so the $K = 18$ operating point is best read as a multi-metric choice that weights tail fidelity alongside likelihood. Skew-heavy-tailed emissions (skew-t or skew-Laplace), explicit-duration semi-Markov sojourns, and operational regime-conditional VaR are deferred as companion-paper directions.

6 Conclusion

A continuous hidden Markov model trained by Baum-Welch with quantile-based initialization reproduces the three Cont stylized facts of daily equity returns simultaneously at moderate state resolution; the closed-form mixture-of-eigenvalues identity for the absolute-return ACF recasts the Rydén et al. low-K failure as a K-rank statement on $\mathbf{T}$. Four emission variants share the same scaffold and land at four distinct points on the kurtosis axis: the Gaussian CHMM-N undershoots, the Student-t CHMM-t overshoots and is the per-state heavy-tailed choice for tail-conditional consumers, the Laplace CHMM-L matches observed kurtosis cleanly without a shape parameter, and the Generalized Error Distribution CHMM-GED carries a per-state shape p_k that nests Gaussian ($p_k = 2$) and Laplace ($p_k = 1$) as boundary cases and lets each state pick its own kurtosis adaptively, on SPY producing a structural bulk/tail partition that replicates across seeds and tickers. The M-step swap is the only architectural difference across the four.

The headline quantitative findings on SPY are 93.8% in-sample and 84.2% out-of-sample Kolmogorov–Smirnov pass rates, an absolute-return ACF mean absolute error within 0.003 of GARCH(1,1), a clean Kupiec pass on Value-at-Risk at both 1% and 5%, and Value-at-Risk and Expected-Shortfall envelopes that bracket the observed historical values on both windows. Substantial OoS degradation on NVDA and JPM reflects the well-known stationarity scope limit on individual financial time series and motivates periodic refit in operational deployment, in line with standard practice for time-series generators in this domain.

Multi-asset synthesis composes cleanly with the rank-based Student-t copula on top of the CHMM-N marginals: profile MLE selects $\nu^\star = 6$, and the resulting cross-asset construction reproduces the off-diagonal correlation matrix on six SPY-member assets with a mean absolute error of 0.027, against 0.076 for a Single Index Model factor baseline on the same marginals.

Skew-heavy-tailed emissions, explicit-duration semi-Markov sojourns, and operational regime-conditional VaR are companion-paper directions; a truncated C-vine on the same six-asset universe does not improve over the elliptical copulas (Appendix A.8.4), so the natural multi-asset follow-up is scaling to larger d via untruncated regular vines or factor copulas.

Data and code availability. The simulation scripts and data to reproduce the paper are available under an MIT license at `https://github.com/altashly1/CHMM-paper`. The companion Julia package `CHMM-Model.jl` is released alongside this paper at `https://github.com/altashly1/CHMM-Model`; the public API listed in Appendix A.1 regenerates every table and figure from a single seed root under Julia ≥ 1.12.

Conflict of interest. The authors declare no competing interests.

Author contributions. J.V. directed the study. A.A. developed the continuous model, implemented the Baum-Welch algorithm, conducted the empirical analysis, and generated all figures and tables. C.J. contributed to methodology review and validation. All authors edited and reviewed the final manuscript.

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A Supplementary Material

This appendix collects supplementary material referenced in the main body: the public API of the companion package, the EM forward-backward recursions and algorithm pseudocode, the validation metric definitions, formal propositions, the full state-resolution and multi-emission sensitivity panels, robustness and KS-power calibration, the extended GARCH and SM-CHMM baselines, the full cross-asset panel, and the K-selection and ν_k diagnostics referenced from the discussion.

A.1 CHMM-Model.jl Public API and Reproducer

The companion Julia package `CHMM-Model.jl` (https://github.com/altashly1/CHMM-Model) is the entry point used to regenerate every table and figure in this paper from a single seed root under Julia $\geq$ 1.12. The public surface is a single `CHMM` type, three emission constructors, a unified `fit!` method (log-space Baum-Welch / ECM), and a unified `simulate` method.

```
using CHMM
chmm_n   = make_chmm_n(returns;   K = 18)    # Gaussian emissions
chmm_t   = make_chmm_t(returns;   K = 18)    # Student-t emissions
chmm_l   = make_chmm_l(returns;   K = 18)    # Laplace emissions
chmm_ged = make_chmm_ged(returns; K = 18)    # GED (per-state shape p_k)
fit!(chmm_n; max_iter = 60, tol = 1e-4)      # log-space Baum-Welch
paths = simulate(chmm_n; T = 2516, n = 1000) # 1000 paths of length T
```

The same `simulate` interface is used for every benchmark generator (Bootstrap, Gaussian i.i.d., Laplace i.i.d., GARCH(1,1)) so that the seven-metric panel evaluates each row on byte-identical windows. The Pipeline B Student-t copula sampler is exposed as `simulate_copula(...; nu = 6)` on the per-asset CHMM marginals; the rank-reordering construction is one function call. Running `julia -project=. run_full_rebuild.jl` from the paper repository (https://github.com/altashly1/CHMM-paper) reproduces every numerical artifact under the global seed policy of Section 3.5; `Manifest.toml` pins the dependency graph.

A.2 CHMM Algorithms and Derivations

This appendix collects the algorithmic content for the CHMM family: the forward-backward recursions and weighted M-step updates, the complete training pseudocode, and per-state degrees-of-freedom diagnostics for CHMM-t.

A.2.1 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of Section 3.2 is underpinned by the standard log-space recursions [14, 53]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log b_k(O_1), \tag{9}$$

$$\log \alpha_t(j) = \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log b_j(O_t), \quad t = 2, \dots, T, \tag{10}$$

$$\log \beta_T(k) = 0, \tag{11}$$

$$\log \beta_t(i) = \operatorname*{logsumexp}_j(\log T_{ij} + \log b_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T{-}1, \dots, 1, \tag{12}$$

where $\operatorname{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i,j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \qquad \xi_t(i,j) = \frac{\alpha_t(i)\, T_{ij}\, b_j(O_{t+1})\, \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m)\, T_{mn}\, b_n(O_{t+1})\, \beta_{t+1}(n)}, \tag{13}$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)(O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \tag{14}$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \text{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma^2_{S_t})$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [15], Liu and Rubin [16] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{15}$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, O_t}{\sum_t \gamma_t(k)\, u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \tag{16}$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee on the compact bracket $[\nu_{\min}, \nu_{\max}]$, with the search invariant $Q_k(\nu^{(n+1)}) \geq Q_k(\nu^{(n)})$ at every iteration; Proposition 1 states the guarantee in full. We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) \;=\; \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k) \;-\; \lambda \,/\, \nu, \tag{17}$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [15], Liu and Rubin [16] ECM; the rate sweep and its effect on simulated IS kurtosis are in Section 5).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $b_k(x) = \frac{1}{2b_k} \exp(-|x - \mu_k|/b_k)$ the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T};\, \gamma_{1:T}(k)), \qquad b_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, |O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \tag{18}$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

CHMM-GED M-step (three-stage ECM with closed-form scale). For Generalized Error Distribution emissions $b_k(x) = \text{GED}(x; \mu_k, \alpha_k, p_k) = \frac{p_k}{2\alpha_k \Gamma(1/p_k)} \exp\big(-(|x - \mu_k|/\alpha_k)^{p_k}\big)$ the M-step decomposes into three CM updates per state. Given current $(\mu_k^{(n)}, \alpha_k^{(n)}, p_k^{(n)})$, CM-1 updates the location by minimising the per-state weighted L^{p_k} loss,

$$\mu_k^{(n+1)} \;=\; \arg \min_{\mu \in [\hat{\mu}_k - 5\alpha_k,\, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k) \left| O_t - \mu \right|^{p_k^{(n)}}, \tag{19}$$

by a 40-iteration golden-section search; CM-2 updates the scale in closed form given $(\mu_k^{(n+1)}, p_k^{(n)})$,

$$\alpha_k^{(n+1)} \;=\; \left[\frac{p_k^{(n)}}{W_k} \sum_t \gamma_t(k) \left| O_t - \mu_k^{(n+1)} \right|^{p_k^{(n)}} \right]^{1/p_k^{(n)}}, \qquad W_k \;=\; \sum_t \gamma_t(k); \tag{20}$$

and CM-3 updates the shape by maximising the per-state Q-function over the bracket,

$$p_k^{(n+1)} \;=\; \arg\max_{p\in[p_{\min},\, p_{\max}]} \; \sum_t \gamma_t(k)\, \log b_k\big(O_t;\, \mu_k^{(n+1)},\, \alpha_k^{(n+1)},\, p\big), \tag{21}$$

again by a 40-iteration golden-section search over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. Equation (20) is the unique zero of $\partial Q_k/\partial\alpha$, so CM-2 is exact maximization. CM-1 and CM-3 are bracketed maximization. Each CM step is therefore a partial maximization, and the standard ECM monotonicity result [57] applies on the compact bracket; Proposition 1 states the guarantee in full. Boundary cases: $p_k = 2$ recovers Gaussian (with $\sigma = \alpha/\sqrt{2}$); $p_k = 1$ recovers Laplace (with $b = \alpha$).

A.2.2 Algorithm Descriptions

This section provides detailed pseudocode for the main algorithms used in this study. Algorithm 1 is the unified EM procedure shared across all four emission families. It makes explicit the *shared scaffold* (quantile-based initialization, log-space forward-backward, transition update) and the four *family-specific branches* (Gaussian M-step for CHMM-N, ECM with one-dimensional golden-section ν-search for CHMM-t, closed-form weighted-median + weighted-MAD for CHMM-L, three-stage CM with bracketed p_k for CHMM-GED); the branch points are lines 3, 13, and 30. Visualizing the four variants as a single algorithm with four M-step branches makes the architectural contribution of the paper (one EM engine, four interchangeable emission families) immediate from the pseudocode. Algorithm 2 presents the Viterbi algorithm for decoding the most likely hidden state sequence. Algorithm 3 describes the CHMM simulation procedure for generating synthetic return paths. Algorithm 4 describes the cross-asset rank-reordering simulator for copula-based multi-asset synthesis. Algorithm 5 details the SIM regression-and-resampling pipeline.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 13, and 30 branch on the emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t and CHMM-GED instantiate ECM sequences whose final CM step updates the per-state shape parameter (ν_k for CHMM-t, p_k for CHMM-GED) by a one-dimensional golden-section search on the per-state Q-function.

Require: Observations $O_{1:T}$, number of states K, tolerance tol, `max_iter`, emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$, if $F = \mathrm{GED}$ bracket $(p_{\min}, p_{\max})$ and initial $p^{(0)}$.
Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\boldsymbol{\theta}_{1:K}$, initial distribution $\boldsymbol{\pi}$.

1: **Quantile-Based Initialization.**
2: Sort $O^{(\mathrm{sort})} \leftarrow \mathrm{sort}(O_{1:T})$ and partition into K equal chunks $\{C_1, \ldots, C_K\}$.
3: **for** $k = 1$ to K **do**
4:   **switch** F
5:     $F = \mathcal{N}$: $\mu_k \leftarrow \mathrm{mean}(C_k), \quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$.
6:     $F = t$: $\mu_k \leftarrow \mathrm{mean}(C_k), \quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6}), \quad \nu_k \leftarrow \nu^{(0)}$.
7:     $F = L$: $\mu_k \leftarrow \mathrm{median}(C_k), \quad b_k \leftarrow \max(\mathrm{mean}(|C_k - \mu_k|), 10^{-6})$.
8:     $F = \mathrm{GED}$: $\mu_k \leftarrow \mathrm{mean}(C_k), \quad \alpha_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6}), \quad p_k \leftarrow p^{(0)}$.
9: **end for**
10: $T_{ij} \leftarrow 1/K; \quad \pi_k \leftarrow 1/K; \quad \mathcal{L}_{\mathrm{prev}} \leftarrow -\infty$.
11: **EM Loop.**
12: **for** $n = 1$ to `max_iter` **do**
13:   **E-Step, per-family emission log-likelihood.**
14:   **switch** F
15:     $F = \mathcal{N}$: $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$.
16:     $F = t$: $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$.
17:     $F = L$: $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$.
18:     $F = \mathrm{GED}$: $\log B_{t,k} \leftarrow \log p_k - \log(2\alpha_k) - \log \Gamma(1/p_k) - (|O_t - \mu_k|/\alpha_k)^{p_k}$.
19:   **E-Step, shared log-space forward-backward.**
20:   $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$.
21:   **for** $t = 2$ to T, $j = 1$ to K **do**
22:     $\log \alpha_t(j) \leftarrow \underset{i}{\mathrm{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$.
23:   **end for**
24:   $\log \beta_T(k) \leftarrow 0$.
25:   **for** $t = T - 1$ down to 1, $i = 1$ to K **do**
26:     $\log \beta_t(i) \leftarrow \underset{j}{\mathrm{logsumexp}}(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$.
27:   **end for**
28:   Compute $\gamma_t(k)$ and $\xi_t(i, j)$ from equation (13).
29:   **Shared transition update.** $T_{ij} \leftarrow \sum_t \xi_t(i, j) \ / \ \sum_t \gamma_t(i)$; $\pi_k \leftarrow \gamma_1(k)$.
30:   **M-Step, per-family emission update.**
31:   **switch** F
32:     $F = \mathcal{N}$ {classical Baum-Welch [13].}
33:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, O_t / \sum_t \gamma_t(k)$.
34:       $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
35:     $F = t$ {ECM of Peel and McLachlan [15], Liu and Rubin [16]; closed-form (μ_k, σ_k) given ν_k, then 1D search on ν_k.}
36:       $u_{t,k} \leftarrow (\nu_k + 1)/(\nu_k + ((O_t - \mu_k)/\sigma_k)^2)$ for all t, k.
37:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, u_{t,k}\, O_t / \sum_t \gamma_t(k)\, u_{t,k}$.
38:       $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
39:       $\nu_k \leftarrow \arg\max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ {40-iter golden-section search.}
40:     $F = L$ {closed-form weighted Laplace MLE.}
41:       $\mu_k \leftarrow \mathrm{WeightedMedian}(O_{1:T},\ \gamma_{1:T}(k))$ {cumulative-weight bracketing with interpolation at the $W_k/2$ crossing.}
42:       $b_k \leftarrow \max\big(\sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k),\ 10^{-6}\big)$.
43:     $F = \mathrm{GED}$ {three-stage ECM: bracketed μ_k, closed-form α_k, bracketed p_k.}
44:       $\mu_k \leftarrow \arg\min_{\mu \in [\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ {40-iter golden-section search.}
45:       $\alpha_k \leftarrow \max\Big(\big[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\big]^{1/p_k},\ 10^{-6}\Big)$ where $W_k = \sum_t \gamma_t(k)$ {closed form, unique zero of $\partial Q_k / \partial \alpha$.}
46:       $p_k \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \log \mathrm{GED}(O_t; \mu_k, \alpha_k, p)$ {40-iter golden-section search.}
47:   **Convergence check.** $\mathcal{L}^{(n)} \leftarrow \underset{k}{\mathrm{logsumexp}}(\log \alpha_T(k))$.
48:   **if** $|\mathcal{L}^{(n)} - \mathcal{L}_{\mathrm{prev}}| < \mathrm{tol}$ **then**
49:     **break**
50:   **end if**
51:   $\mathcal{L}_{\mathrm{prev}} \leftarrow \mathcal{L}^{(n)}$.
52: **end for**
53: **return** $\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi}$ {$\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, b_k) for $F = L$; (μ_k, α_k, p_k) for

Algorithm 2 Viterbi decoding for the most likely hidden state sequence

Require: Observations $O_{1:T}$, Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$
Ensure: Most probable state sequence $S^*_{1:T}$
for $k = 1$ to K **do**
$\log \delta_1(k) \leftarrow \log(1/K) + \log \mathcal{N}(O_1 \mid \mu_k, \sigma_k^2)$.
end for
for $t = 2$ to T **do**
for $j = 1$ to K **do**
$\text{vals}(i) \leftarrow \log \delta_{t-1}(i) + \log T_{ij}$ for all i.
$\log \delta_t(j) \leftarrow \max_i \text{vals}(i) + \log \mathcal{N}(O_t \mid \mu_j, \sigma_j^2)$.
$\psi_t(j) \leftarrow \arg\max_i \text{vals}(i)$.
end for
end for
$S^*_T \leftarrow \arg\max_k \log \delta_T(k)$.
for $t = T - 1$ down to 1 **do**
$S^*_t \leftarrow \psi_{t+1}(S^*_{t+1})$.
end for
return $S^*_{1:T}$

Algorithm 3 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M, Number of paths P
Ensure: Simulated return paths $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$
for $p = 1$ to P **do**
Sample initial state $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$.
for $t = 1$ to M **do**
Emit observation $\hat{G}^{(p)}_t \sim \mathcal{N}(\mu_{S_t}, \sigma^2_{S_t})$.
if $t < M$ **then**
Transition to next state $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$.
end if
end for
end for
return $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

Algorithm 4 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (positive semi-definite (PSD), unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

for $p = 1$ to P **do**
  Draw copula sample $\mathbf{U}^{(p)} \in [0,1]^{T \times d}$ from $\mathcal{C}(\Sigma)$:
    $L \leftarrow \text{Cholesky}(\Sigma), \quad Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$ i.i.d.
    $Y \leftarrow ZL^\top$
  if $\mathcal{C} = \text{Gaussian}$ **then**
    $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$ componentwise
  else
    Draw $W \sim \chi_\nu^2/\nu$ i.i.d. for $t = 1, \dots, T$; $\quad X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$
    $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$ componentwise
  end if
  for $j = 1$ to d **do**
    Simulate $\tilde{g}_{j,1:T}^{(p)}$ from $\mathcal{M}_j$ via Algorithm 3.
    Compute order statistics $\tilde{g}_{j,(1)}^{(p)} \le \dots \le \tilde{g}_{j,(T)}^{(p)}$.
    Compute ranks $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$ for $t = 1, \dots, T$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$ for $t = 1, \dots, T$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

Algorithm 5 Single Index Model (SIM) construction and simulation

Require: IS return matrix $\mathbf{R} \in \mathbb{R}^{T \times d}$, market-column index m, market CHMM $\mathcal{M}_m$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

Ordinary least squares (OLS) fit. For $j \neq m$:
  $\hat{\beta}_j \leftarrow \text{Cov}(\mathbf{R}_{:,j}, \mathbf{R}_{:,m})/\text{Var}(\mathbf{R}_{:,m})$
  $\hat{\alpha}_j \leftarrow \overline{\mathbf{R}_{:,j}} - \hat{\beta}_j \overline{\mathbf{R}_{:,m}}$
  $\hat{\eta}_{j,t} \leftarrow R_{t,j} - \hat{\alpha}_j - \hat{\beta}_j R_{t,m}, \quad R_j^2 \leftarrow 1 - \sum_t \hat{\eta}_{j,t}^2 / \sum_t (R_{t,j} - \overline{\mathbf{R}_{:,j}})^2$
for $p = 1$ to P **do**
  Simulate market path $G_{m,1:T}^{(p)}$ from $\mathcal{M}_m$ via Algorithm 3.
  $\hat{G}_{m,t}^{(p)} \leftarrow G_{m,t}^{(p)}$.
  for $j \neq m$ **do**
    Draw residual indices $\{\iota_t\}_{t=1}^T \sim \text{Uniform}(\{1, \dots, T\})$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,\iota_t}$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

A.2.3 Per-State Shape Diagnostics: CHMM-t and CHMM-GED

This appendix backs the discussion in Section 5 of the per-state shape allocation under CHMM-t and CHMM-GED on SPY IS at $K = 18$ (seed = 20260420).

CHMM-t per-state ν_k. Figure 6 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit. Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

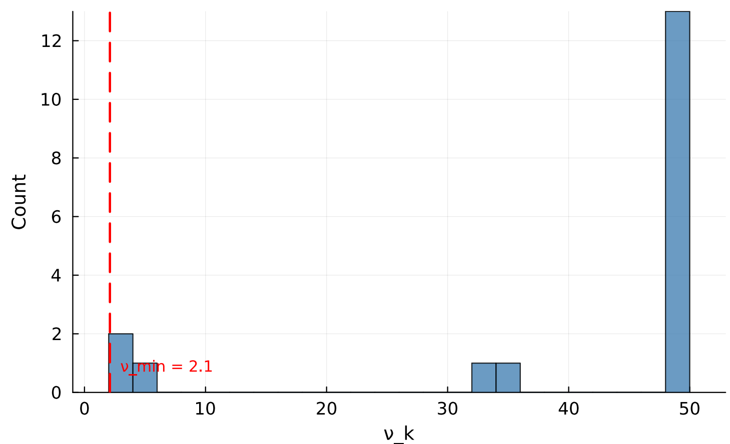
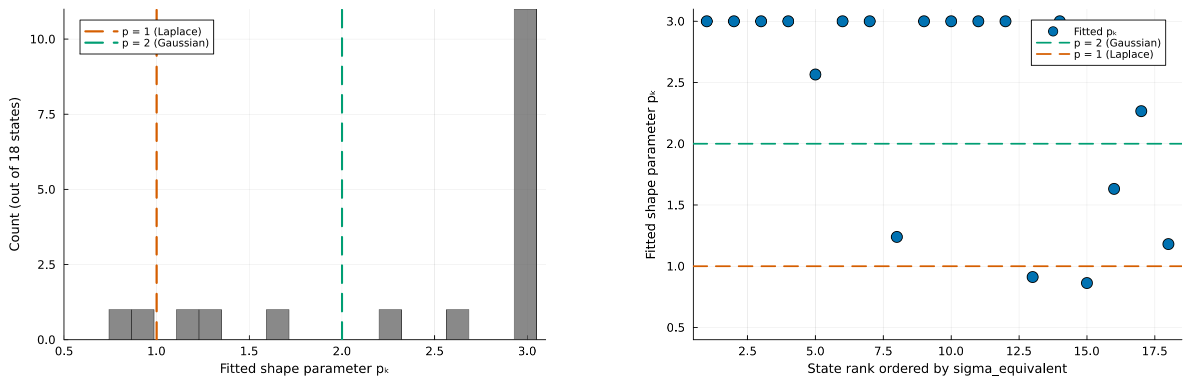


Figure 6. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

CHMM-GED per-state p_k (Gaussian-bulk / Laplace-tail bimodality). Figure 7 plots the analogous per-state p_k histogram for the $K = 18$ CHMM-GED fit, alongside the p_k ranking by variance-equivalent state σ. The fit is bimodal: thirteen states attain the upper ECM bracket at $p_k = 3.0$ (Gaussian-shape and beyond), one state lands at intermediate shape $p_k = 1.6$, and four states cluster in the Laplace-shape regime $p_k \in [0.86, 1.24]$. The smallest fitted p_k on SPY is 0.86, well clear of every $p_{\min}$ tested in the bracket sweep below, so the lower bracket is non-binding. Crucially the four Laplace-shape states are concentrated in the high-volatility tail of the state ordering: ranks 13, 15, and 18 (by variance-equivalent σ) all have $\hat{p}_k$ near 1. This is the data-driven analog of a hand-classified Gaussian-bulk / Laplace-tail hybrid: CHMM-GED finds the partition by ECM, not by a priori state assignment.

The bimodal partition replicates across 10 Monte Carlo seeds (every seed yields the same partition counts to within one state) and across the cross-asset universe (Appendix A.7).

Table 6 reports the seven-metric panel for CHMM-t fits whose ECM bracket lower bound is lifted from $\nu_{\min} = 2.1$ through $\{2.5, 3.0, 4.0\}$, plus the Gaussian limit ($\nu = \infty$, equivalent to CHMM-N). Lifting the bracket does not reduce the IS kurtosis overshoot: the simulated kurtosis moves within the 12.5–14.0 band across bracket floors and only collapses to 5.05 in the full Gaussian limit. The overshoot is therefore a feature of the Student-t emission family at $K = 18$ and not a lower-bracket artefact.



(a) Histogram of fitted $\hat{p}_k$.

(b) $\hat{p}_k$ vs. volatility rank.

Figure 7. Per-state shape diagnostics for the CHMM-GED fit at $K = 18$ on SPY IS. Panel (a): histogram of fitted $\hat{p}_k$ across the eighteen states. The dashed reference lines mark the Laplace shape ($p = 1$) and Gaussian shape ($p = 2$); $\hat{p}_k$ values cluster in two regimes (bulk near the upper bracket, tail near $p = 1$). Panel (b): $\hat{p}_k$ versus state rank ordered by variance-equivalent σ (rank 1 is the calmest state, rank 18 is the highest-volatility state). The high-volatility ranks 13, 15, 18 sit at the Laplace line; the bulk of states sit at the Gaussian-or-beyond line.

Table 6. CHMM-t $\nu_{\min}$ bracket sensitivity at $K = 18$ on SPY (seed = 20260420, 1,000 paths). AD = Anderson–Darling pass rate; W_1 = Wasserstein-1; H = Hellinger distance (definitions in Appendix A.3).

$\nu_{\min}$	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	Kurt OoS	ACF-MAE	W_1	H
2.1	95.7	90.2	86.2	82.0	13.95	10.08	0.0546	0.100	0.0759
2.5	95.8	90.6	84.2	79.1	15.12	10.34	0.0547	0.100	0.0759
3.0	93.8	89.5	85.3	80.4	12.25	9.65	0.0548	0.099	0.0760
4.0	95.5	89.5	85.0	80.3	13.25	9.39	0.0550	0.101	0.0759
∞ (Gauss)	94.1	86.7	84.2	75.5	5.03	4.46	0.0510	0.110	0.0809
Observed	–	–	–	–	7.68	5.29	–	–	–

Table 7 reports the analogous bracket sensitivity for CHMM-GED on $p_{\max} \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ (with $p_{\min} = 0.5$ fixed) and on $p_{\min} \in \{0.5, 0.7, 0.85\}$ (with $p_{\max} = 3.0$ fixed). The headline reads cleanly. The Gaussian-like state count is invariant at 13/18 across every upper bracket; only the redistribution between intermediate and Laplace-like states moves slightly with $p_{\max}$, and the total heavy-shape count stays in $\{4, 5\}$. Tightening $p_{\max}$ from 4.0 to the canonical 3.0 costs about 4 nats of incomplete-data log-likelihood. The $p_{\min}$ sweep is fully degenerate: every $p_{\min} \in [0.5, 0.85]$ produces an identical fit because the smallest observed $\hat{p}_k$ on SPY is 0.86, so the lower bracket is non-binding.

Table 7. CHMM-GED bracket sensitivity at $K = 18$ on SPY (seed = 20260420, 1,000 paths). $\hat{p}_{\min}$, $\hat{p}_{\mathrm{med}}$, $\hat{p}_{\max}$ are the across-state minimum, median, and maximum of the fitted $\hat{p}_k$. nG / nI / nL / nS are the partition counts (Gaussian-like $\hat{p}_k \geq 1.85$, intermediate $1.30 \leq \hat{p}_k < 1.85$, Laplace-like $0.85 \leq \hat{p}_k < 1.30$, super-Laplace $\hat{p}_k < 0.85$). LL is the incomplete-data log-likelihood at convergence.

Sweep	$p_{\min}$	$p_{\max}$	LL	$\hat{p}_{\min}$	nG	nI	nL	nS	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	ACF-MAE	W_1	H
$p_{\max}$ sweep	0.5	2.0	−4732.0	0.89	13	0	5	0	95.8	91.0	83.8	81.2	5.02	0.0541	0.099	0.0781
	0.5	2.5	−4725.3	0.87	13	2	3	0	96.1	92.4	84.3	80.1	5.09	0.0549	0.098	0.0778
	0.5	**3.0**	−4720.3	0.86	13	1	4	0	96.3	92.7	83.8	81.3	5.05	0.0546	0.098	0.0779
	0.5	3.5	−4717.5	0.86	13	1	4	0	96.3	91.8	86.7	82.4	5.11	0.0544	0.099	0.0782
	0.5	4.0	−4716.3	0.86	13	1	4	0	95.9	91.3	84.4	80.6	5.18	0.0543	0.099	0.0785
$p_{\min}$ sweep	0.50	3.0	−4720.3	0.86	13	1	4	0	96.3	92.7	83.8	81.3	5.05	0.0546	0.098	0.0779
	0.70	3.0	−4720.3	0.86	13	1	4	0	96.3	92.7	83.8	81.3	5.05	0.0546	0.098	0.0779
	0.85	3.0	−4720.3	0.86	13	1	4	0	96.3	92.7	83.8	81.3	5.05	0.0546	0.098	0.0779

A.3 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in Section 3.5. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [50, 51] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \tag{22}$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|. \tag{23}$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [64],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| \, du. \tag{24}$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^{B} (\sqrt{p_i} - \sqrt{q_i})^2}, \tag{25}$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}^{(p)}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of Section 4.5), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} = \frac{1}{P} \sum_{p=1}^{P} \frac{2}{d(d-1)} \sum_{1 \le i < j \le d} \left| \hat{\Sigma}^{(p)}_{\text{sim},\,ij} - \hat{\Sigma}_{\text{obs},\,ij} \right|.$$

A.3.1 Continuous Ranked Probability Score and Diebold-Mariano

The OoS CRPS column of Table 1 reports a proper-scoring-rule complement to the KS pass rate. We use the unbiased sample CRPS estimator: for an ensemble of N simulated paths $\{x_i\}_{i=1}^{N}$ at time t and observed value y_t,

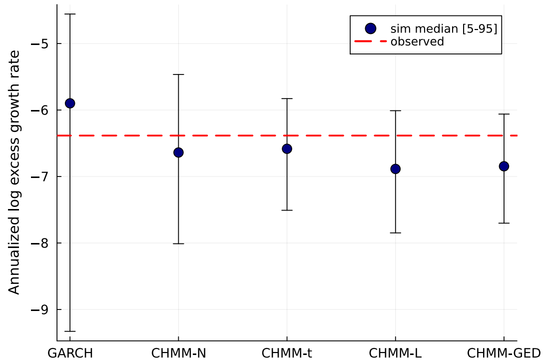
$$\widehat{\text{CRPS}}_t = \frac{1}{N} \sum_{i=1}^{N} |x_i - y_t| - \frac{1}{N(N-1)} \sum_{1 \le i < j \le N} |x_i - x_j|. \tag{26}$$

The double sum is computed in $O(N \log N)$ via the sorted-ensemble identity $\sum_{i<j}(x_{(j)} - x_{(i)}) = \sum_i x_{(i)}(2i - N - 1)$. The ensemble at each t is the cross-section of the $N = 1{,}000$ unconditional simulated paths used throughout Section 4; this yields a marginal-predictive CRPS suitable for an unconditional generative-fidelity assessment, distinct from the conditional one-step-ahead CRPS used in forecasting.

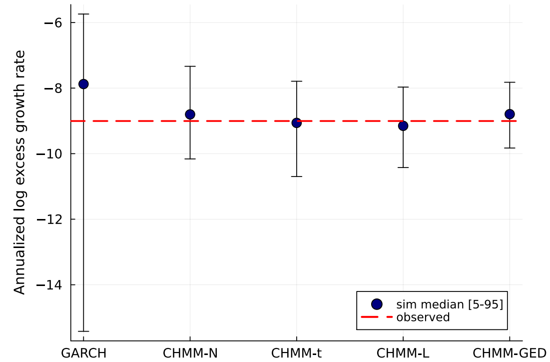
For pairwise comparison we report a Diebold-Mariano test on the per-t loss differential $d_t = \widehat{\text{CRPS}}^{A}_t - \widehat{\text{CRPS}}^{B}_t$ with a Newey-West HAC variance under a Bartlett kernel and bandwidth $h = \lfloor T^{1/3}_{\text{OoS}} \rfloor = 8$. Two-sided p-values are reported under the standard-normal null. The body claim of Section 4.3 is: CHMM ties or beats every benchmark on mean OoS CRPS, with Gaussian i.i.d. the only significantly worse comparator.

A.3.2 VaR / ES Envelope Visualization (Companion to Table 4)

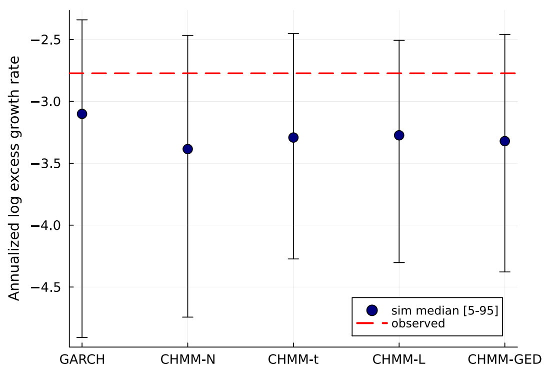
Figure 8 visualizes the median and $[5\%, 95\%]$ envelopes across simulated paths whose numeric values are reported in Table 4.



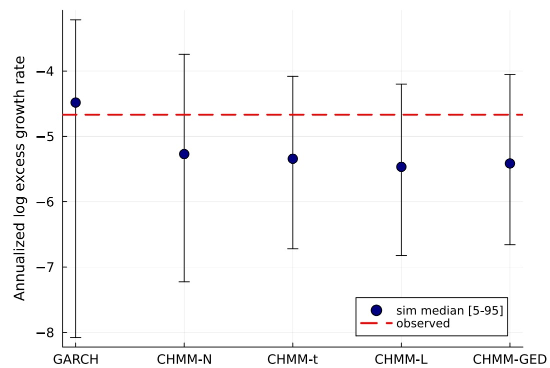
(a) IS VaR ($\alpha = 0.01$).



(b) IS ES ($\alpha = 0.01$).



(c) OoS VaR ($\alpha = 0.05$).



(d) OoS ES ($\alpha = 0.05$).

Figure 8. VaR / ES envelope diagnostic (SPY, $N_{\text{paths}} = 1{,}000$; global seed policy in Section 3.5). Each panel shows the simulated median (dot) and $[5\%, 95\%]$ envelope (error bar) across paths against the observed historical value (dashed red line); panels (a)/(b) are the IS window at $\alpha = 0.01$, panels (c)/(d) are the OoS window at $\alpha = 0.05$, and the x-axis in each panel lists the five generators kept for the volatility-aware conditional-coverage comparison (GARCH(1,1), CHMM-N, CHMM-t, CHMM-L, CHMM-GED). The i.i.d. bootstrap is omitted here because its envelope is by construction matched to the empirical marginal and does not discriminate among volatility-aware models. The y-axis is the annualized log excess growth rate (daily log return scaled by $1/\Delta t$ with $\Delta t = 1/252$), consistent with the $R_{\text{IS}}/R_{\text{OoS}}$ convention used throughout the paper. Headline: every observed red line falls inside the corresponding simulated envelope, confirming that the CHMM family passes the coverage leg of the back-test complementary to the Kupiec likelihood-ratio leg in Section 4.6.

A.4 Formal Theoretical Statements

The following propositions are referenced in Section 3.8 and stated formally here for reference; each is a one-line specialisation of a standard result and the proof sketch is one sentence.

Proposition 1 (ECM monotonicity for CHMM-t and CHMM-GED (and EM monotonicity for CHMM-N, CHMM-L))**.** *Under compactness of the parameter space (μ_k, σ_k or b_k or α_k, $\nu_k \in [\nu_{\min}, \nu_{\max}]$, $p_k \in [p_{\min}, p_{\max}]$ each in a closed bounded interval, $T_{ij} \geq \epsilon_T > 0$), the two-block CM step for CHMM-t of Section 3.7, namely the closed-form weighted updates of $(\mathbf{T}, \boldsymbol{\pi}, \mu_k, \sigma_k)$ at fixed ν_k, followed by a golden-section maximisation of the per-state Q-function over ν_k, satisfies*

$Q(\theta^{(n+1)} \mid \theta^{(n)}) \geq Q(\theta^{(n)} \mid \theta^{(n)})$, *and consequently* $\mathcal{L}(\theta^{(n+1)}) \geq \mathcal{L}(\theta^{(n)})$ *where* $\mathcal{L}$ *is the incomplete-data log-likelihood. The same monotonicity holds for the closed-form CHMM-N and CHMM-L M-steps under the standard EM guarantee, and for the three-stage CHMM-GED CM sequence (bracketed* μ_k*, closed-form* α_k*, bracketed* p_k*) of equations (19)–(21).*

Proof sketch. Each CM step does not decrease Q by definition (closed-form steps are exact maximisers, golden-section steps are bracketed maxima within the prescribed interval); the standard EM inequality $\mathcal{L}(\theta^{(n+1)}) - \mathcal{L}(\theta^{(n)}) \geq Q(\theta^{(n+1)} \mid \theta^{(n)}) - Q(\theta^{(n)} \mid \theta^{(n)})$ [56, 57] converts Q-monotonicity into log-likelihood monotonicity. Compactness gives boundedness above, hence convergence.

Proposition 2 (Identifiability up to label permutation)**.** *Under irreducibility and aperiodicity of* $\mathbf{T}$ *(Assumption 1) and the emission contrast condition (Assumption 2), the CHMM parameter is identifiable up to label permutation in all four emission families considered (CHMM-N, CHMM-t with* $\nu_k \in [\nu_{\min}, \nu_{\max}]$*, CHMM-L, and CHMM-GED with* $p_k \in [p_{\min}, p_{\max}]$*).*

Proof sketch. The HMM identifiability theorem of Allman et al. [58] requires $\mathbf{T}$ of full rank and per-state emissions linearly independent in the space of densities on $\mathbb{R}$. Full-rank $\mathbf{T}$ follows from irreducibility [55]; linear independence of Gaussian, bracketed-ν Student-t, Laplace, and bracketed-p GED emissions with distinct location-scale-shape parameters follows from Yakowitz and Spragins [59].

Proposition 3 (MLE consistency)**.** *Under Assumptions 1–2 and the assumption that the true parameter lies in the interior of a compact* Θ*, the MLE* $\hat{\theta}_T = \arg\max_{\theta\in\Theta} \mathcal{L}_T(\theta)$ *is consistent:* $\hat{\theta}_T \xrightarrow{p} \theta_0$ *as* $T \to \infty$*, modulo label permutation.*

Proof reference. Bickel et al. [60] (Theorem 1) and Douc et al. [61] apply directly under our assumption set: irreducibility supplies the mixing condition, finite second moment the moment condition, compactness the parameter-space condition.

Proposition 4 (Marginal preservation under rank reordering)**.** *Let* $\tilde{g}_{j,1:T}$ *be a sample of size* T *from the per-asset CHMM marginal of asset* j*, and let the rank-reordered output be* $\hat{g}^{(p)}_{j,t} = \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$ *where* $r^{(p)}_{j,t}$ *is the rank of the copula sample. Then for every asset* j *the empirical CDF of* $\{\hat{g}^{(p)}_{j,t}\}_{t=1}^{T}$ *equals that of* $\{\tilde{g}^{(p)}_{j,t}\}_{t=1}^{T}$*.*

Proof sketch. The map $t \mapsto r^{(p)}_{j,t}$ is a permutation almost surely, so the multiset of values is preserved; the empirical CDF depends only on the multiset.

A.5 Spectral ACF Identity: Extended Derivation

The closed-form spectral identity (8) is the central technical contribution of Section 3.8 and is invoked throughout Sections 4–5. The main-body derivation is intentionally compressed; we restate it here as a self-contained theorem with a step-by-step proof, and record the lag-zero behaviour and the complex / non-diagonalisable extensions that the main body does not.

Theorem 1 (Mixture-of-eigenvalues identity for the absolute-return ACF)**.** *Let* $\{(s_t, G_t)\}_{t\in\mathbb{Z}}$ *be a stationary CHMM on* $\{1, \ldots, K\}$ *with irreducible aperiodic transition matrix* $\mathbf{T}$ *(Assumption 1) and per-state emissions of finite second moment (Assumption 2). Write* $\bar{\boldsymbol{\pi}}$ *for the unique stationary distribution,* $\mathbf{D} = \operatorname{diag}(\bar{\boldsymbol{\pi}})$*,* $m_k = \mathbb{E}[|G_t| \mid s_t = k]$*,* $\mathbf{m} = (m_1, \ldots, m_K)^\top$*,* $\mu = \mathbb{E}|G_t| = \bar{\boldsymbol{\pi}}^\top \mathbf{m}$*, and* $\sigma^2_{|G|} = \operatorname{Var}(|G_t|)$*. Assume* $\mathbf{T}$ *is diagonalisable with right eigenvectors* $\mathbf{v}_k$ *and left eigenvectors* $\mathbf{w}_k$

biorthonormal, $\mathbf{w}_k^\top \mathbf{v}_l = \delta_{kl}$*, indexed so that* $\lambda_1 = 1 > |\lambda_2| \geq \cdots \geq |\lambda_K|$*. Then for every integer lag* $\tau \geq 1$*,*

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k\, \lambda_k^{\tau}, \qquad w_k \;=\; \frac{(\mathbf{m}^\top \mathbf{D}\, \mathbf{v}_k)\,(\mathbf{w}_k^\top \mathbf{m})}{\sigma_{|G|}^2}.$$

Proof. The proof is organised in six steps.

Step 1 (Definitions). The autocorrelation of $|G_t|$ at lag $\tau \geq 0$ is, by stationarity,

$$\rho_{|G|}(\tau) \;=\; \frac{\mathrm{Cov}(|G_t|,\, |G_{t+\tau}|)}{\sigma_{|G|}^2} \;=\; \frac{\mathbb{E}[|G_t|\,|G_{t+\tau}|] - \mu^2}{\sigma_{|G|}^2}.$$

The task reduces to evaluating $\mathbb{E}[|G_t|\,|G_{t+\tau}|]$ in closed form for $\tau \geq 1$.

Step 2 (Conditioning on the latent state pair). For $\tau \geq 1$, by the law of total expectation conditional on $(s_t, s_{t+\tau})$,

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \sum_{i,j=1}^{K} \mathbb{P}(s_t = i, s_{t+\tau} = j)\, \mathbb{E}\big[|G_t|\,|G_{t+\tau}| \,\big|\, s_t = i, s_{t+\tau} = j\big]\,.$$

The HMM emission structure gives the conditional independence $G_t \perp G_{t+\tau} \mid (s_t, s_{t+\tau})$ for $\tau \geq 1$: each emission depends only on its current state, and integrating the intermediate states $s_{t+1:t+\tau-1}$ out of the full-sequence factorisation $\big(\prod_u b_{s_u}(G_u)\big)\, p(s_{t+1:t+\tau-1} \mid s_t, s_{t+\tau})$ leaves the endpoint emissions decoupled. Hence

$$\mathbb{E}\big[|G_t|\,|G_{t+\tau}| \,\big|\, s_t = i, s_{t+\tau} = j\big] \;=\; \mathbb{E}[|G_t| \mid s_t = i]\, \mathbb{E}[|G_{t+\tau}| \mid s_{t+\tau} = j] \;=\; m_i\, m_j.$$

The joint state probability factors by stationarity and the Markov property, $\mathbb{P}(s_t = i, s_{t+\tau} = j) = \bar{\pi}_i\, (\mathbf{T}^\tau)_{ij}$. Substituting and noting $[\mathbf{D}\mathbf{T}^\tau]_{ij} = \bar{\pi}_i (\mathbf{T}^\tau)_{ij}$,

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mathbf{m}^\top \mathbf{D}\, \mathbf{T}^\tau\, \mathbf{m}, \qquad \tau \geq 1. \tag{27}$$

Step 3 (Spectral preliminaries). For irreducible aperiodic $\mathbf{T}$, Perron–Frobenius applied to row-stochastic matrices gives $\lambda_1 = 1$ as a simple eigenvalue with all other eigenvalues strictly inside the unit disk [55]. The right eigenvector for λ_1 is $\mathbf{v}_1 = \mathbf{1}$, since $\mathbf{T}\mathbf{1} = \mathbf{1}$ from the row-sum constraint, and the left eigenvector is $\mathbf{w}_1 = \bar{\boldsymbol{\pi}}$, since $\bar{\boldsymbol{\pi}}^\top \mathbf{T} = \bar{\boldsymbol{\pi}}^\top$ by definition of the stationary distribution. Biorthonormality $\mathbf{w}_1^\top \mathbf{v}_1 = \bar{\boldsymbol{\pi}}^\top \mathbf{1} = 1$ is automatic since $\bar{\boldsymbol{\pi}}$ is a probability vector.

Step 4 (Spectral decomposition of $\mathbf{T}^\tau$). Diagonalisability with biorthonormality gives the dyadic expansion

$$\mathbf{T}^\tau \;=\; \sum_{k=1}^{K} \lambda_k^\tau\, \mathbf{v}_k \mathbf{w}_k^\top \;=\; \mathbf{1}\, \bar{\boldsymbol{\pi}}^\top \;+\; \sum_{k=2}^{K} \lambda_k^\tau\, \mathbf{v}_k \mathbf{w}_k^\top,$$

isolating the dominant ($k = 1$) projection from the contractive spectrum. The dominant projector $\mathbf{P}_1 = \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is the rank-one stationary projection: $\mathbf{P}_1 \mathbf{x} = (\bar{\boldsymbol{\pi}}^\top \mathbf{x})\, \mathbf{1}$.

Step 5 (Evaluation of the bilinear form). Substituting the spectral decomposition into (27),

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mathbf{m}^\top \mathbf{D}\,\mathbf{1}\,\bar{\boldsymbol{\pi}}^\top \mathbf{m} \;+\; \sum_{k=2}^{K} \lambda_k^\tau\,(\mathbf{m}^\top \mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top \mathbf{m}).$$

The $k=1$ term simplifies to μ^2: $\mathbf{D1}$ is the column vector $\bar{\boldsymbol{\pi}}$, so $\mathbf{m}^\top\mathbf{D1} = \bar{\boldsymbol{\pi}}^\top\mathbf{m} = \mu$, and likewise $\bar{\boldsymbol{\pi}}^\top\mathbf{m} = \mu$, giving the product $\mu \cdot \mu = \mu^2$. Denoting the $k \geq 2$ scalar coefficients

$$c_k \;=\; (\mathbf{m}^\top \mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top \mathbf{m}),$$

we obtain

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mu^2 \;+\; \sum_{k=2}^{K} c_k\,\lambda_k^\tau, \qquad \tau \geq 1.$$

Step 6 (Autocovariance and autocorrelation). Subtracting μ^2 from Step 1 cancels the dominant term, leaving

$$\mathrm{Cov}(|G_t|,\,|G_{t+\tau}|) \;=\; \sum_{k=2}^{K} c_k\,\lambda_k^\tau, \qquad \tau \geq 1,$$

which is equation (7). Dividing by $\sigma^2_{|G|} = \gamma(0)$ gives $\rho_{|G|}(\tau) = \sum_{k\geq 2} w_k\,\lambda_k^\tau$ with $w_k = c_k/\sigma^2_{|G|}$, equation (8). □

Lag zero, the sum of weights, and the within-state white-noise contribution. The bilinear identity (27) fails at $\tau = 0$, where $G_t = G_{t+\tau}$ and the conditional product is the per-state second moment $M_i = \mathbb{E}[G_t^2 \mid s_t = i]$ rather than m_i^2. The eigenvalue completeness $\sum_{k=1}^K \mathbf{v}_k\mathbf{w}_k^\top = I$ instead gives the identity

$$\sum_{k=2}^{K} c_k \;=\; \mathbf{m}^\top\mathbf{D}\,\mathbf{m} \;-\; \mu^2 \;=\; \sum_{i=1}^{K} \bar{\pi}_i\,m_i^2 \;-\; \mu^2 \;=\; \mathrm{Var}\big(m_{s_t}\big),$$

the between-state variance of $|G_t|$, equal to the variance of the conditional-mean random variable m_{s_t} under $\bar{\boldsymbol{\pi}}$. By the law of total variance, $\sigma^2_{|G|} = \mathrm{Var}(m_{s_t}) + \mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)]$, so

$$\sum_{k=2}^{K} w_k \;=\; \frac{\mathrm{Var}(m_{s_t})}{\sigma^2_{|G|}} \;\in\; [0,1],$$

with equality to 1 if and only if every per-state distribution is degenerate at m_i. The residual $1 - \sum_{k\geq 2} w_k = \mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)]\,/\,\sigma^2_{|G|}$ is the within-state variance ratio: it contributes to $\sigma^2_{|G|}$ but not to $\mathrm{Cov}(|G_t|, |G_{t+\tau}|)$ at any positive lag, and is the source of the discontinuity from $\rho_{|G|}(0) = 1$ to $\rho_{|G|}(1) = \sum_{k\geq 2} w_k\,\lambda_k$ at the first lag. The substantive consequence for moderate-K CHMM is unchanged: at every $\tau \geq 1$ the autocorrelation lies in the span of the geometric decays $\{\lambda_k^\tau\}_{k\geq 2}$, and the slow-decay envelope is governed by the largest non-unit eigenvalue.

Squared-return analogue. Replacing $|G_t|$ with G_t^2 and $\mathbf{m}$ with $\mathbf{M} = (M_1, \ldots, M_K)^\top$ in Steps 1–6 above is termwise valid for $\tau \geq 1$ (the conditional independence and stationarity arguments are unaffected), giving the squared-return analogue

$$\rho_{G^2}(\tau) \;=\; \sum_{k=2}^{K} \tilde{w}_k\,\lambda_k^\tau, \qquad \tilde{w}_k \;=\; \frac{(\mathbf{M}^\top\mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top\mathbf{M})}{\sigma^2_{G^2}}.$$

The non-unit spectrum of $\mathbf{T}$ that determines $\rho_{|G|}(\tau)$ also determines $\rho_{G^2}(\tau)$; only the linear projection of the per-state moment vector onto the eigenmodes changes.

Remark (complex eigenvalues). For real $\mathbf{T}$ without self-adjoint structure, eigenvalues outside $\mathbb{R}$ occur in complex conjugate pairs with conjugate eigenvectors. A pair $(\lambda_k, \bar{\lambda}_k)$ at $\lambda_k = re^{i\theta}$ contributes a real damped-oscillation term of the form $A\,r^\tau \cos(\theta\tau) + B\,r^\tau \sin(\theta\tau)$ to $\gamma(\tau)$, with (A, B) determined from $(\operatorname{Re} c_k, \operatorname{Im} c_k)$. Theorem 1 as stated remains valid; only the interpretation of the individual modes shifts from monotone geometric to damped sinusoidal. For $\mathbf{T}$ estimated on equity returns at the operating point of Section 4, the slow-decay mode at $\lambda_2 \approx \exp(-1/\tau^*)$ is real and positive, and the mono-exponential interpretation of the dominant timescale used in Section 3.8 carries through.

Remark (non-diagonalisable T). Diagonalisability is generic: for $\mathbf{T}$ estimated by Baum–Welch in the interior of a non-degenerate parameter region, the eigenvalues are almost surely distinct, and the sample paths used in this paper realise this generic case at every K in the sweep. Should $\mathbf{T}$ admit a non-trivial Jordan block of size J, the dyadic expansion of Step 4 acquires polynomial-in-τ prefactors $\binom{\tau}{j}\lambda_k^{\tau-j}$ for $j = 0, \ldots, J-1$ on the off-diagonal blocks, and $\rho_{|G|}(\tau)$ becomes a finite sum of polynomial-times-geometric terms with the same exponential rate $\max_{k\geq 2}|\lambda_k|$. The substantive content of Theorem 1, that the absolute-return ACF is determined entirely by the non-unit spectrum of $\mathbf{T}$, is unchanged.

A.6 State-Resolution and Multi-Emission Sensitivity

This appendix reports the full K-sweep underlying the headline operating point of Section 4, the multi-emission family extension confirming that $K = 18$ is stable under Student-t and Laplace emissions, the EM convergence curves at representative K, the IS and OoS visual comparison panels at non-selected $K \in \{3, 6, 12, 21\}$, the multi-emission figure panels at $K = 18$, and the Rydén $K = 2$ replication.

A.6.1 Full State-Resolution Sensitivity Table

Table 8 reports the full K-sweep referenced in Section 4.2 of the main text. Observed excess kurtosis is 7.68 IS and 5.29 OoS at all K.

Table 8. State resolution sensitivity for SPY CHMM-N (1,000 paths, $\alpha = 0.05$). Every metric is computed on the identical simulation pipeline used for Table 1 in the main text. IS coverage was 100% at every K; OoS coverage is reported separately. ACF-MAE columns: $|G_t|$ measures the volatility-clustering reproduction; G_t measures linear-autocorrelation reproduction (parallel to $|G_t|$, both averaged over lags 1–252).

	KS (%)		AD (%)		Kurtosis		ACF-MAE				
K	IS	OoS	IS	OoS	Obs	Sim	$\lvert G_t\rvert$	G_t	W_1	H	Cov OoS (%)
3	89.0	77.2	78.9	68.6	7.68	3.82	0.0463	0.0240	0.131	0.0839	94.9
6	92.9	78.7	84.3	74.3	7.68	5.38	0.0496	0.0238	0.116	0.0829	90.9
9	93.0	81.4	84.4	74.4	7.68	4.63	0.0492	0.0238	0.118	0.0810	91.9
12	94.9	81.6	85.8	75.0	7.68	4.57	0.0500	0.0238	0.113	0.0801	91.9
15	93.9	81.0	84.8	73.5	7.68	4.89	0.0508	0.0238	0.114	0.0804	88.9
18	94.8	85.5	88.2	79.7	7.68	5.04	0.0515	0.0237	0.110	0.0807	91.9
21	**95.6**	**88.0**	**89.1**	**78.8**	7.68	3.78	0.0534	**0.0236**	0.112	0.0801	**99.0**

A.6.2 Multi-Emission State-Resolution Sensitivity

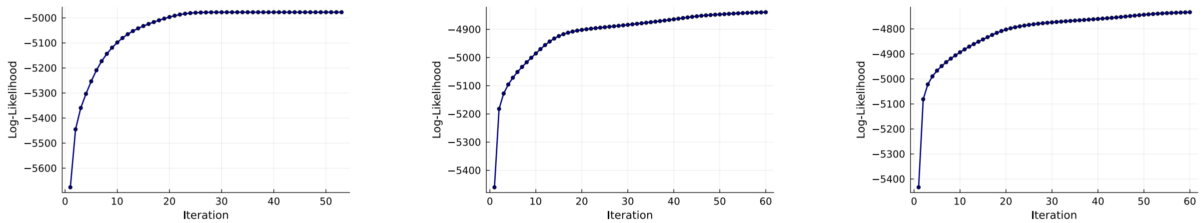
Table 9 extends Table 8 across the four emission families, confirming that the qualitative operating point $K = 18$ is stable when the Gaussian component is replaced with Student-t, Laplace, or GED emissions. For CHMM-t the per-state degrees of freedom ν_k are refit by ECM-plus-golden-section at every K; for CHMM-L the weighted-median location and weighted MAD scale are closed form and impose no additional cost; for CHMM-GED the three-stage CM update of Section 3.7 refits (μ_k, α_k, p_k) at every K with p_k bracketed in $[0.5, 3.0]$. Three patterns extend from Table 8 to the new families. First, the distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band for all four families: CHMM-t reaches its highest IS KS at $K = 18$ (94.6%), CHMM-L reaches its highest IS KS at $K = 18$ (94.2%) and its highest IS AD at $K = 18$ (91.4%), and CHMM-GED reaches its highest IS KS at $K = 12$ (97.6%) and the highest entry across the entire panel. Once a family enters the plateau the operating point is flat with respect to K, so we report the main-text results at the same $K = 18$ used for CHMM-N. Second, the absolute-return ACF-MAE remains within the narrow 0.046–0.058 band across the four families, and the raw-return ACF-MAE remains essentially constant at ≈ 0.024 across all 28 (family, K) combinations. Both temporal axes are properties of the regime structure rather than the emission family: $|G_t|$ ACF reproduction is governed by the eigenvalue spectrum of $\mathbf{T}$ (Section 3.8), and raw-G_t ACF is null in expectation under any HMM with i.i.d. within-state emissions. Third, simulated kurtosis behaves as expected from the component families: the Gaussian sweep peaks around 5.0 at the high-K end, the Laplace sweep lands squarely in the 5.2–6.7 range (closest to the observed 7.68), the Student-t sweep climbs sharply above 13 across the entire sweep because the ECM search assigns small ν_k to a subset of tail states, and the GED sweep mirrors the Gaussian curve at 4.7–5.5 because the bulk of $\hat{p}_k$ values pin near the upper bracket and the Laplace-like tail states contribute too little stationary mass to lift the aggregate moment.

Table 9. Multi-emission state resolution sensitivity for SPY (1,000 paths, $\alpha = 0.05$). CHMM-N: Gaussian; CHMM-t: Student-t with per-state ν_k refit by ECM-plus-golden-section; CHMM-L: Laplace with weighted-median location and weighted-MAD scale; CHMM-GED: Generalized Error Distribution with per-state shape $p_k \in [0.5, 3.0]$ via three-stage ECM (Section 3.7). Every metric is computed on the identical simulation pipeline used for Table 1. Observed excess kurtosis is 7.68 IS and 5.29 OoS. ACF-MAE columns parallel the main-text Table 1: $|G_t|$ for volatility clustering, G_t (raw) for linear autocorrelation.

		KS (%)		AD (%)			ACF-MAE			
Family	K	IS	OoS	IS	OoS	Kurt (IS sim)	$\|G_t\|$	G_t	W_1 IS	H IS
CHMM-N	3	89.7	80.1	79.2	70.7	3.86	0.0467	0.0240	0.128	0.0833
	6	93.8	79.1	87.2	72.8	5.29	0.0504	0.0238	0.114	0.0825
	9	94.9	83.0	84.3	74.8	4.58	0.0490	0.0238	0.115	0.0806
	12	93.7	82.7	85.5	75.0	4.56	0.0501	0.0237	0.115	0.0804
	15	93.6	83.0	87.0	75.1	4.89	0.0509	0.0238	0.114	0.0803
	18	94.7	83.4	88.5	76.6	4.97	0.0511	0.0237	0.108	0.0805
	21	**95.1**	**88.0**	**88.2**	**79.7**	3.80	0.0532	0.0236	0.110	0.0796
CHMM-t	3	92.2	82.9	88.0	79.1	33.68	0.0538	0.0235	0.108	0.0759
	6	91.8	80.4	86.7	74.8	22.74	0.0527	0.0235	0.108	0.0772
	9	95.4	84.0	89.9	78.0	16.60	0.0535	0.0234	0.102	0.0752
	12	95.2	85.0	89.8	78.4	14.98	0.0543	0.0234	0.100	0.0757
	15	94.7	83.7	88.8	79.5	16.36	0.0539	0.0234	0.101	0.0758
	18	**96.0**	**84.5**	**91.7**	**80.8**	13.19	0.0551	**0.0234**	**0.099**	**0.0759**
	21	94.5	83.3	88.5	78.1	14.66	0.0539	0.0234	0.102	0.0758
CHMM-L	3	80.5	62.7	81.2	66.5	5.24	0.0531	0.0235	0.113	0.0855
	6	88.3	76.3	83.1	71.8	5.86	0.0510	0.0235	0.112	0.0821
	9	90.7	77.2	84.2	72.2	5.92	0.0519	0.0235	0.112	0.0823
	12	93.8	79.3	88.7	76.7	6.17	0.0541	0.0234	0.105	0.0793
	15	92.5	78.3	88.2	75.5	5.98	0.0539	0.0234	0.107	0.0794
	18	**94.2**	**80.5**	**91.4**	**80.6**	**6.74**	0.0566	0.0234	0.104	0.0795
	21	91.5	79.6	87.5	77.7	5.95	0.0539	0.0234	0.107	0.0807
CHMM-GED	3	90.3	78.4	85.5	73.6	5.45	0.0531	0.0236	0.109	0.0818
	6	91.8	79.0	84.9	73.5	5.22	0.0518	0.0235	0.108	0.0805
	9	94.6	81.6	89.1	75.9	5.03	0.0535	0.0234	0.101	0.0787
	12	**97.6**	**84.2**	**94.0**	**81.2**	4.67	0.0563	0.0234	**0.094**	**0.0775**
	15	94.2	82.0	87.4	76.5	5.14	0.0535	0.0235	0.104	0.0784
	18	96.3	83.8	92.7	81.3	5.05	0.0546	0.0235	0.098	0.0779
	21	94.1	85.3	88.7	81.2	4.82	0.0542	0.0235	0.102	0.0781

A.6.3 Baum-Welch Convergence

Figure 9 shows the Baum-Welch convergence curves for representative state resolutions. All models converged within the 60-iteration budget, with the log-likelihood monotonically increasing (as guaranteed by the EM algorithm). Lower K values typically converged faster (15–25 iterations), while higher K values required 25–40 iterations to reach the convergence tolerance of $|\Delta\mathcal{L}| < 10^{-4}$. The monotonic increase confirms that the quantile-based initialization provided a valid starting point from which the EM algorithm could improve without encountering degenerate solutions.

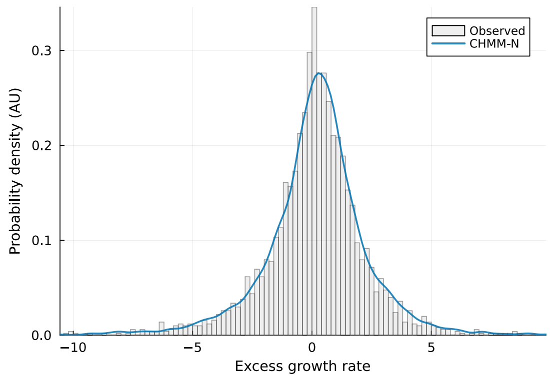


(a) $K = 3$ **(b)** $K = 12$ **(c)** $K = 18$ (selected)

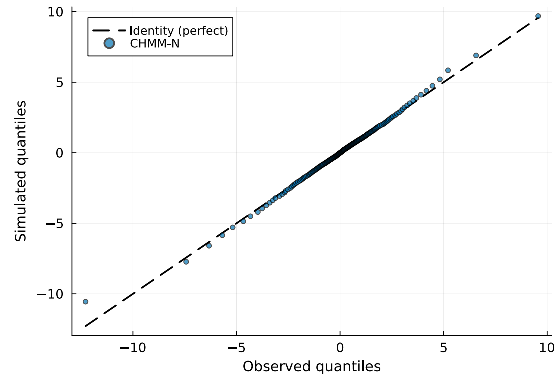
Figure 9. Baum-Welch convergence for the SPY CHMM-N fit (Pipeline A, IS) at selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus well before `max_iter` $= 60$.

A.6.4 IS Comparison at Non-Selected State Resolutions

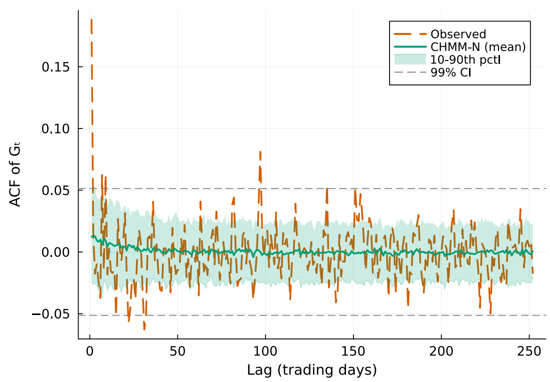
Figures 10–13 show the IS model comparison panels for $K \in \{3, 6, 12, 21\}$, complementing the $K = 18$ comparison in Figure 4 of the main text. Across all resolutions, the CHMM's simulated density closely overlays the observed density, and the ACF of absolute returns tracks the slow decay pattern. The visible improvement from $K = 3$ to $K = 18$ is concentrated in the tails of the Q-Q panel and the steady-state level of the ACF panel. $K = 9$ and $K = 15$ panels are omitted because they interpolate smoothly between the shown resolutions; their numerical metrics are in Table 8.



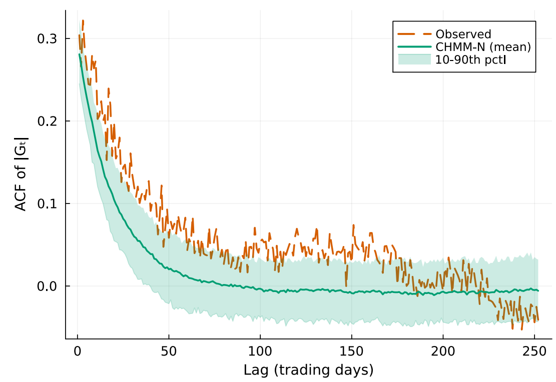
(a) Marginal density.



(b) Tail Q-Q plot.

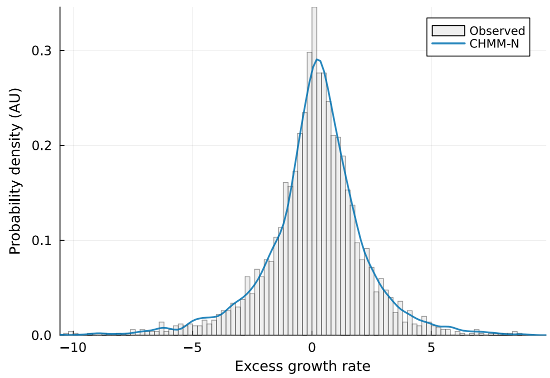


(c) ACF of raw G_t.

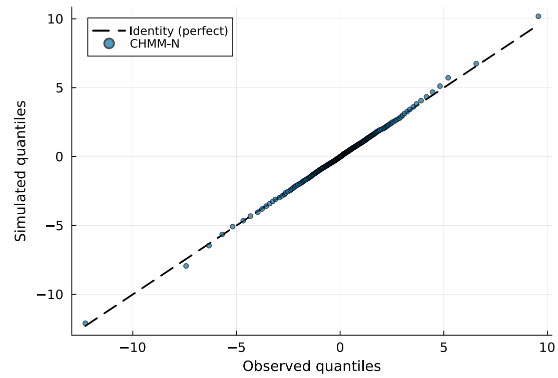


(d) ACF of $|G_t|$.

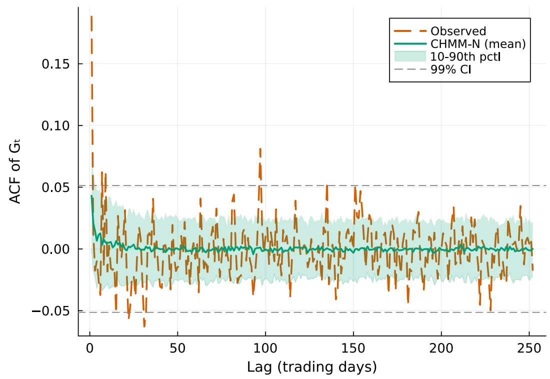
Figure 10. IS SPY CHMM-N comparison at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4). The $K = 18$ counterpart is Figure 4 of the main text.



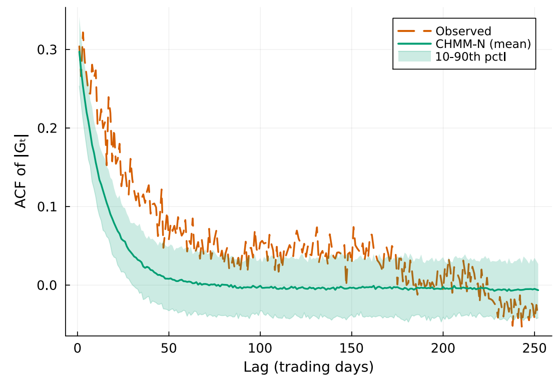
(a) Marginal density.



(b) Tail Q-Q plot.

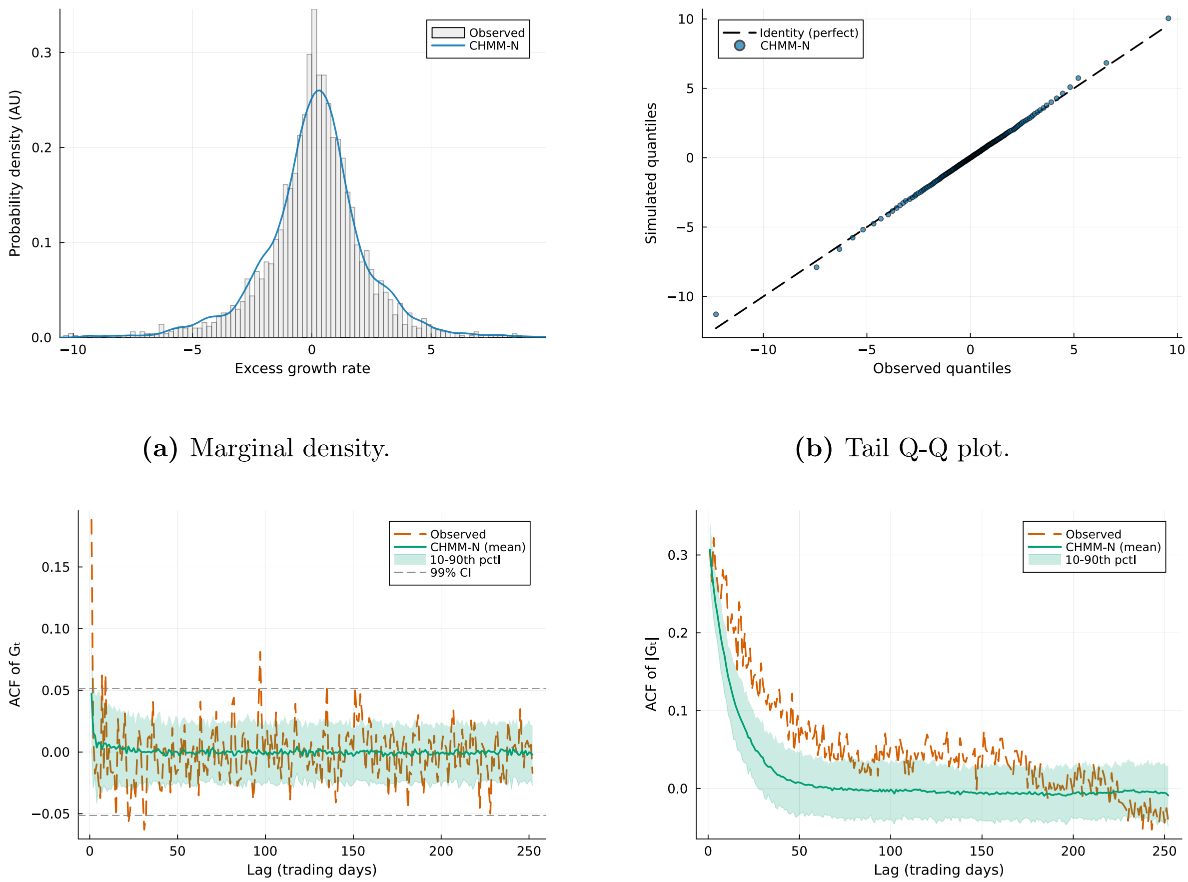


(c) ACF of raw G_t.



(d) ACF of $|G_t|$.

Figure 11. IS SPY CHMM-N comparison at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).



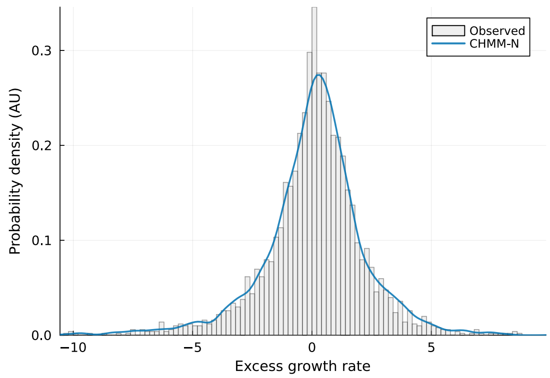
(a) Marginal density.

(b) Tail Q-Q plot.

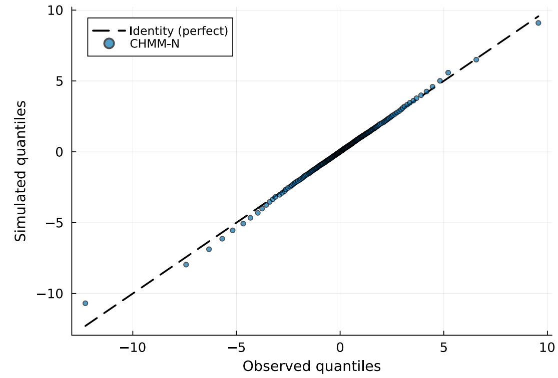
(c) ACF of raw G_t.

(d) ACF of $|G_t|$.

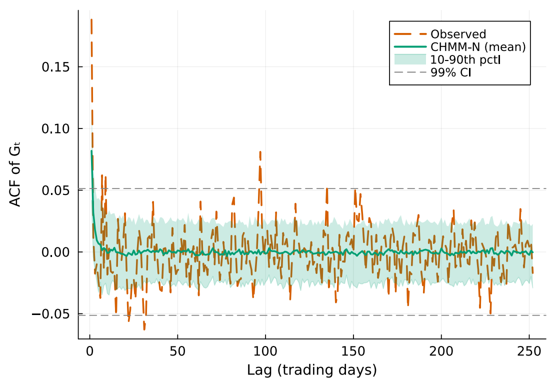
Figure 12. IS SPY CHMM-N comparison at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).



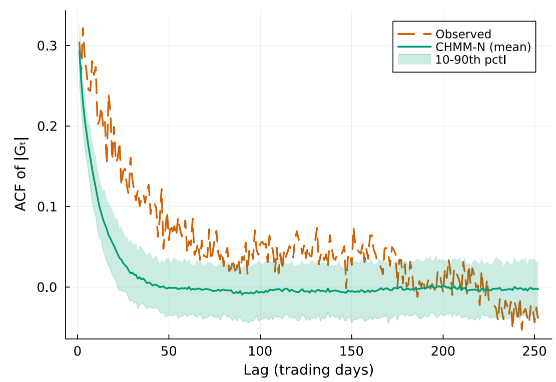
(a) Marginal density.



(b) Tail Q-Q plot.



(c) ACF of raw G_t.

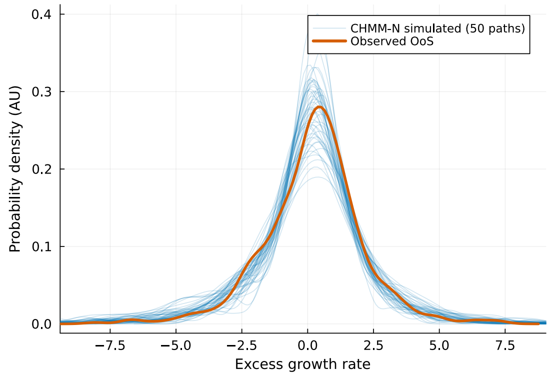


(d) ACF of $|G_t|$.

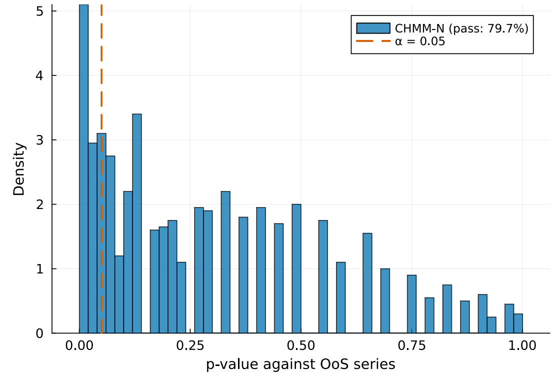
Figure 13. IS SPY CHMM-N comparison at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

A.6.5 OoS Validation at Non-Selected State Resolutions

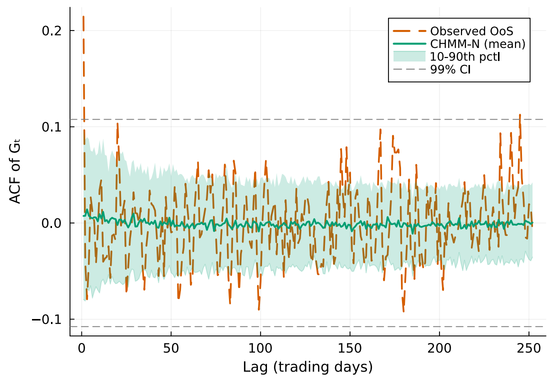
Figures 14–17 show the OoS validation panels for $K \in \{3, 6, 12, 21\}$. KS pass rates remain above 91% at all resolutions; the $K = 18$ panel is reproduced in Figure 5 of the main text. As with the IS panels, $K = 9$ and $K = 15$ are omitted; numerical OoS KS pass rates are in Table 8.



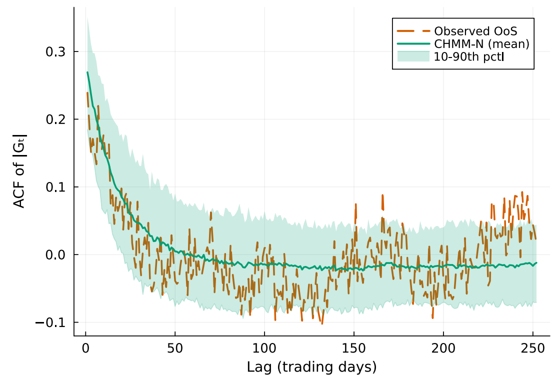
(a) OoS density fan.



(b) KS p-value histogram.

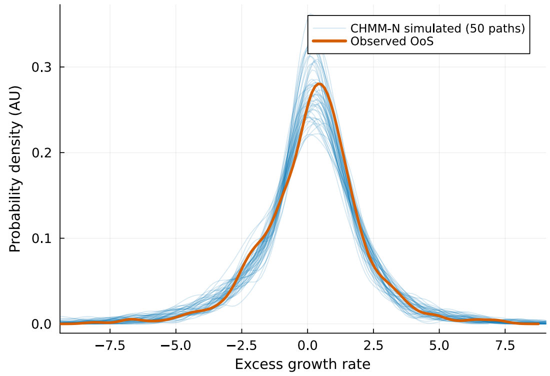


(c) OoS ACF of raw G_t.

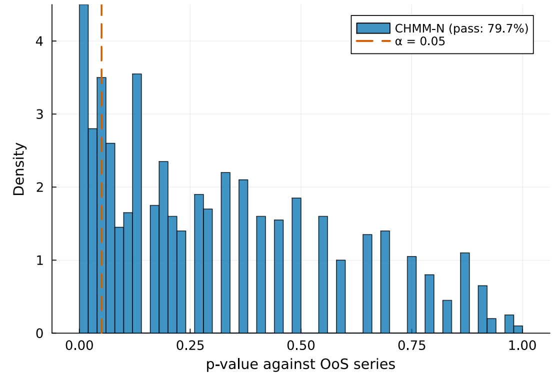


(d) OoS ACF of $|G_t|$.

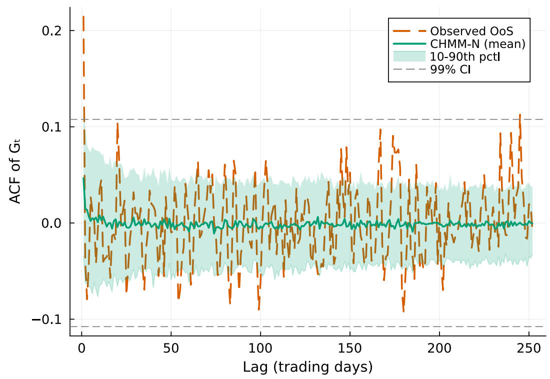
Figure 14. OoS SPY CHMM-N validation at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5). The $K = 18$ counterpart is Figure 5 of the main text.



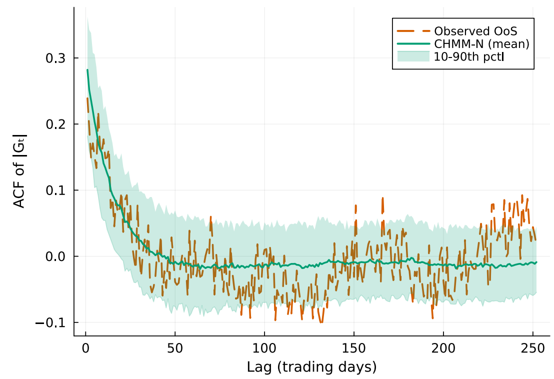
(a) OoS density fan.



(b) KS p-value histogram.

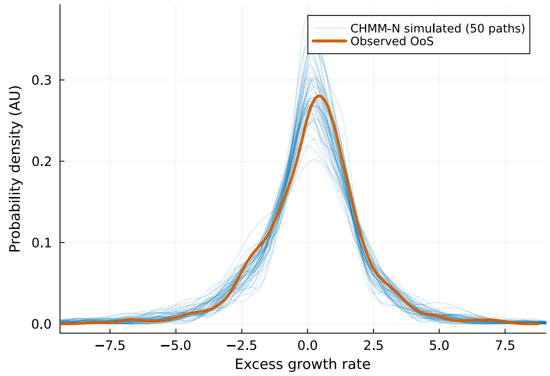


(c) OoS ACF of raw G_t.

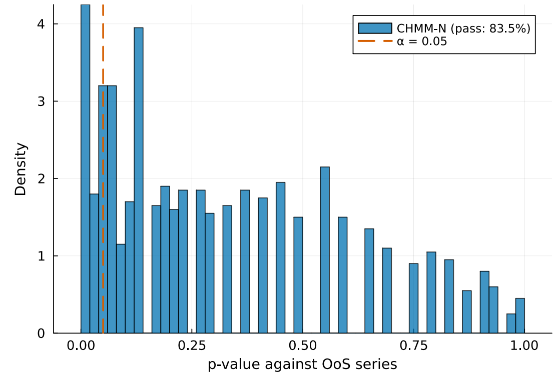


(d) OoS ACF of $|G_t|$.

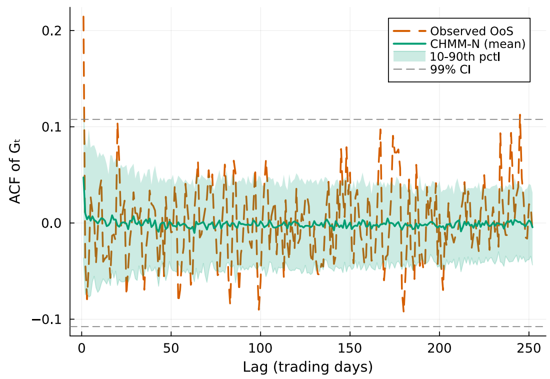
Figure 15. OoS SPY CHMM-N validation at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).



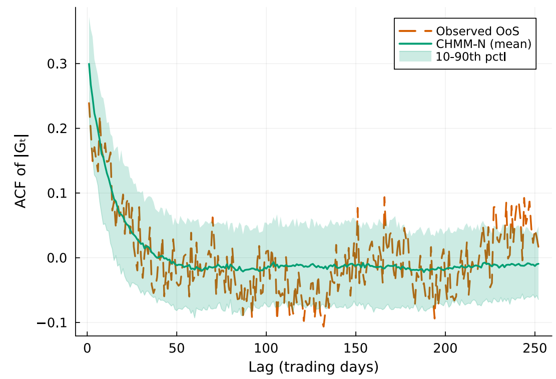
(a) OoS density fan.



(b) KS p-value histogram.

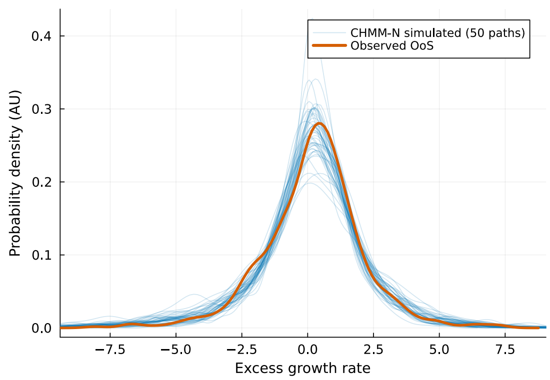


(c) OoS ACF of raw G_t.

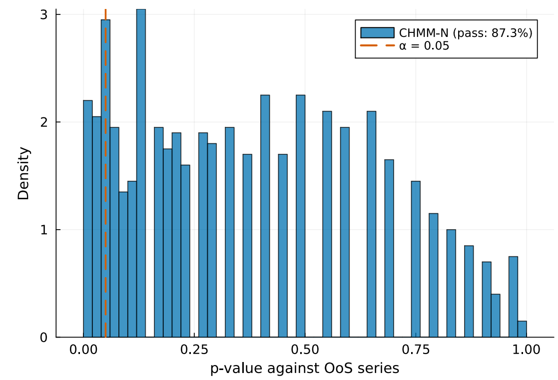


(d) OoS ACF of $|G_t|$.

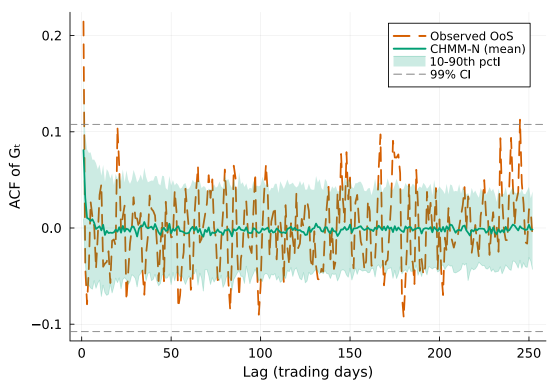
Figure 16. OoS SPY CHMM-N validation at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).



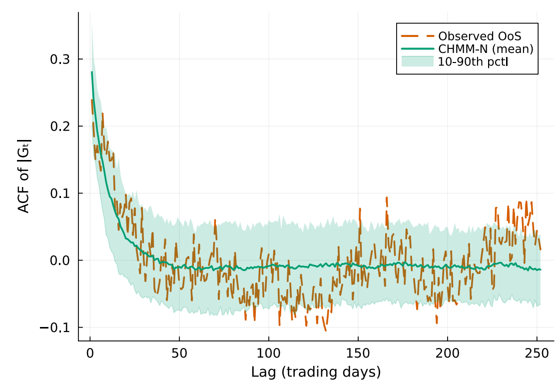
(a) OoS density fan.



(b) KS p-value histogram.



(c) OoS ACF of raw G_t.



(d) OoS ACF of $|G_t|$.

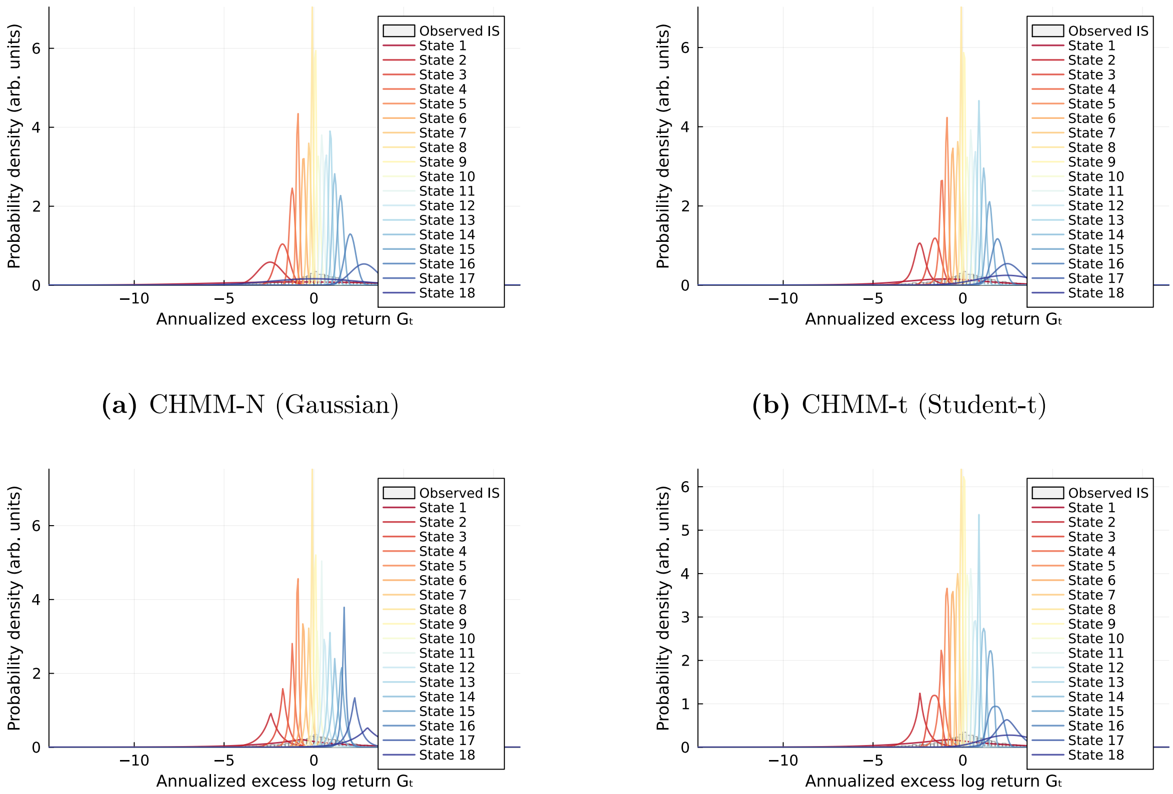
Figure 17. OoS SPY CHMM-N validation at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).

A.6.6 Multi-Emission Figures at $K = 18$

This appendix reproduces the standard diagnostic suite (emission PDFs, residence times, IS comparison, OoS validation, and convergence) for the two heavy-tailed emission families (CHMM-t and CHMM-L) at the selected state resolution $K = 18$; transition matrices and stationary distributions are discussed in prose only below, since they are visually indistinguishable across families at $K = 18$. The Gaussian counterparts are already shown as Figures 4 and 5 in the main text and as Figure 9 above in this appendix. The figures confirm that the shared forward–backward scaffold produces internally-similar regime structure across the four families: the near-uniform interior of the transition matrix, the K-component dense emission cover, the longer central-state residence times, and the monotone log-likelihood convergence are features of the CHMM framework and not of the Gaussian component specifically.

Emission PDFs. Figure 18 compares the learned $K = 18$ emissions for the four families. The CHMM-t panel's per-state ν_k allocation is visible in the narrower modes of certain tail states; the CHMM-L panel's Laplace components have sharper peaks and slower exponential tail decay than the Gaussians; the CHMM-GED panel mixes both, with bulk states near the Gaussian shape ($\hat{p}_k$

near the upper bracket) and a small set of tail states near the Laplace shape ($\hat{p}_k$ near 1).



(a) CHMM-N (Gaussian)

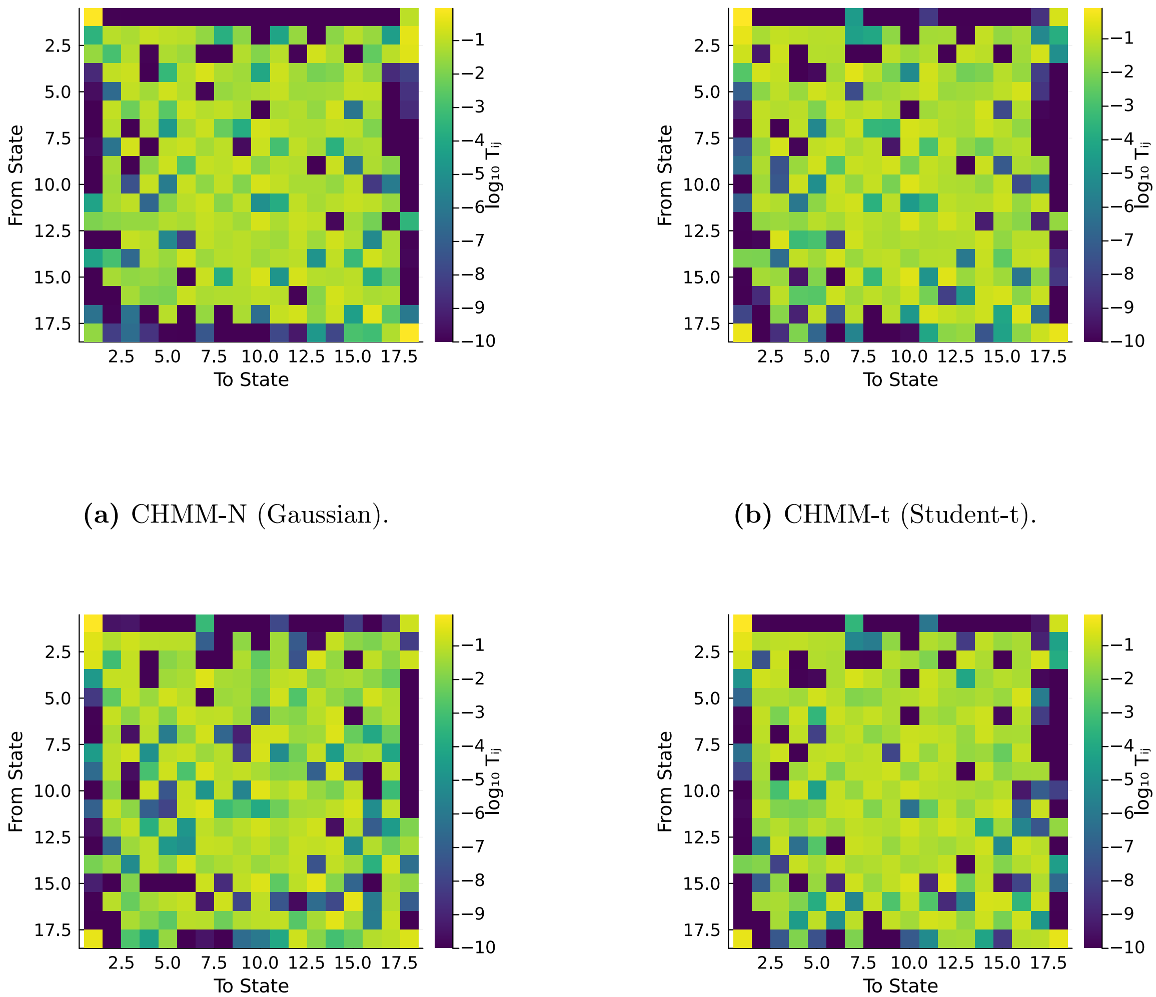
(b) CHMM-t (Student-t)

(c) CHMM-L (Laplace)

(d) CHMM-GED (per-state p_k)

Figure 18. Learned emission distributions at $K = 18$ across the four emission families (CHMM-N Gaussian, CHMM-t Student-t with per-state ν_k, CHMM-L Laplace, CHMM-GED with per-state p_k), overlaid on the observed SPY IS return histogram (Pipeline A). Heavier-tailed emission families (t, L) give individual components more tail mass, which is what closes the kurtosis gap of the Gaussian CHMM; CHMM-GED interpolates between Gaussian and Laplace per state.

Transition matrices and stationary distributions. Figure 19 shows the $\log_{10}$ transition-matrix heatmaps at $K = 18$ across the four emission families. The four panels are visually indistinguishable: the same near-uniform interior, the same characteristic boundary states (rows 1 and K) with a small handful of low-mass entries, and self-transition probabilities of comparable magnitude. The eigenvalue spectrum of $\mathbf{T}$ that drives the absolute-return ACF identity (8) is therefore set by the K-rank regime structure rather than by the emission family. This is consistent with the ACF-MAE being stable across the family axis (Table 9). The per-family residence times $\tau_k = 1/(1 - T_{kk})$, which magnify small differences in the self-transition probabilities each family recovers, are compared in Figure 20.



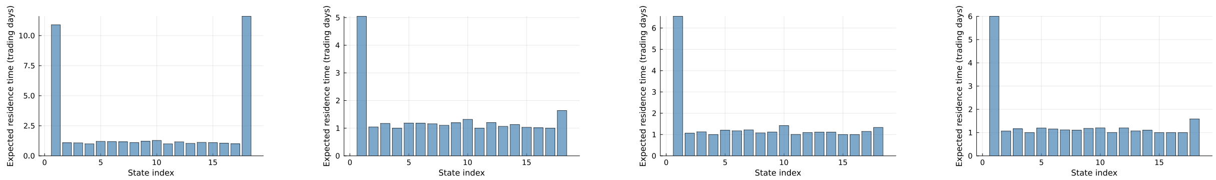
(a) CHMM-N (Gaussian).

(b) CHMM-t (Student-t).

(c) CHMM-L (Laplace).

(d) CHMM-GED (per-state p_k).

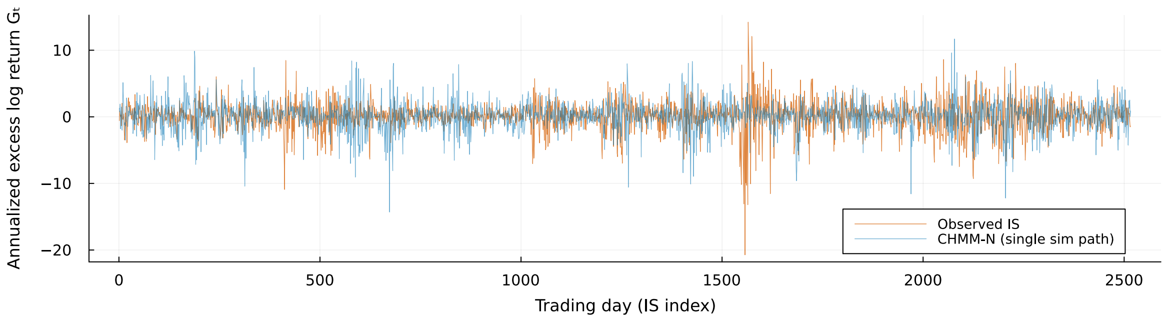
Figure 19. $\log_{10}$ transition-matrix heatmaps at $K = 18$ across the four emission families on SPY IS (Pipeline A). Color scale is $\log_{10} T_{ij}$ with a 10^{-10} floor for the near-zero entries. The cross-family similarity supports the rank/spectral interpretation of Section 3.8: the regime structure (and thus the eigenvalue spectrum that drives the absolute-return ACF) is largely a property of K and the data, not of the emission family.



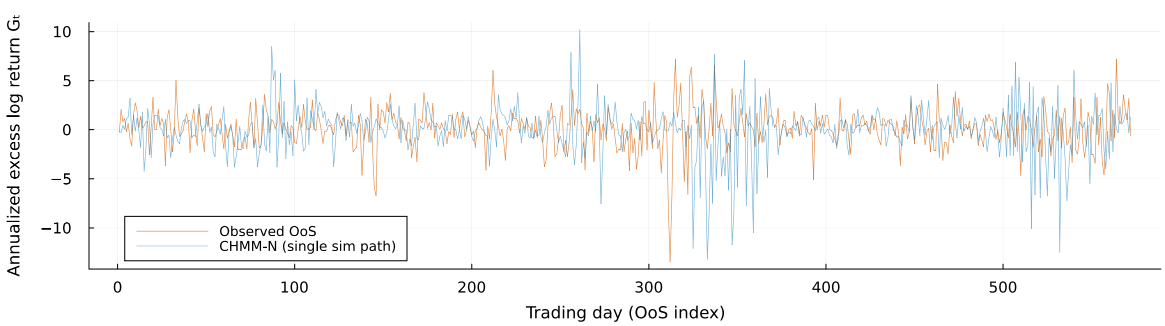
(a) CHMM-N **(b)** CHMM-t **(c)** CHMM-L **(d)** CHMM-GED

Figure 20. Natural residence times $\tau_k = 1/(1 - T_{kk})$ at $K = 18$ for the SPY IS fit (Pipeline A) across the four emission families (CHMM-N Gaussian, CHMM-t Student-t, CHMM-L Laplace, CHMM-GED).

Sample simulated trajectories. Figure 21 overlays one simulated SPY return path against the observed series under CHMM-N at $K = 18$, separately for the IS and OoS windows. Both panels show the simulated path tracking the volatility-clustering envelope of the observed series at the same overall amplitude. CHMM-N is shown as the headline reference; per-family marginal differences are governed by the emission family rather than by the trajectory representation, and are visualised more directly in Figure 18.



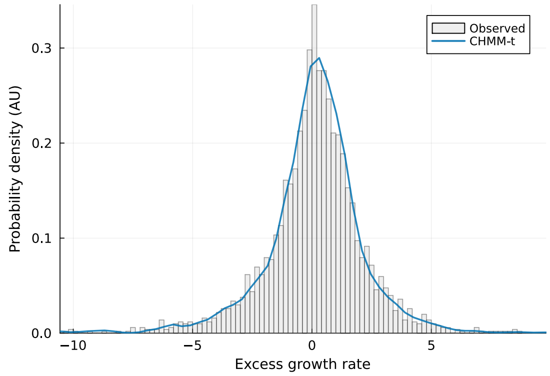
(a) In-sample window, full $T_{\text{IS}} = 2{,}516$ trading days.



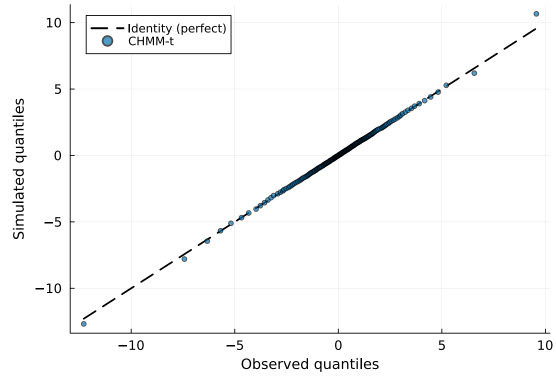
(b) Out-of-sample window, full $T_{\text{OoS}} = 572$ trading days.

Figure 21. Sample simulated return trajectory at $K = 18$ for CHMM-N on SPY (Pipeline A). Orange is the observed series; blue is one simulated path drawn from the fit. Panel (a): full in-sample window. Panel (b): full out-of-sample window. Both panels share the same y-axis convention (annualized excess log return G_t).

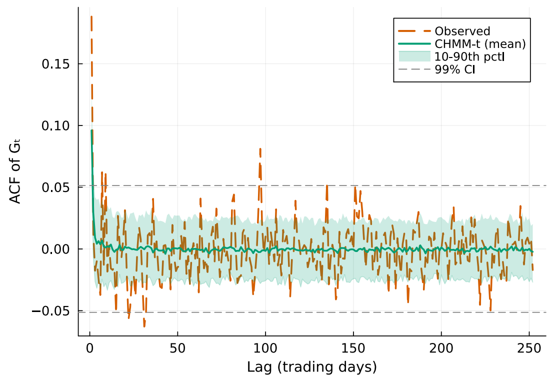
IS comparison. Figures 22–24 reproduce the four-panel IS comparison (marginal density, tail Q-Q, raw-return ACF, $|G_t|$ ACF) at $K = 18$ for the three non-Gaussian emission families; the CHMM-N panel is Figure 4 in the main text.



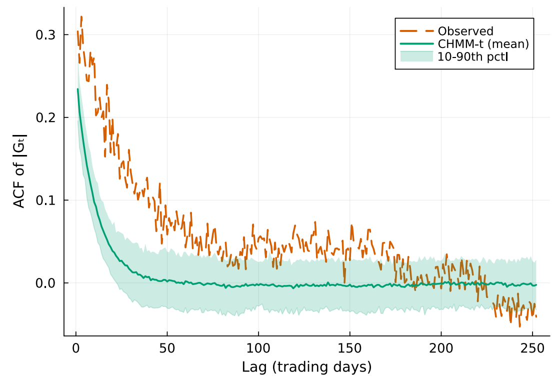
(a) Marginal density.



(b) Tail Q-Q plot.

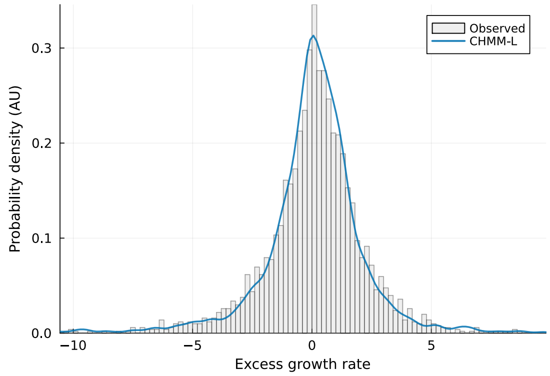


(c) ACF of raw G_t.

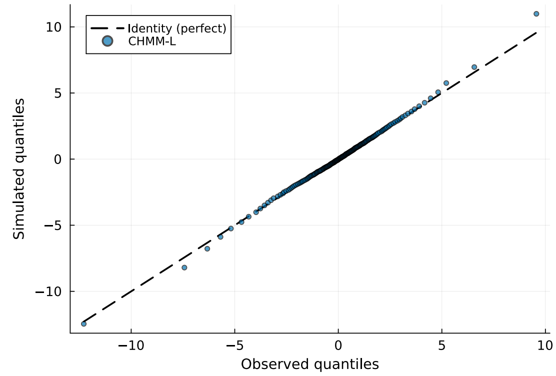


(d) ACF of $|G_t|$.

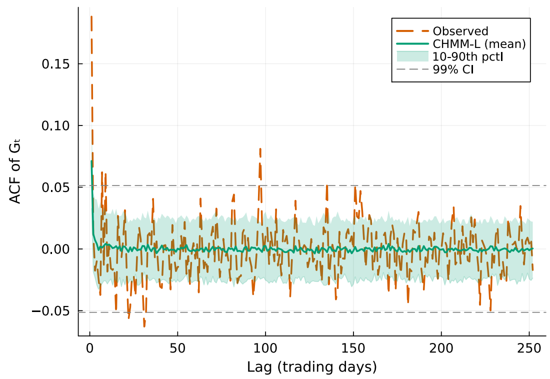
Figure 22. IS SPY model comparison at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) marginal density with KS pass rate, (b) tail Q-Q plot, (c) ACF of raw G_t (linear autocorrelation, with the 99% Bartlett band), and (d) ACF of $|G_t|$ (volatility clustering). The CHMM-N counterpart is Figure 4; the CHMM-L and CHMM-GED counterparts are Figures 23 and 24.



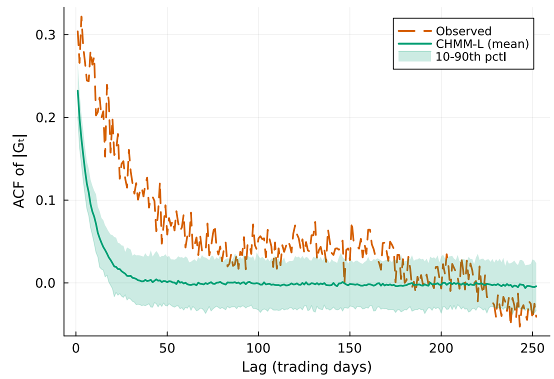
(a) Marginal density.



(b) Tail Q-Q plot.

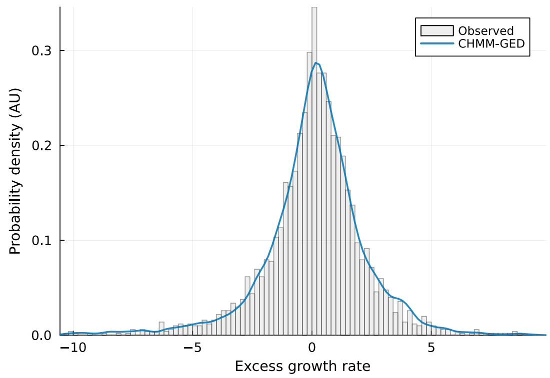


(c) ACF of raw G_t.

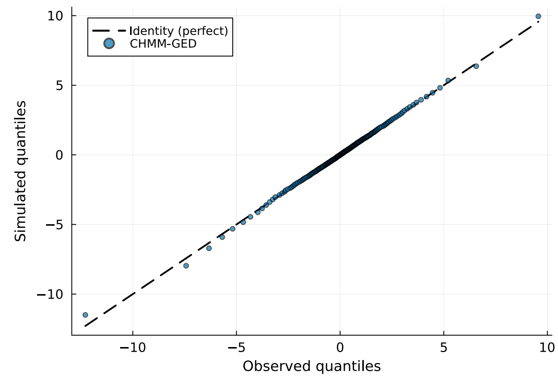


(d) ACF of $|G_t|$.

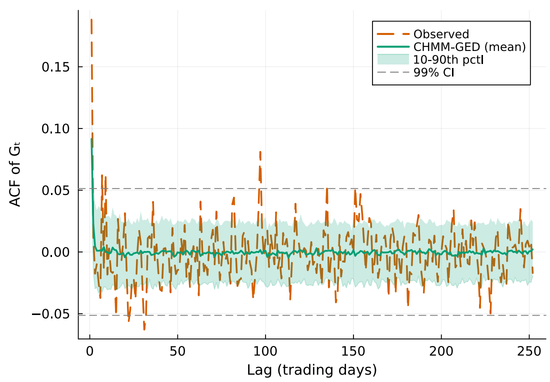
Figure 23. IS SPY model comparison at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 22.



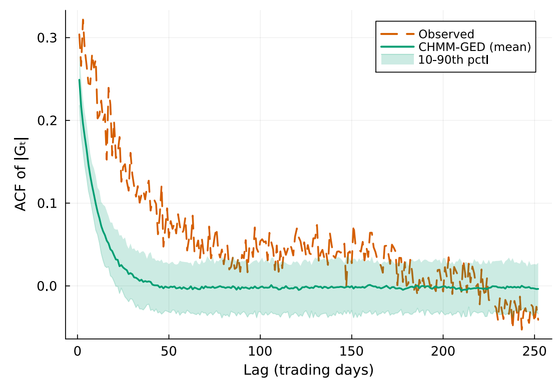
(a) Marginal density.



(b) Tail Q-Q plot.



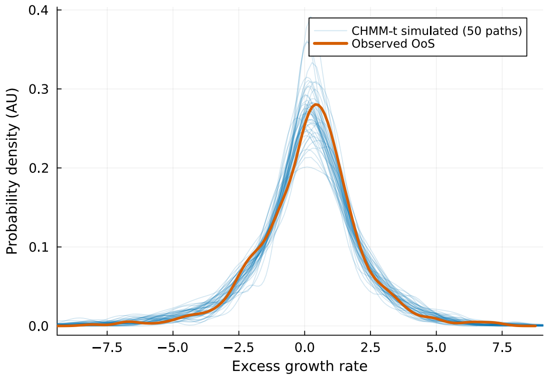
(c) ACF of raw G_t.



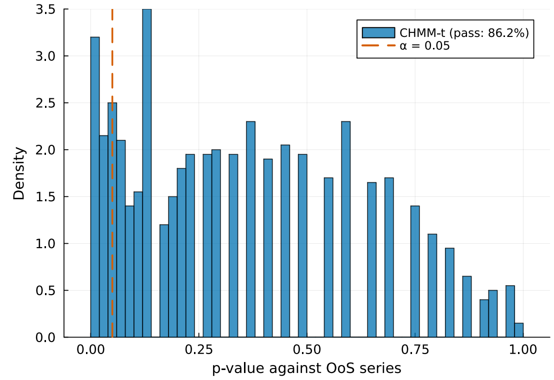
(d) ACF of $|G_t|$.

Figure 24. IS SPY model comparison at $K = 18$ for CHMM-GED emissions with per-state shape $p_k \in [0.5, 3.0]$ (Pipeline A, 1,000 simulated paths). Panels as in Figure 22. The fitted $\hat{p}_k$ partition is reported in Figure 7 and Appendix A.2.3.

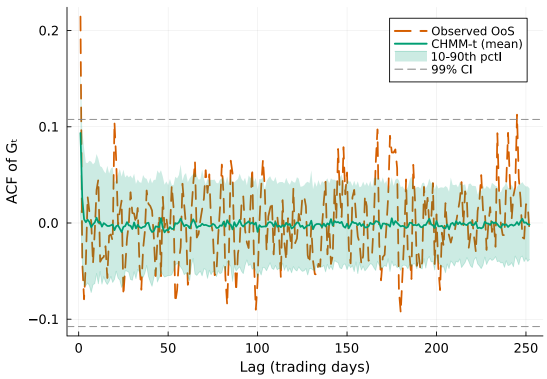
OoS validation. Figures 25–27 show the OoS validation panels for CHMM-t, CHMM-L, and CHMM-GED at $K = 18$. The CHMM-t panel demonstrates the OoS kurtosis alignment noted in the main text, the CHMM-L panel shows the tighter density fan produced by the heavier Laplace tails relative to the Gaussian, and the CHMM-GED panel shows the density fan tightening that follows from the data-chosen p_k partition.



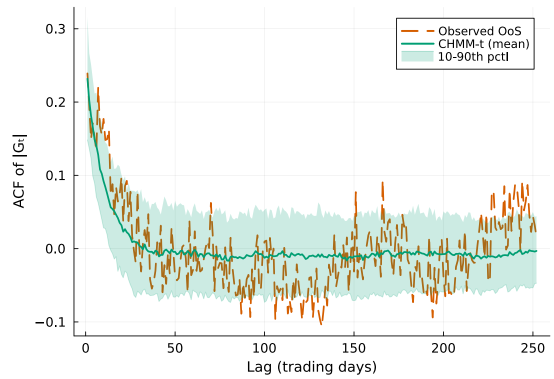
(a) OoS density fan.



(b) KS p-value histogram.

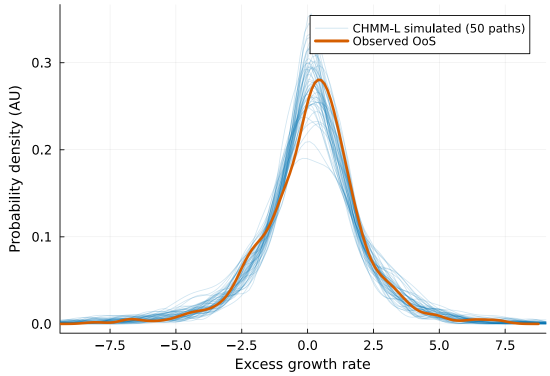


(c) OoS ACF of raw G_t.

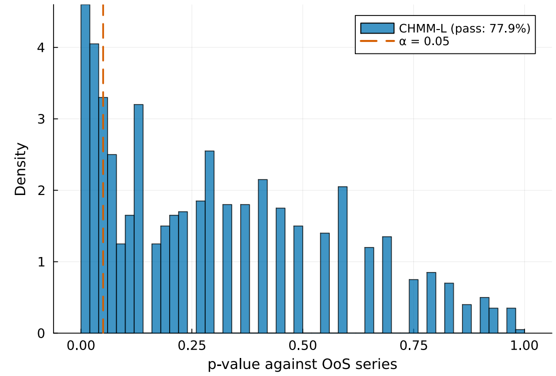


(d) OoS ACF of $|G_t|$.

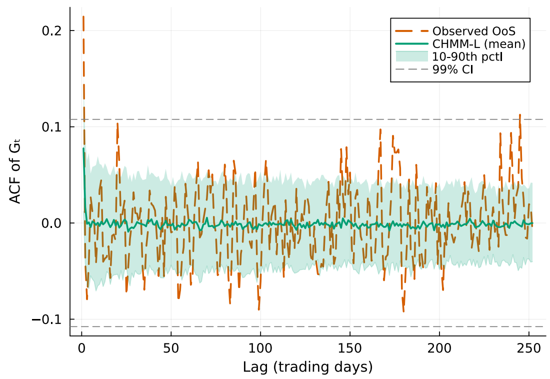
Figure 25. OoS SPY validation at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) OoS density fan, (b) KS p-value histogram with the $\alpha = 0.05$ threshold, (c) OoS ACF of raw G_t (linear autocorrelation, with the 99% Bartlett band), and (d) OoS ACF of $|G_t|$ (volatility clustering). The CHMM-N counterpart is Figure 5; the CHMM-L and CHMM-GED counterparts are Figures 26 and 27.



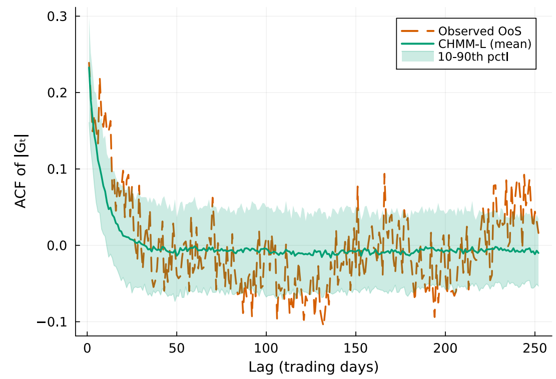
(a) OoS density fan.



(b) KS p-value histogram.

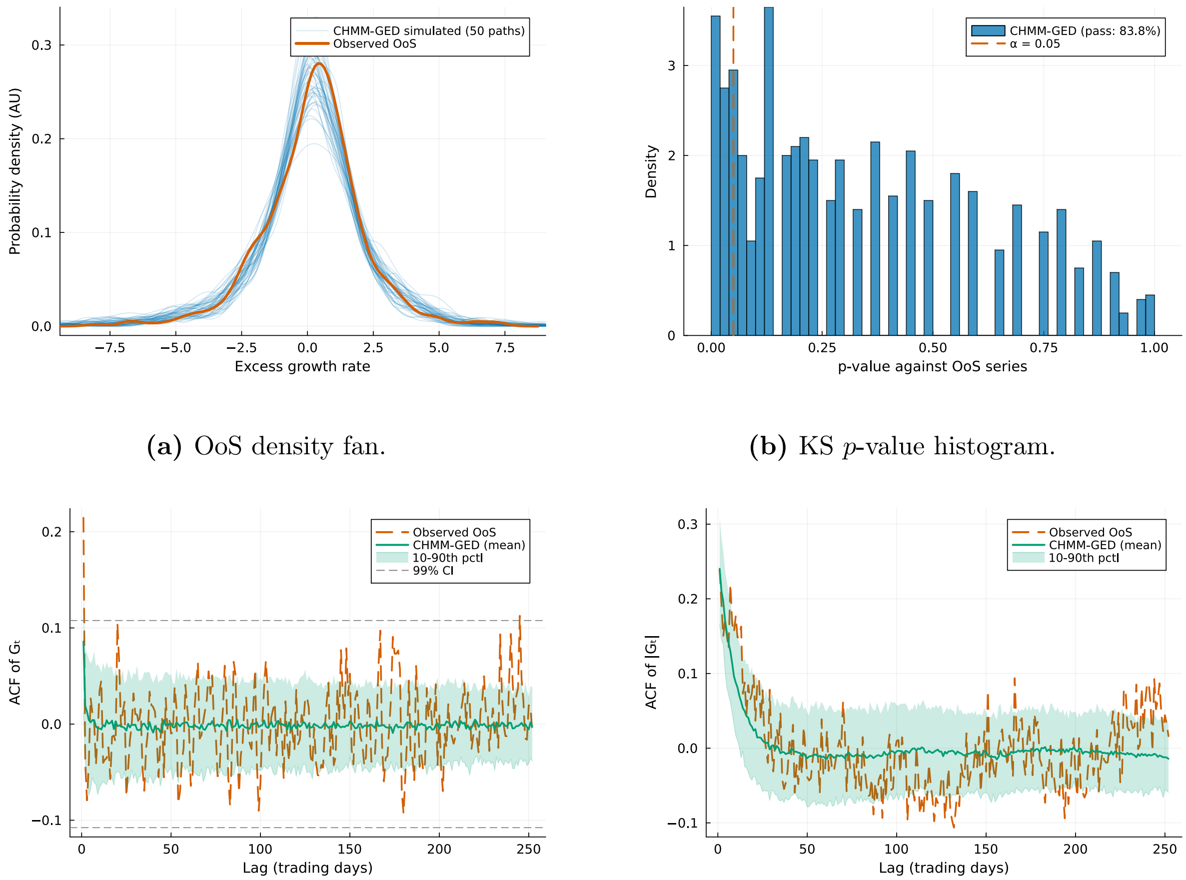


(c) OoS ACF of raw G_t.



(d) OoS ACF of $|G_t|$.

Figure 26. OoS SPY validation at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 25.



(a) OoS density fan.

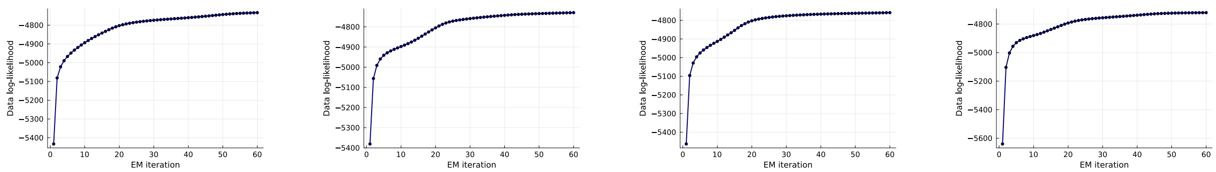
(b) KS p-value histogram.

(c) OoS ACF of raw G_t.

(d) OoS ACF of $|G_t|$.

Figure 27. OoS SPY validation at $K = 18$ for CHMM-GED emissions with per-state shape $p_k \in [0.5, 3.0]$ (Pipeline A, 1,000 simulated paths). Panels as in Figure 25.

Convergence. Figure 28 shows the Baum-Welch / ECM / weighted-EM log-likelihood histories at $K = 18$ for all four families. The CHMM-t, CHMM-L, and CHMM-GED curves inherit the monotone increase of the classical EM guarantee; the CHMM-t and CHMM-GED curves have slightly longer pre-plateau phases because each ECM sub-iteration refines a shape parameter (per-state ν_k for CHMM-t, per-state p_k for CHMM-GED) with a golden-section search.



(a) CHMM-N **(b)** CHMM-t **(c)** CHMM-L **(d)** CHMM-GED

Figure 28. Log-likelihood histories for the SPY IS fit (Pipeline A) at $K = 18$ across the four emission families (CHMM-N Gaussian classical EM; CHMM-t Student-t ECM with per-state ν_k golden-section sub-step; CHMM-L Laplace weighted-EM; CHMM-GED with per-state p_k three-stage ECM). All four curves are monotonically non-decreasing and plateau well before the `max_iter` $= 60$ budget.

A.6.7 Rydén K = 2 Replication

Table 10 reports the direct replication of Rydén et al.'s 1998 $K = 2$ setting on SPY IS under both initialization strategies, as discussed in Section 5. The quantile-based initialization used in the main body is row 1; five independent random-seed initializations are rows 2–6. Random init here draws per-state (μ_k, σ_k) from a moderate perturbation of the sample moments, which is representative of the random-init practice common in the 1990s HMM literature.

Table 10. Rydén replication: $K = 2$ CHMM-N on SPY IS under quantile vs random initialization (seed sub-index varies the random init).

Init	KS IS (%)	AD IS (%)	ACF-MAE	Kurt IS	KS OoS (%)
Quantile (main body)	79.0	71.5	0.0501	2.57	72.4
Random seed = 1	78.7	70.7	0.0502	2.57	73.2
Random seed = 7	75.3	69.6	0.0498	2.56	74.0
Random seed = 42	77.6	70.0	0.0498	2.59	71.8
Random seed = 100	76.2	69.4	0.0500	2.59	72.3
Random seed = 2024	76.4	69.8	0.0499	2.54	72.9

The key observation is that ACF-MAE is essentially constant at ≈ 0.050 across all six rows, matching the $K = 18$ CHMM-N of the main body. IS KS pass rate remains below the $K = 18$ operating point (75.3–79.0%) because two Gaussian components cannot tile the full multi-regime return distribution. The combination is consistent with the reading that the binding low-K constraint in the Rydén et al. [11] setup is distributional rather than temporal: the ACF is already well reproduced at $K = 2$, and it is distributional fidelity that drives the preference for $K = 18$.

A.7 Robustness, Test Power, Multi-Seed, and Auxiliary Baselines

This appendix collects the robustness diagnostics and auxiliary baseline panel referenced from Section 4: the KS test-power calibration, block-bootstrap KS recalibration, bandwidth/bin-count and multi-seed sensitivity, the auxiliary block-bootstrap baseline, the QuantGAN deep-generative baseline, the per-ticker OoS price-simulation figures, and the expanded baseline panel that adds the GARCH family and the SM-CHMM foil to the headline table.

A.7.1 OoS KS Test-Power Calibration

Table 11 reports the two-sample KS test-power calibration used to anchor the OoS KS pass rates of Section 4.3. For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\mathrm{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table 11. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\mathrm{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\mathrm{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\mathrm{IS}}, \hat{\sigma}_{\mathrm{IS}}$) at length T_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in Section 4.3 (80–84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

A.7.2 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the headline IS KS pass-rate metric: (i) the asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM and GARCH all carry volatility clustering, so the asymptotic null distribution may not apply; and (ii) the binary "pass / fail at $\alpha = 0.05$" reduces a continuous p-value to a one-bit summary. Table 12 reports two recalibrations addressing each concern.

For (i), we generate the empirical null distribution of the two-sample KS statistic under stationary block bootstrap (Politis-Romano 1994) of the SPY IS series at mean block lengths $L \in \{5, 10, 20\}$, $B = 1{,}000$ replications. The resulting 95% block-bootstrap critical values are 0.0306, 0.0338, and 0.0350 respectively, all *smaller* than the asymptotic critical value $1.36\sqrt{2/n_{\mathrm{IS}}} = 0.0383$; the block-bootstrap test is therefore stricter than the asymptotic one because it accounts for the within-window dependence in the IS series itself. For each generator we then compute the per-path KS statistic and report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

For (ii), we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator's paths are from the observed IS series.

Two patterns emerge that the asymptotic pass-rate metric of the main panel does not surface. First, the cross-generator ordering on the mean p-value scale (bootstrap 0.82, CHMM-t 0.57, CHMM-GED 0.55, CHMM-N 0.55, CHMM-L 0.50, GARCH 0.05) is much more compressed for CHMM-vs-bootstrap than the 94.8–99.6% asymp pass rates suggest: the CHMM family lands at mean p-value ~ 0.5, the natural value for paths that are statistically indistinguishable from the observed series. Second, GARCH(1,1)'s drop from 26.8% asymp $\rightarrow$ 6.8–17.4% block-aware confirms concern (i): the asymptotic test is mildly lenient on volatility-clustering generators with the wrong marginal, and the block-aware test catches GARCH's misspecification more sharply. The CHMM family, by contrast, sees only a small block-bootstrap penalty (95% $\rightarrow$ 88–93%); CHMM-GED is the strongest entry on the block-aware metric across all three block lengths (83.2%, 90.2%, 93.2% at $L = 5, 10, 20$).

Table 12. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed 20260422 with $N_{\text{paths}} = 500$. "asymp pass" is the original Section 4.3 metric (two-sample KS p-value ≥ 0.05). "mean pv" is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. "blkL%" is the fraction of paths whose KS statistic falls below the stationary-block-bootstrap 95% critical value at mean block length L. Block-bootstrap critical values: 0.0306 ($L = 5$), 0.0338 ($L = 10$), 0.0350 ($L = 20$); asymptotic critical value 0.0383.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑	blk20% ↑
i.i.d. bootstrap	99.6	0.820	0.352	0.999	97.6	99.4	99.6
GARCH(1,1)	26.8	0.048	0.000	0.227	**6.8**	**13.6**	**17.4**
CHMM-N	94.8	0.546	0.047	0.976	81.2	89.4	91.6
CHMM-t	**95.8**	**0.571**	0.060	0.987	82.4	89.2	91.6
CHMM-L	94.8	0.498	0.048	0.948	81.0	88.2	90.0
CHMM-GED	95.6	0.547	0.064	0.968	**83.2**	**90.2**	**93.2**

A.7.3 Bandwidth, Bin-Count, and Multi-Seed Robustness

Multi-seed Monte Carlo on headline metrics. The paper reports headline statistics under a single global seed. We re-run the CHMM-N / -t / -L / -GED pipeline at $K = 18$ across ten alternative global seeds (base 20260422, increment 100) with $N_{\text{paths}} = 250$ per seed and report mean $\pm$ std for the headline metrics in Table 13. Seed-to-seed variability is small: IS KS std is 1.2–1.8%; OoS KS std is 1.8–3.6%; simulated IS kurtosis std is 0.06 for CHMM-N, 0.05 for CHMM-L, 0.09 for CHMM-GED, and 1.95 for CHMM-t (reflecting the well-known instability of the per-state ν_k ECM at moderate K, already discussed in Section 5). Kupiec $\text{LR}_{\text{uc}}^{1\%}$ has std up to 1.25 for CHMM-N because the breach count moves on a coarse 0–10 grid at $T_{\text{OoS}} = 572$. The single-seed headline numbers in Table 1 sit comfortably inside the multi-seed bands. CHMM-GED has the highest mean IS KS pass rate among the four families (95.9%), consistent with its single-seed headline value.

Table 13. Multi-seed Monte Carlo on headline metrics. Ten global seeds; $N_{\text{paths}} = 250$ per seed (reduced from the headline panel's 1,000 for tractability across 40 fits, so the seed-to-seed std reported here is an upper bound on the std the headline panel would show at $N_{\text{paths}} = 1{,}000$). Both ACF-MAE columns parallel Table 1: $|G_t|$ for volatility clustering, G_t for linear autocorrelation.

Model	IS KS %	OoS KS %	sim Kurt	ACF-MAE $\lvert G_t \rvert$	ACF-MAE G_t	$\text{LR}_{\text{uc}}^{1\%}$	$\text{LR}_{\text{uc}}^{5\%}$
CHMM-N	94.7 ± 1.2	82.6 ± 1.8	5.03 ± 0.06	0.0467 ± 0.0006	0.0173 ± 0.0001	2.12 ± 1.25	3.83 ± 0.00
CHMM-t	94.6 ± 1.2	$\mathbf{85.6 \pm 1.9}$	13.95 ± 1.95	0.0518 ± 0.0003	0.0170 ± 0.0001	0.54 ± 0.15	3.19 ± 0.34
CHMM-L	94.4 ± 1.8	80.8 ± 3.6	6.71 ± 0.05	0.0537 ± 0.0003	0.0171 ± 0.0001	3.26 ± 0.00	2.71 ± 0.59
CHMM-GED	$\mathbf{95.9 \pm 1.2}$	83.2 ± 2.7	5.15 ± 0.09	0.0513 ± 0.0003	0.0171 ± 0.0001	2.93 ± 0.71	3.75 ± 0.25

A.7.4 Block-Bootstrap Baseline

Table 14 reports the full seven-metric panel for the stationary block bootstrap at block length $b = 10$ trading days, the temporally-aware non-parametric baseline added to Table 1.

Table 14. Stationary block bootstrap (block length $b = 10$ trading days) on SPY (1,000 paths, seed = 20260420).

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	99.2	96.4	7.07	0.0568	0.092	0.0534	100.0
OoS	87.5	86.4	5.34	0.0512	0.225	0.1570	86.9

A.7.5 QuantGAN Deep-Generative Baseline

Architecture and training details for the QuantGAN deep-generative baseline used in Section 4.3 are summarised here. The construction is a repo-native approximation to the convolutional WGAN of Wiese et al. [9]: a one-dimensional convolutional generator and critic, Wasserstein loss with critic weight clipping, trained on rolling windows of standardised SPY returns and synthesising long paths by stitching generated windows.

- **Generator.** Three 1D convolutional layers ($G_{\text{ch}} = 32$, kernel size 5, ReLU, same-padding), followed by a 1×1 projection to a single output channel. Input is i.i.d. Gaussian noise of latent dimension $L = 8$ at every position of the rolling window.
- **Critic.** Three 1D convolutional layers ($D_{\text{ch}} = 32$, kernel size 5, leaky-ReLU slope 0.2, same-padding), global mean pooling along the time axis, and a single linear scalar output.
- **Loss.** Wasserstein loss with critic weight clipping at ± 0.01 [65]. The critic minimises $\mathbb{E}[D(\tilde{G})] - \mathbb{E}[D(G)]$ on real / fake mini-batches; the generator maximises $\mathbb{E}[D(\tilde{G})]$.
- **Optimiser.** Adam, learning rate 1.0×10^{-4} for both networks, batch size 64.
- **Training schedule.** 15 epochs, 120 outer steps per epoch, 4 critic updates per generator update, training on rolling windows of length $W = 64$ standardised by the IS mean and standard deviation.
- **Synthesis.** Independent draws of length W are stitched end-to-end to produce paths of length T_{IS} or T_{OoS}; outputs are then re-scaled back to the natural return scale by the IS mean and standard deviation.
- **Implementation.** Julia with `Flux.jl` [66] and `Optimisers.Adam` state. Random seed for weight initialisation, training shuffling, and per-path sampling: 20260422, with deterministic per-path sub-seeds.

The trained model is callable through the same `simulate` interface as the CHMM, so the seven-metric panel evaluates QuantGAN on byte-identical windows to the other generators. The QuantGAN row is reported alongside the extended-GARCH and SM-CHMM rows in Table 16; the deep-generative row reaches 0% IS and OoS KS pass rates and undershoots IS kurtosis decisively (2.1 vs. observed 7.7), consistent with the documented difficulty of GAN-based generators on financial-time-series volatility-clustering benchmarks [34, 35].

A.7.6 Per-Ticker OoS Price-Simulation Figures and Full Path-Level Metrics (Unconditional CHMM)

This appendix collects (i) the full per-family path-level probabilistic-forecast metrics for an auxiliary OoS price-simulation diagnostic on the six SPY-member tickers and (ii) the per-ticker price-fan and terminal-distribution figures for NVDA, JNJ, JPM, AAPL, and QQQ. Plotting conventions: blue shaded bands are the 5–95% and 25–75% quantile envelopes of the 500 simulated paths, blue solid line is the simulated median, red solid line is the observed OoS price track. Terminal-histogram panels show the cross-path distribution of $\widetilde{P}_T$ at T_{OoS} in gray, with the observed terminal P_T^{obs} as a red vertical line and the simulated median as a blue dashed line. For brevity, the figures show CHMM-N only (the representative family; differences across the four emission families in the price-level metrics are small, see Table 15).

Table 15. Full path-level probabilistic-forecast metrics for OoS equity-price simulation, all (ticker, family) cells ($K = 18$, 500 paths, $T_{\mathrm{OoS}} = 572$; global seed policy in Section 3.5). $\overline{C}_{0.90}$: fraction of OoS steps at which the observed price falls inside the simulated 5–95% band (calibration); $\mathrm{MAPE}_{0.50}$: mean absolute percentage error between observed path and simulated median (sharpness); $\overline{\mathrm{CRPS}}$: horizon-average bias-corrected sample CRPS of Gneiting and Raftery [43]; $\overline{\mathrm{CRPS}}/S_0$: CRPS scaled by S_0 to remove price-level effects. Lower MAPE and CRPS, higher $\overline{C}_{0.90}$, are better. **Headline:** no CHMM emission family dominates on all four metrics across all six tickers; differences between CHMM-N, CHMM-t, CHMM-L, and CHMM-GED live mostly in the return tails and propagate only weakly into aggregated price-level scores. CHMM-GED is the strongest entry on JNJ (lowest CRPS, lowest MAPE) and effectively ties CHMM-t on QQQ; the other four tickers show no clear ordering.

Ticker	Family	$\overline{C}_{0.90}$	$\mathrm{MAPE}_{0.50}$	$\overline{\mathrm{CRPS}}$	$\overline{\mathrm{CRPS}}/S_0$
SPY	CHMM-N	1.000	0.100	39.25	0.0835
	CHMM-t	1.000	0.112	42.66	0.0908
	CHMM-L	1.000	0.102	38.68	0.0823
	CHMM-GED	1.000	0.116	44.54	0.0948
NVDA	CHMM-N	0.698	0.339	30.62	0.6424
	CHMM-t	0.636	0.383	34.77	0.7293
	CHMM-L	0.666	0.411	36.58	0.7674
	CHMM-GED	0.682	0.375	33.26	0.6977
JNJ	CHMM-N	1.000	0.101	12.55	0.0780
	CHMM-t	1.000	0.098	12.62	0.0784
	CHMM-L	1.000	0.101	12.65	0.0786
	CHMM-GED	1.000	0.095	12.21	0.0759
JPM	CHMM-N	1.000	0.195	32.23	0.1881
	CHMM-t	1.000	0.241	40.38	0.2357
	CHMM-L	1.000	0.211	35.25	0.2057
	CHMM-GED	0.993	0.253	43.73	0.2552
AAPL	CHMM-N	1.000	0.108	21.30	0.1156
	CHMM-t	1.000	0.075	18.90	0.1025
	CHMM-L	1.000	0.077	18.82	0.1021
	CHMM-GED	1.000	0.089	19.28	0.1046
QQQ	CHMM-N	0.998	0.073	30.48	0.0763
	CHMM-t	0.995	0.079	31.63	0.0792
	CHMM-L	1.000	0.118	41.12	0.1029
	CHMM-GED	0.997	0.080	31.76	0.0795

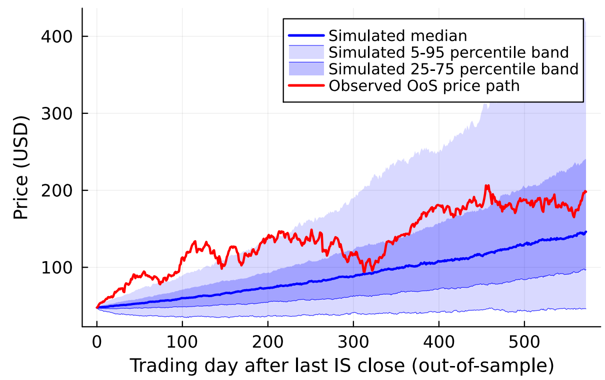
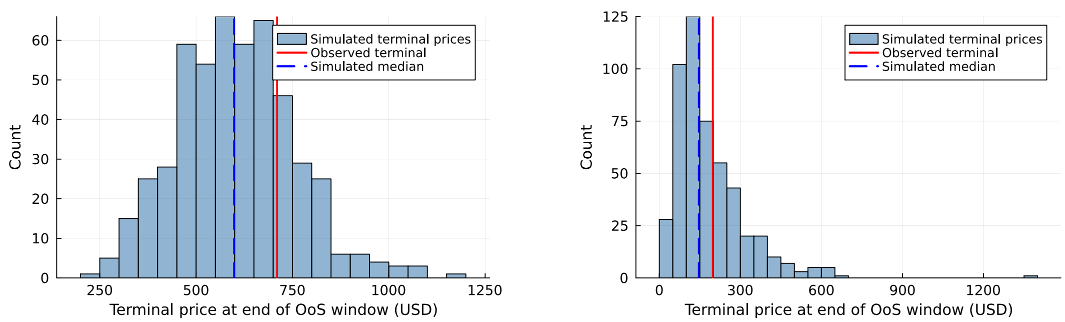


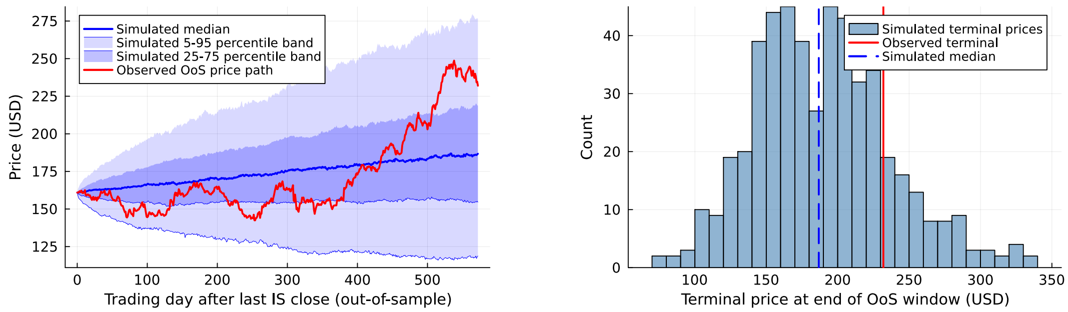
Figure 29. NVDA OoS price fan, CHMM-N ($S_0 = 47.67 \to P_T^{\mathrm{obs}} = 198.07$, a $4.15\times$ rally). The observed path exits the 25–75% interquartile band by mid-horizon but remains inside the 5–95% envelope; horizon-wide coverage under CHMM-N is 0.70 (CHMM-t: 0.64, CHMM-L: 0.67; Table 15), the lone calibration outlier in the panel.



(a) SPY terminal.

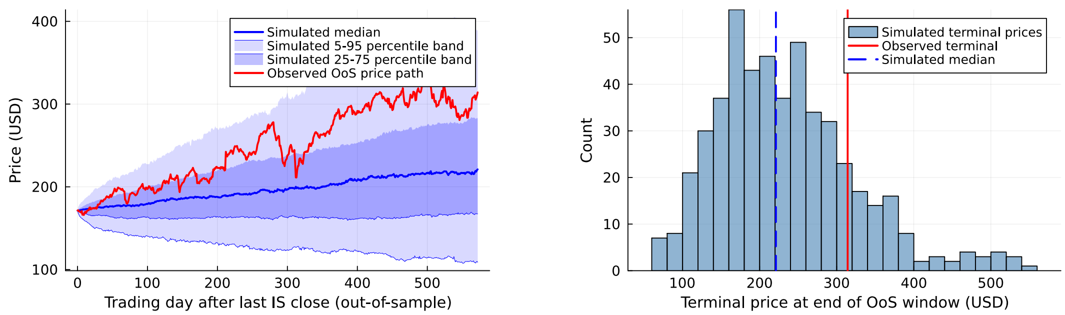
(b) NVDA terminal.

Figure 30. OoS terminal-price distributions under CHMM-N for SPY and NVDA. The right-skew is characteristic of log-returns aggregated to prices; the SPY terminal sits close to the simulated median, while the NVDA terminal sits in the upper-right of the simulated distribution (the realised rally is in the upper tail of the CHMM-N terminal density).



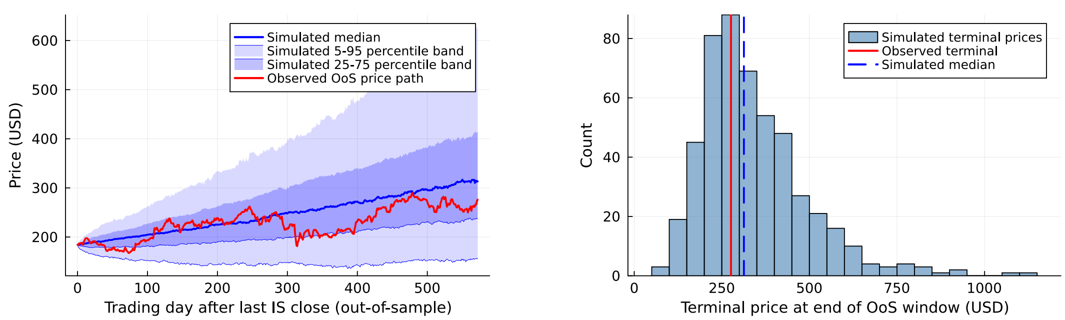
(a) Price fan. **(b)** Terminal distribution.

Figure 31. JNJ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as above).



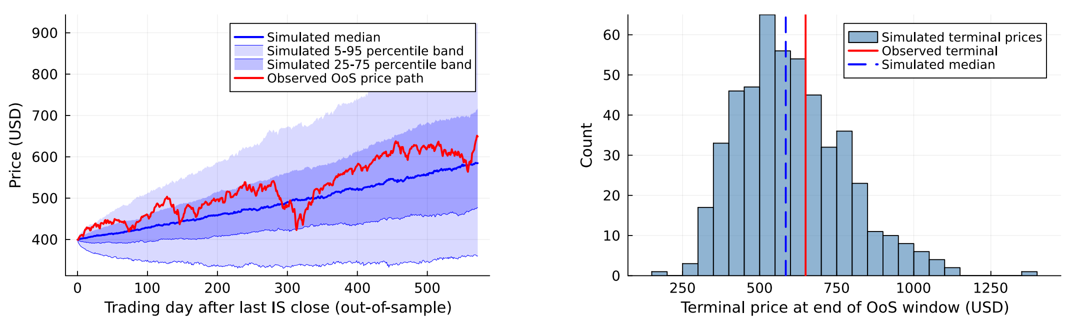
(a) Price fan. **(b)** Terminal distribution.

Figure 32. JPM OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as above).



(a) Price fan. **(b)** Terminal distribution.

Figure 33. AAPL OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as above).



(a) Price fan.

(b) Terminal distribution.

Figure 34. QQQ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as above).

A.7.7 Expanded Baseline Panel: GARCH Family, Semi-Markov CHMM, and QuantGAN

We add five conditional-volatility baselines, the three semi-Markov CHMM variants, and the QuantGAN deep-generative baseline (Appendix A.7.5) to the single GARCH(1,1) Gaussian row of Table 1. The conditional-volatility models are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window (T_{IS} = 2,516); simulation follows the same recursion used in estimation. Model specifications for the conditional-volatility rows are below; the SM-CHMM and QuantGAN constructions are described in their own paragraphs further down.

EGARCH(1,1) Gaussian. The exponential GARCH of Nelson [29] evolves $\log \sigma_t^2$ in levels:

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha \left(|z_{t-1}| - \sqrt{2/\pi} \right) + \gamma z_{t-1}, \qquad z_t = (r_t - \mu)/\sigma_t, \quad z_t \sim \mathcal{N}(0,1). \tag{28}$$

The log-variance formulation removes the GARCH positivity constraints, and the leverage term γz_{t-1} captures the asymmetric response to negative returns.

GJR-GARCH(1,1) Gaussian. The threshold GARCH of Glosten et al. [30] / Zakoïan [31] augments GARCH(1,1) with an indicator on negative innovations:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}\{\varepsilon_{t-1} < 0\} + \beta \sigma_{t-1}^2, \tag{29}$$

with $\varepsilon_t = r_t - \mu$. Stationarity requires $\alpha + \beta + \gamma/2 < 1$; positive γ indicates leverage.

GARCH(1,1) Student-t innovations. A direct tail-aware competitor to CHMM-t: the GARCH(1,1) variance recursion is unchanged, but innovations follow a standardised Student-t with $\nu > 2$ degrees of freedom rescaled to unit variance. The log-likelihood uses the Student-t density:

$$r_t - \mu = \sigma_t \cdot z_t, \qquad z_t = \eta_t \sqrt{(\nu - 2)/\nu}, \qquad \eta_t \sim t_\nu. \tag{30}$$

The fitted $\hat{\nu} = 6.89$ on SPY IS sits inside the moderate-tail regime the equity-risk literature typically reports.

HAR-RV on daily squared returns. The heterogeneous-autoregressive model of Corsi [33] models realised variance as a mixture of daily, weekly, and monthly components:

$$\log \mathrm{RV}_t = \beta_0 + \beta_d \log \mathrm{RV}_{t-1} + \beta_w \log \overline{\mathrm{RV}}_{t-5:t-1} + \beta_m \log \overline{\mathrm{RV}}_{t-22:t-1} + \eta_t. \tag{31}$$

In the absence of intraday bars we use daily squared demeaned returns as the RV proxy; simulation samples $\eta_t \sim \mathcal{N}(0, \hat{\sigma}_\eta^2)$ and rolls forward the recursion, then draws $r_t - \mu = \sqrt{\mathrm{RV}_t}\, z_t$ with $z_t \sim \mathcal{N}(0,1)$.

MS-GARCH $K = 3$. A three-regime extension of the Markov-switching GARCH(1,1) ($G_t = \mu + \sigma_t z_t$ with regime-specific recursion $\sigma_t^2(k) = \omega_k + \alpha_k (G_{t-1} - \mu)^2 + \beta_k \sigma_{t-1}^2(k)$). Parameter layout is three regime triples $(\omega_k, \alpha_k, \beta_k)$ plus six transition-matrix logits (softmax-normalised per row, enforcing a non-trivial diagonal); fit by grid-initialised Nelder-Mead. States are canonicalised by ascending unconditional variance, producing the calm / medium / stress decomposition $\sigma = [0.10, 1.77, 10.04]$ on SPY IS.

Table 16. Extended baseline panel. SPY IS / OoS, $T_{\text{IS}} = 2{,}516$, $T_{\text{OoS}} = 572$, $N_{\text{paths}} = 1{,}000$, $\alpha = 0.05$ for KS. Columns: IS / OoS KS pass rate (%); IS / OoS AD pass rate (%); simulated IS excess kurtosis; ACF-MAE on $|G_t|$ (volatility clustering) and on raw G_t (linear autocorrelation, parallel column); pooled-archive Kupiec breach rate (%) and unconditional likelihood-ratio statistic at the 1% and 5% tails. Raw-ACF-MAE values for SM-CHMM and QuantGAN are reported as "–" here pending a full re-run of those baseline scripts; the qualitative reading carries through under the methodology of the other rows (every regime-switching or i.i.d.-emission generator produces a near-zero raw ACF in expectation).

	KS (%) ↑		AD (%) ↑		Kurt	ACF-MAE		1% tail		5% tail	
Model	IS	OoS	IS	OoS	IS	$\lvert G_t \rvert$	G_t	br%	LR_{uc}	br%	LR_{uc}
GARCH(1,1) Gaussian	28.0	59.4	12.5	59.5	7.0	0.0309	0.0173	0.9	0.10	4.0	1.23
EGARCH(1,1)	6.3	34.9	4.1	41.7	2.8	0.0430	0.0173	1.0	0.01	3.5	3.03
GJR-GARCH(1,1)	40.1	57.7	28.7	61.4	7.6	0.0330	0.0173	1.2	0.27	4.7	0.10
GARCH(1,1)-t	**57.3**	**80.8**	**36.6**	65.3	15.1	0.0316	0.0173	1.0	0.01	4.9	0.01
HAR-RV	0.0	0.0	0.0	0.0	189	0.0566	0.0173	0.2	5.99	3.5	3.03
MS-GARCH $K = 2$	27.7	38.7	24.0	44.4	4.7	0.0367	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 3$	36.1	33.1	28.8	38.0	4.1	**0.0284**	0.0173	1.0	0.01	4.2	0.82
CHMM-N ($K = 18$)	94.1	81.8	**36.6**	**73.0**	5.0	0.0509	0.0237	0.7	1.58	4.0	3.83
CHMM-t ($K = 18$)	95.6	**85.7**	33.4	69.8	14.4	0.0549	0.0234	1.2	0.58	5.0	3.83
CHMM-L ($K = 18$)	93.6	80.8	30.1	65.4	6.6	0.0567	0.0234	1.4	3.26	5.6	3.03
CHMM-GED ($K = 18$)	**95.2**	84.3	34.1	71.0	5.2	0.0548	0.0234	0.9	0.58	4.4	3.83
SM-CHMM-N ($K = 18$)	79.2	66.7	19.5	57.1	4.7	0.0598	–	1.1	0.01	5.4	0.82
SM-CHMM-t ($K = 18$)	77.1	63.4	17.2	52.3	13.9	0.0625	–	1.4	0.10	5.7	1.74
SM-CHMM-L ($K = 18$)	80.4	69.1	21.0	58.9	6.4	0.0581	–	1.4	3.26	5.5	3.03
QuantGAN	0.0	0.0	0.0	0.0	2.1	0.0591	–	1.0	0.01	3.3	3.83

MS-GARCH $K = 2$ (Haas-Mittnik-Paolella 2004). The path-independent two-regime variant of Haas et al. [19]. Regime stickiness is high: $p_{11} = 0.915$ (calm) and $p_{22} = 0.547$ (stress), with unconditional standard deviations 0.97 (calm) and 12.67 (stress); the calm regime is near-GARCH behaviour and the stress regime is a rare, transient high-variance mode. Common $\mu = 0.094$, log-likelihood $-4{,}965$.

Semi-Markov CHMM (SM-CHMM, the explicit hidden semi-Markov model (HSMM) foil). The conceptual foil of the present paper, drawing on the explicit-duration HSMM construction of the companion volatility paper [67]: a hidden semi-Markov extension of the CHMM in which the geometric sojourn distribution of the Markov chain is replaced by a per-state explicit-duration family chosen by AIC from {Pareto, negative binomial, geometric}. Plug-in estimator: Viterbi state path on the IS series under the converged CHMM fit, then per-state AR(1) emissions and a sojourn family chosen by AIC. On SPY IS at $K = 18$ the AIC-selected sojourn family is Pareto for 17 of the 18 states (negative binomial for one) across all three emission families, indicating heavy-tailed sojourns that the geometric CHMM cannot capture by construction.

Table 16 reports the IS / OoS distributional fidelity, kurtosis, ACF-MAE, and Kupiec breach diagnostics for the extended GARCH family, the SM-CHMM rows, and the QuantGAN deep-generative row, alongside the headline CHMM panel. Three structural observations.

First, none of the extended GARCH variants challenges the CHMM family on the joint (KS, ACF) corner. GARCH(1,1)-t is the strongest extended row on KS (57.3% IS, 80.8% OoS) but its IS excess kurtosis of 15.1 overshoots the observed 7.7 by a factor of two and its ACF-MAE of 0.0316 is below the CHMM's only because the GARCH variance recursion is hard-wired to track conditional volatility; CHMM at $K = 18$ matches GARCH on ACF-MAE within 0.02 while gaining 36 to 36.5 percentage points on IS KS. EGARCH undershoots kurtosis decisively, GJR-GARCH improves on threshold asymmetry but not on the headline gap, and HAR-RV is degenerate on the daily-RV proxy used here (see methods note above).

Second, the SM-CHMM rows are weaker than the headline CHMM rows on every metric in the panel: IS KS 77–80% vs CHMM 93–94%, OoS KS 63–69% vs CHMM 79–86%. The geometric-sojourn restriction of the standard CHMM is therefore not the bottleneck for the headline KS / ACF / kurtosis story. The Viterbi-AR(1) plug-in estimator used here is one specific HSMM construction; a maximum-likelihood explicit-duration HSMM under a Pareto sojourn family may close part of the gap, but the present empirical evidence is that the spectral mechanism of Section 3.8 dominates the sojourn-family extension on this universe and these horizons.

Third, the QuantGAN deep-generative row (Appendix A.7.5) is the weakest entry in the panel on every distributional metric. IS and OoS KS pass rates collapse to 0% at $\alpha = 0.05$ across all 1,000 simulated paths, IS and OoS AD pass rates are also 0%, and simulated IS excess kurtosis of 2.1 undershoots the observed 7.7 by more than a factor of three. ACF-MAE of 0.0591 is comparable to the i.i.d. baselines of Table 1 (0.0628), so the GAN does not recover absolute-return autocorrelation either. The convolutional WGAN reproduces some surface-level visual properties of the return series but the stitched-window simulation scheme has no mechanism for modelling either the heavy-tailed marginal or the persistent absolute-return autocorrelation; this matches the documented failure mode of GAN-based generators on financial-time-series volatility-clustering benchmarks [34, 35]. The runner is a repo-native convolutional WGAN with critic weight clipping (Appendix A.7.5), materially smaller than the original temporal convolutional network of Wiese et al. [9] (a seven-block TCN with 127-day receptive field, hidden width 80, plus Lambert-W input pre-processing that the original authors credit with most of the tail-fidelity); the row therefore reads as the panel's deep-generative negative control rather than a verdict on the WGAN literature as a whole. The QuantGAN row complements the i.i.d. Gaussian negative control of the headline Table 1.

A.8 Cross-Asset Dependence: Full Pipeline B Panel

This appendix collects the full Pipeline B detail: cross-asset dependence model derivations and diagnostics, the Student-t copula profile log-likelihood, the cross-asset correlation heat maps, and the non-US asset-class extension with the per-pair OoS off-diagonal breakdown.

A.8.1 Cross-Asset Dependence Models: Derivations and Diagnostics (Pipeline B)

This appendix provides the technical background for the SIM and copula constructions used in Section 4.5 of the main text. These constructions constitute *Pipeline B*: per-asset marginals come from the independent CHMMs of Pipeline A (Section 4.4) and joint dependence is overlaid on top. Pipeline A stays purely univariate; Pipeline B is the sole multi-asset pipeline in the paper.

Sklar's theorem and the rank-reordering simulator. Sklar's theorem [37, 38] states that any d-dimensional joint CDF $H(g_1, \ldots, g_d)$ with continuous marginals $F_1, \ldots, F_d$ can be written as $H = C(F_1, \ldots, F_d)$ for a unique copula C. The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j-th column is an i.i.d. sample from F_j, and $\mathbf{U}$ is an independent sample from C, then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$'s produces a sample from H, up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [41].

Kendall's τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T\times d}$, we estimate the pairwise Kendall's τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [40, 49]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν. The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0,1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \dots, x_d)^\top$, is

$$\begin{aligned}\log c_t(\mathbf{u}; \Sigma, \nu) \;=\; & \log\Gamma\left(\tfrac{\nu+d}{2}\right) + (d-1)\log\Gamma\left(\tfrac{\nu}{2}\right) - d\log\Gamma\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2}\log|\Sigma| \\ & - \tfrac{\nu+d}{2}\log\left(1 + \tfrac{\mathbf{x}^\top\Sigma^{-1}\mathbf{x}}{\nu}\right) + \sum_{j=1}^{d} \tfrac{\nu+1}{2}\log\left(1 + \tfrac{x_j^2}{\nu}\right). \end{aligned} \tag{32}$$

Summing over $t = 1, \dots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's-τ estimate. We evaluate equation (32) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 4.5 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d-dimensional Gaussian copula with correlation Σ, we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν, we draw $W \sim \chi^2_\nu/\nu$ independent of Y and form the multivariate-t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 17 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 4.5. QQQ's high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ's lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 20.

Table 17. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\mathrm{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2{,}516$).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	−0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

A.8.2 Student-t Copula Profile Log-Likelihood (Pipeline B)

Figure 35 plots the profile log-likelihood of the Student-t copula used inside Pipeline B, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in Section 4.5. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood

(6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature.

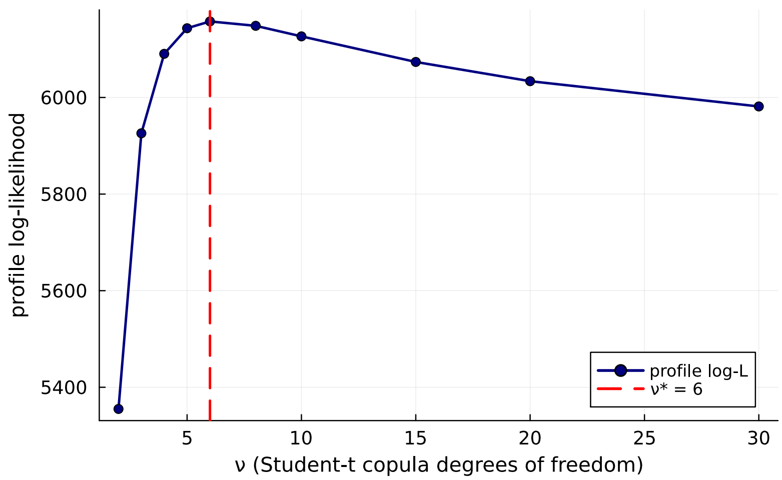
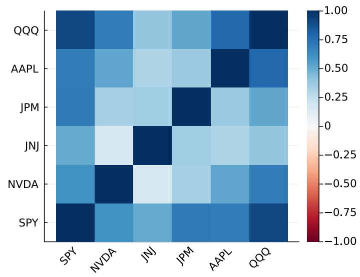


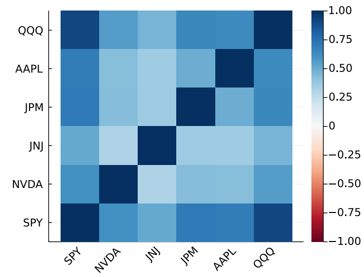
Figure 35. Profile log-likelihood of the Student-t copula used inside Pipeline B on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the Pipeline A per-asset fits while ν is varied.

A.8.3 Cross-Asset Correlation Heat Maps (Pipeline B, Companion to Table 20)

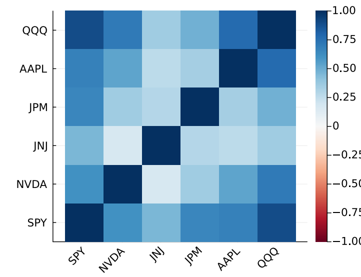
Figure 36 visualises the observed and path-averaged simulated correlation matrices for the dependence models of Section 4.5. The panel ordering is SIM, Gaussian copula, Student-t copula, and a truncated C-vine variant with SPY as root, each reported quantitatively in Table 20 and (for the C-vine) Section A.8.4. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.



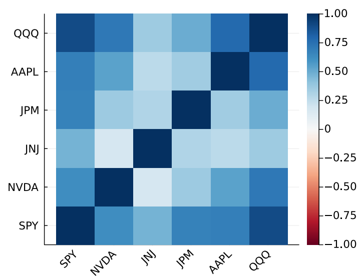
(a) Observed.



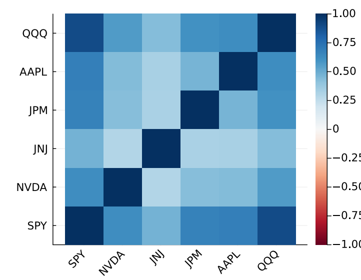
(b) Single Index Model.



(c) Gaussian copula.



(d) Student-t copula, $\nu^* = 6$.



(e) Truncated C-vine.

Figure 36. Cross-asset IS correlation reproduction across dependence models (Pipeline B, companion to Table 20). Panel (a): *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{\text{IS}} = 2{,}516$). Panel (b): path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d): path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e): path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS}. All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 3; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

A.8.4 Truncated C-vine Copula on CHMM Marginals (Pipeline B)

The truncated C-vine variant in panel (e) of Figure 36 uses SPY as the root and selects each pair-copula family edge-wise from {Gaussian, Student-t, Clayton, Gumbel} by Akaike Information Criterion (AIC); on the six-asset universe every selected pair is Student-t, with $\hat{\rho}$ and $\hat{\nu}$ as in Table 18. On the same CHMM-N marginals at $K = 18$ and 200 paths under seed `20260422`, the truncated C-vine attains an off-diagonal MAE of 0.068 on IS and 0.235 on OoS (Table 20), strictly above both elliptical copulas on the IS column and on OoS off-diag MAE, and only modestly below the SIM baseline. The reason is the level-1 truncation: a root-centred C-vine models only the SPY-vs-asset bivariate copulas and treats the remaining pairs as conditionally independent given SPY, which is a stronger restriction than either elliptical copula and only a small relaxation of the SIM factor structure. Untruncated regular vines or factor copulas are the natural way to recover non-root cross-pair structure at larger d, but at $d = 6$ with $T_{\text{IS}} = 2{,}516$ the additional flexibility does not justify the extra estimation cost.

Table 18. Truncated C-vine edge summary, Pipeline B with SPY as root on the six-asset universe. Pair family selected edge-wise by AIC over {Gaussian, Student-t, Clayton, Gumbel}; on this universe every selected pair is Student-t. $\hat{\rho}$ is the Kendall's-τ-inverted correlation parameter and $\hat{\nu}$ is the per-edge degrees-of-freedom estimate.

Child	Family	$\hat{\rho}$	$\hat{\nu}$
NVDA	Student-t	0.637	5
JNJ	Student-t	0.483	4
JPM	Student-t	0.686	4
AAPL	Student-t	0.698	4
QQQ	Student-t	0.911	3

A.8.5 Non-US Asset Class Extension and Per-Pair OoS Off-Diagonal Breakdown

The body universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) is six US-listed equity tickers from one country-index family. To probe whether the cross-asset construction generalises off the equity cluster we extend the universe with GLD (SPDR Gold Trust), a US-listed ETF whose underlying is physically-backed gold, i.e. a non-equity commodity asset class.

GLD Pipeline-A univariate fidelity. Fitting CHMM-N, CHMM-t, and CHMM-L on the GLD return series at $K = 18$ produces 100% IS KS pass rate for all three families and observed IS excess kurtosis 3.6 matched closely by CHMM-L (2.2); on OoS, however, the KS pass rate collapses to 0% for all three families against an observed OoS excess kurtosis of 6.5. Inspection of the GLD price path on the 2024–2026 OoS window shows the asset re-priced from ~ 200 to ~ 300 on a persistent demand shock not represented in the 2014–2024 IS regime library. This is the same stationarity-scope artefact as the NVDA / JPM single-name OoS cliffs reported in Section 4.4, on a non-equity asset; the limitation is not equity-specific.

Pipeline-B: 7-ticker Student-t copula off-diagonal MAE. Refitting the Student-t copula on the augmented seven-ticker universe at $\nu^\star = 6$ gives IS off-diagonal MAE of 0.029 (six-ticker baseline: 0.027) and OoS off-diagonal MAE of 0.179 (six-ticker baseline: 0.209). Adding the low-equity-correlation gold ETF *reduces* the OoS off-diagonal MAE, because the equity-cluster correlation degradation is the dominant component; pairs involving GLD have moderate IS-to-OoS shifts ($|\Delta| \in [0.04, 0.19]$) while the largest equity-cluster pairs (JNJ-QQQ, SPY-JNJ, NVDA-JNJ) have $|\Delta| > 0.48$ on OoS.

Per-pair OoS gap localisation. Table 19 reports the per-pair off-diagonal absolute deviation between observed and path-averaged simulated correlation matrices on IS and OoS, sorted by OoS $|\Delta|$. The OoS gap concentrates in pairs involving JNJ (the defensive single-name proxy): JNJ's IS correlation with the broad equity factor (QQQ, SPY) collapses on OoS, in some pairs flipping sign, and the IS-fixed copula has no mechanism for tracking that shift. The gold-ETF pairs (SPY-GLD, JPM-GLD, JNJ-GLD, QQQ-GLD) sit in the middle of the ranking with $|\Delta|$ between 0.10 and 0.19 on OoS. The dominant cause of the OoS off-diagonal gap is therefore neither equity-cluster size nor copula-degree mis-specification but a single-name regime shift in the JNJ-equity-factor relationship that the IS-fitted dependence layer cannot anticipate.

Table 19. Per-pair off-diagonal correlation MAE on the seven-ticker universe (Pipeline B, Student-t copula on CHMM-N marginals at $K = 18$, 200 paths, $\nu^\star = 6$). For each ticker pair (i, j) the table reports $|\Sigma_{\mathrm{sim,avg}}[i, j] - \Sigma_{\mathrm{obs}}[i, j]|$ on IS and OoS, ranked by OoS $|\Delta|$. Only the ten largest-OoS-gap pairs are shown for compactness.

Pair	IS $\lvert\Delta\rvert$	OoS $\lvert\Delta\rvert$	OoS − IS
JNJ-QQQ	0.027	0.544	0.517
SPY-JNJ	0.037	0.509	0.473
NVDA-JNJ	0.017	0.489	0.473
NVDA-AAPL	0.011	0.278	0.268
JNJ-JPM	0.046	0.261	0.216
JNJ-AAPL	0.042	0.244	0.202
AAPL-QQQ	0.003	0.227	0.224
JPM-GLD	0.027	0.187	0.160
SPY-GLD	0.038	0.127	0.089
JNJ-GLD	0.015	0.117	0.101

A.9 Pre-OoS Validation K-Selection and ν_k Diagnostics

Pre-OoS validation K-selection. The $K = 18$ operating point in the main panel was chosen via a multi-objective criterion combining information-criterion (IC) rank with IS and OoS distributional pass rates, which uses the same 2024–2026 window later used for VaR. As a clean re-selection that decouples the two, we fit CHMM-N on 2014-01-03 through 2021-12-31 ($n = 2{,}013$) and score every K in the sweep by the forward-algorithm held-out log-likelihood on 2022-01-03 through 2024-01-03 ($n = 502$); the 2024–2026 window is not touched. Held-out log-lik, BIC, HQC, and CAIC all select $K^\star = 3$ (per-observation log-lik -2.5345 at $K = 3$ vs. -2.5694 at $K = 18$); AIC selects $K^\star = 6$; held-out two-sample KS selects $K^\star = 9$. None of the six criteria that avoid the OoS window select $K = 18$. The $K = 18$ choice is best read as a multi-metric one that weights tail fidelity (kurtosis, AD) and downstream VaR behaviour alongside likelihood; a parallel main panel at $K = 3$ would re-rank some rows in CHMM's favour on ACF-MAE and against it on tail fidelity. The 2022–2023 validation slice is itself partly a structural-break period (the rate-hike cycle), so any model with substantial IS-specific structure generalises worse on it.

Penalised-ECM rate sweep. The bracket sensitivity is reported with its table in Section A.2.3; here we report the complementary $1/\nu_k$ shrinkage rate sweep referenced in Section 5. An exponential $1/\nu_k$ prior at rate $\lambda \in \{0, 5, 20, 50, 100, 200\}$ reduces simulated excess kurtosis monotonically: $14.30, 11.19, 8.43, 5.41, 3.96, 3.67$. The recommended rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 (close to the observed 7.69) at a 1pp IS KS cost. Only the $\nu_{\min}$ state parameter actually moves ($2.1 \to 4.89$); upper bracket and median remain at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under unpenalised ECM.

Cross-asset summary. The full per-construction Pipeline B panel is in Section A.8; for reference, the headline ordering is summarised in Table 20.

Table 20. Cross-asset dependence summary (Pipeline B; CHMM-N marginals at $K = 18$; 200 paths; seed `20260422`). Median per-asset KS pass rate (%) and off-diagonal MAE of the simulated correlation matrix vs. observed. Bold marks the column winner. The Student-t copula at $\nu^\star = 6$ is the strongest dependence layer on the two IS columns; on OoS the Gaussian copula attains marginally lower off-diagonal MAE (0.202 vs. 0.209) while the truncated C-vine sits between the elliptical copulas and the SIM baseline. Full per-asset detail is in Section A.8; the C-vine edge summary is in Table 18.

	Median IS KS	Median OoS KS	Off-diag MAE IS	Off-diag MAE OoS
Single Index Model	77.0	87.0	0.076	0.252
Gaussian copula	97.8	**87.5**	0.030	**0.202**
Student-t copula ($\nu^\star = 6$)	**98.8**	85.8	**0.027**	0.209
Truncated C-vine (root SPY)	97.2	86.0	0.068	0.235